

Kinematics and Dynamics of Machines MEL3020

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Course Content

Title	Kinematics and Dynamics of Machines	Number	MEL3XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-2 [4]
Offered for	B.Tech (ME)	Type	Program Compulsory (PC)
Prerequisite	Engineering Mechanics		

Objectives

1. To cover the kinematics and dynamics of planar single degree freedom mechanisms
2. To develop skills for designing and analyzing linkages, cams, gears and other mechanisms
3. To teach students concepts of Newtonian dynamic analysis of mechanisms and machines

Contents

Part I: Kinematics [21 Lectures]

Kinematics and Mechanism [2 Lectures]: Links, Kinematic pairs, Kinematics diagram and inversion, mobility, range of movements, Equivalent linkage

Motion characteristics of Mechanism [5 Lectures]: Displacement analysis of planar mechanisms analytical methods, Plane motion of a rigid body, Velocity analysis: analytical and Instantaneous Centre of Velocity methods, Coriolis component of acceleration, Acceleration analysis

Mechanism Synthesis [4 Lectures]: Function, Path, and Motion Generation, Three Position Dimensional Synthesis

Cam follower mechanism [5 Lectures]: Classification of Cam and followers, nomenclature, description and analysis of follower motion, pressure angle, Determine of basic dimensions and synthesis of cam profiles

Gear and gear train [5 Lectures]: Spur gears and gear terminology, Fundamental law of gearing, Involute profile, Interference and undercutting, Minimum number of teeth, Contact ratio, Helical gears on parallel shafts, Bevel helical, Spiral and Worm gears. Simple, compound, epicyclic gear trains, planetary gear trains.

Course Content

Part I: Dynamics [21 Lectures]

Introduction to Dynamics [2 Lectures]: Dynamics lumped-parameter models of mechanical systems, Newton's Law, Inertia and D'Alembert Principle, Energy Method

Dynamics of Rigid bodies [6 Lectures]: Solution Methods: superposition and simultaneous equation solution, Newtonian Approach, Shaking force and Shaking Moments, Applications to Linkage and Single and Multi-cylinder engines, Controlling Input Torque - Flywheel Design

Balancing [5 Lectures]: Static unbalance, static balancing machines, dynamics balancing, applications to balancing of linkage, single and multi-cylinder engines

Vibration Analysis [4]: Introduction, Basic features of vibratory systems, single degree freedom systems

Dynamics of Cams [3 Lectures]: Rigid and elastic body cam system, analysis of eccentric cams, analysis of elastic cam systems, lumped models of cams and vibration analysis.

Gyroscope [1 Lecture]: Gyroscopic Torque

Books:

- Kinematics, Dynamics, and Design of Machinery. Waldron and Kinzel
- Kinematics and Dynamics of Machinery– R L Norton
- Theory of Machines and Mechanisms – Shigley

Evaluations

Evaluation policy for the course:

Components	Weightage (%)	Remarks
Quizzes	10	Written (4)
Minor	20	Written Examination
Major	40	Written Exam
Projects	5	Hardware/ simulation submissions
Lab (regular assessment)	15	Lab work and quiz
Lab (final exam)	10	Lab activity/ quiz/ viva

Overview of Engineering

Overview of Mechanical Engineering

Definitions

Kinematics

Dynamics

Machines and Mechanisms

Analysis and Design (synthesis)

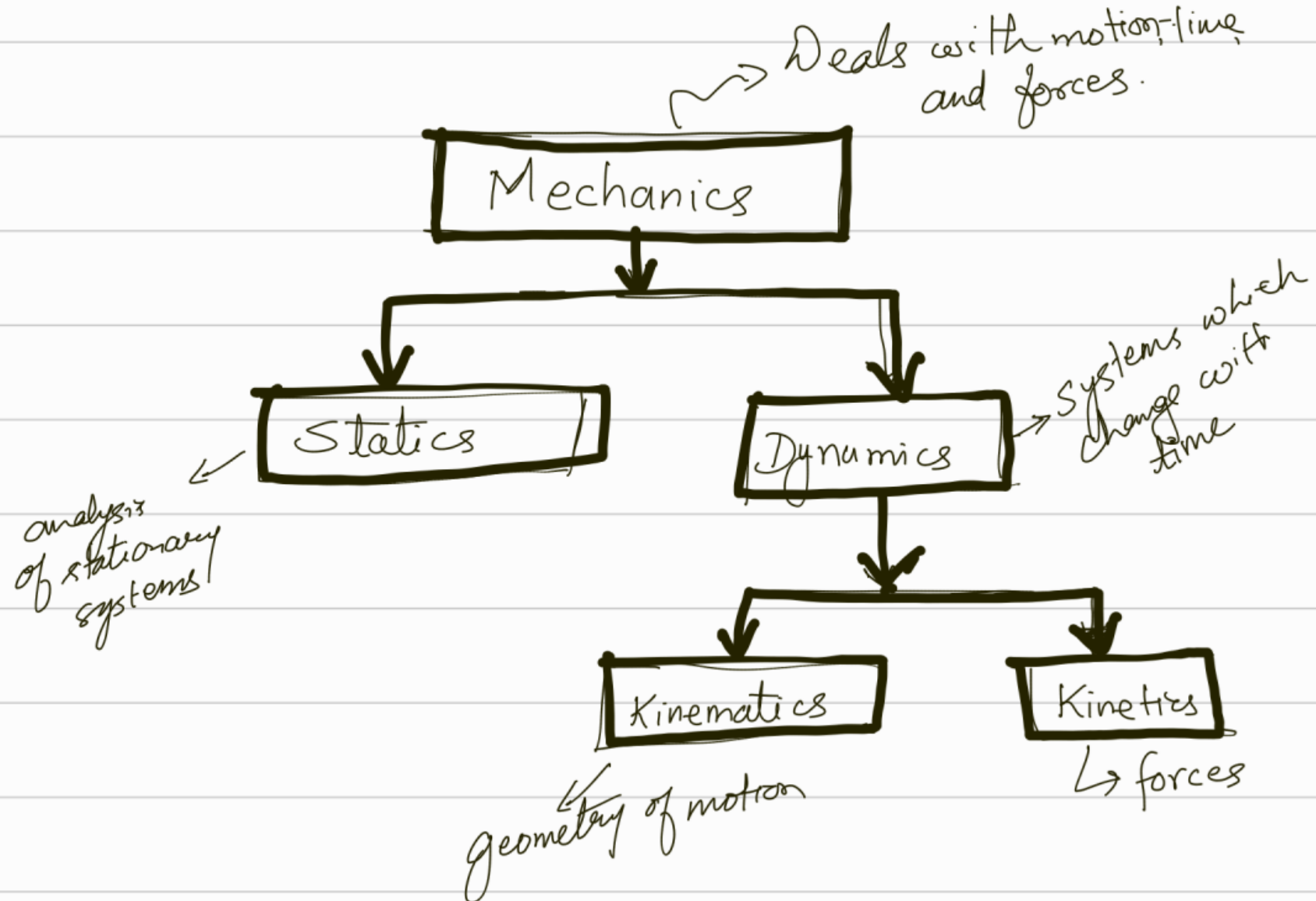
Links

Joints or Kinematic Pairs (lower pairs and higher pairs)

Kinematic line diagram

Degrees of freedom

Mechanics



Kinematics and Dynamics

Kinematics is the study of motion without studying the forces that might cause the motion.

→ Study of position, displacement, rotation, speed, velocity, acceleration, and jerk.

Dynamics is the study of motion as well as the forces that cause the motion.

→ Apply Newton's second law of motion. Find accelerations from Kinematics and then find forces using Newton's second law.

Example: A point mass undergoes a plane circular motion with a given angular velocity ω and given angular acceleration α , find the linear velocity and acceleration of the mass.

Mechanisms and Machines

Mechanism: “Mechanisms are assemblages of rigid members (links) connected together by joints. Mechanisms transfer motion and mechanical work from one or more actuators to one or more “output” members.”

Reuleaux’s (रूलो) definition: “Mechanisms are assemblage of resistant bodies connected by movable joints to form a closed kinematic chain with one link fixed and having the purpose of transforming motion.”

Example: Gear Mechanism, slider-crank mechanism

Machine:

A **machine** applies mechanical force, energy, or power for useful purposes. A machine can be composed of several mechanisms.

Reuleaux’s definition: “A machine is a combination of resistant bodies so arranged that by their means the mechanical forces of nature can be compelled to do work accompanied by certain determinate motions.”

Simple machines: lever, pulley

Complex machines: Excavator, IC Engine

Definitions

Mechanism Synthesis/ Mechanism Design: Design or synthesis is the process of prescribing the sizes, shapes material composition, and arrangement of parts so that the resulting machine will perform the prescribed task.

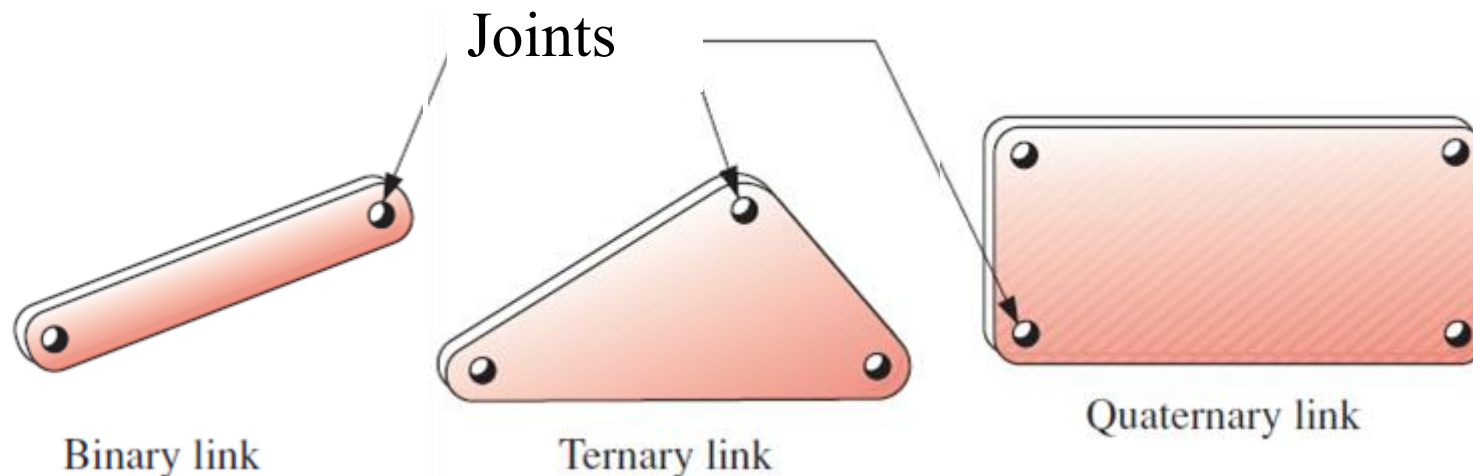
Mechanism Analysis: Analysis is the process of studying the motion of an existing or proposed design of a mechanism and to evaluate its suitability for a task.

Links

- The word link is used to designate a machine part or a component of a mechanism. A link is assumed to be completely rigid.

Types of links

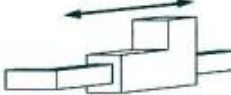
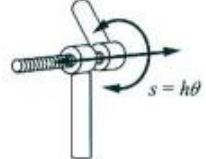
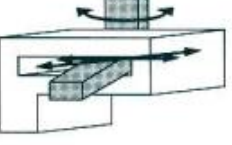
- **Binary link** - *one with two joints.*
- **Ternary link** - *one with three joints.*
- **Quaternary link** - *one with four joints.*



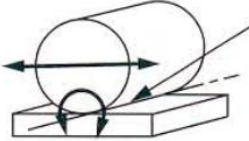
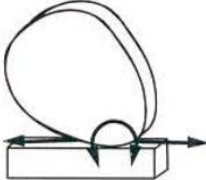
Joins

- A **joint** (also called **kinematic pairs**) is *a connection between two or more links which allows some motion between the connected links.*
- A kinematic joint is formed by direct contact between surfaces of two members.
- Classification of kinematic pairs by Reuleaux (रूलो), 1876
 - Lower pairs: contact between two rigid members occurs at every point of one or more surface segments.
 - Higher pairs: contacts occur only at isolated points or along line segments.
- All other things being equal a higher pair will produce higher contact stresses than will a lower pair.
- The **number of degrees of freedom of a joint** is equal to the number of independent coordinates needed to specify uniquely the position of one link relative to the other constrained by the joint.

Types of Joints – Lower pairs

Connectivity (Number of degrees of freedom)	Names	Letter symbol	Typical form	Sketch symbol
1	Revolute Hinge Turning pair	R		 (Planar)  (Spatial)
1	Prismatic joint Slider Sliding pair	P		 (Planar)  (Spatial)
1	Screw joint Helical joint Helical pair	H		 (Spatial)
2	Cylindrical joint Cylindrical pair	C		 (Spatial)
3	Spherical joint Ball joint Spherical pair	S		 (Spatial)
3	Planar joint Planar pair	P _L		 (Spatial)

Types of Joints – higher pairs

Connectivity (Number of degrees of freedom)	Names	Typical form	Comments
1	Cylindrical roller		Roller rotates about this line at this instant in its motion. Roller does not slip on the surface on which it rolls.
2	Cam pair		Cam rolls and slides on follower.
3	Rolling ball		Ball rolls without slipping.
4	Ball in cylinder		Ball can rotate about any axis through its center and slide along cylinder axis.
5	Spatial point contact		Body can rotate about any axis through the contact point and slide in any direction in the tangent plane.

Planar Linkage and Kinematic Chain

- A **kinematic chain** is any assemblage of rigid links connected by kinematic joints.
- A **frame** or base member is a link that is fixed. That is, it has zero degrees of freedom relative to the fixed coordinate system.
- A **linkage** is a closed kinematic chain with one link selected as a frame (fixed).
- A **closed chain** is one in which the links and joints form one or more closed circuits (you can come back to the same initial point while moving from one joint to another). Each link is connected to at least two other links.
- A **planar linkage** is one in which the velocities of all points in all members are directed parallel to a plane, called the plane of motion.
- The only lower pair joints that are compatible with planar motion are revolute and prismatic joints.
- Axes of rotation of all revolute joints is perpendicular to the plane of motion and the direction of sliding of all prismatic joints are parallel to the plane of motion.

Kinematic Diagram

Constraint Analysis

- Degrees of freedom of a rigid body
 - Connectivity (DOF) of a joint
 - Mobility of a mechanism
-
- Degrees of freedom of a rigid body
- “The number of degrees of freedom of a rigid body is the number of independent coordinates needed to specify uniquely the position of that body relative to a given reference frame.”

DOF of a rigid body in 2D and 3D.

Constraint Analysis

- Connectivity (DOF) of a joint

“The number of degrees of freedom of a joint is the minimum number of independent parameters required to specify the position of one link relative to the other connected by the joint.”

Constraint Analysis

- Degrees of freedom of a mechanism (mobility)

“The number of degrees of freedom (mobility) of a mechanism is the number of independent parameters/ coordinates needed to uniquely specify the position of all links relative to a particular member chosen as the base or frame.”

Definitions

- Kinematics:
- Dynamics:
- Mechanism:
- Machine:
- Analysis:
- Synthesis:
- Rigid Body:
- Linkage:
- Higher pair:
- Lower pair:
- Links:
- Joints:
- Kinematic pair:

For the linkage with $r_1 = 1$, $r_2 = 2$, $r_3 = 3.5$, $r_4 = 4$, and $\theta_1 = 0$ shown in Fig. 5.9, compute the corresponding values of θ_3 and θ_4 for each of the solution branches when the driving crank is in the positions $\theta_2 = 0, \pi/2, \pi$, and $-\pi/2$. Units for the lengths are not explicitly given in this example because the angular results are independent of the units for the lengths.

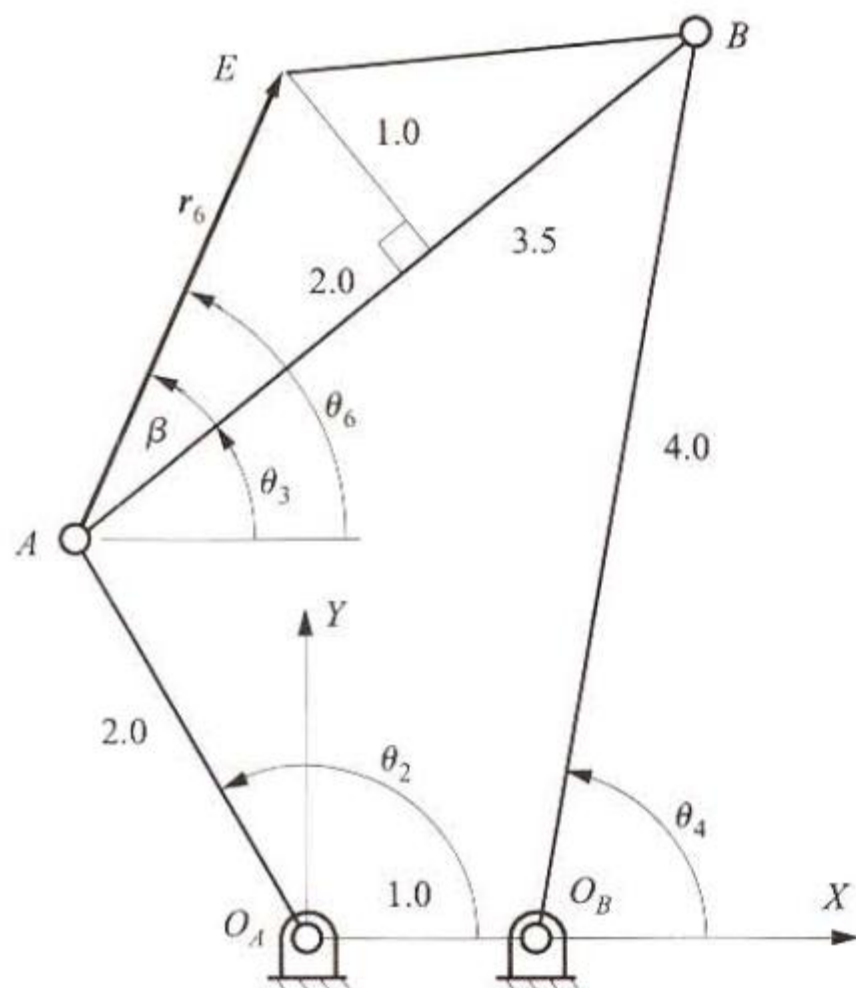
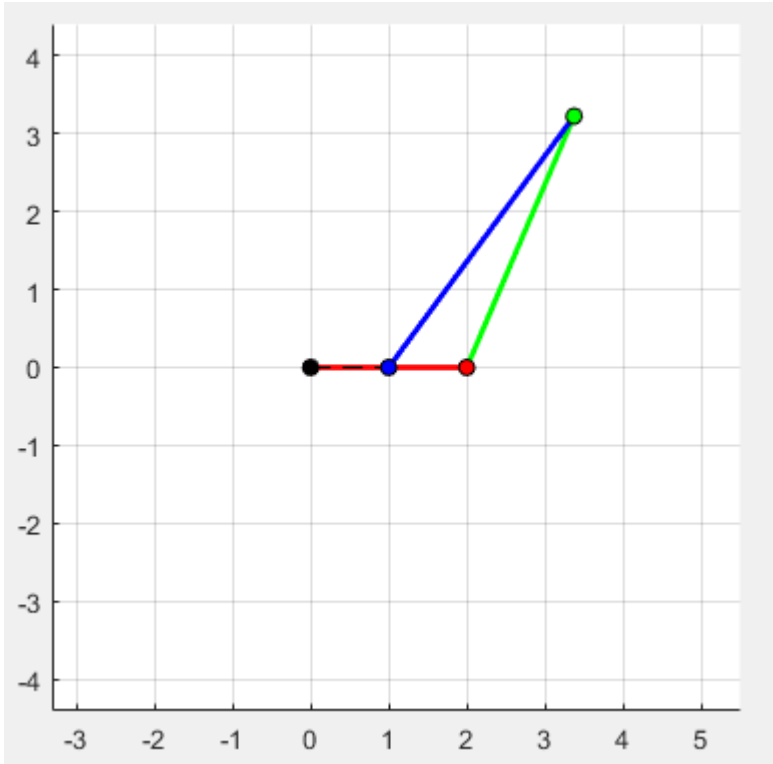
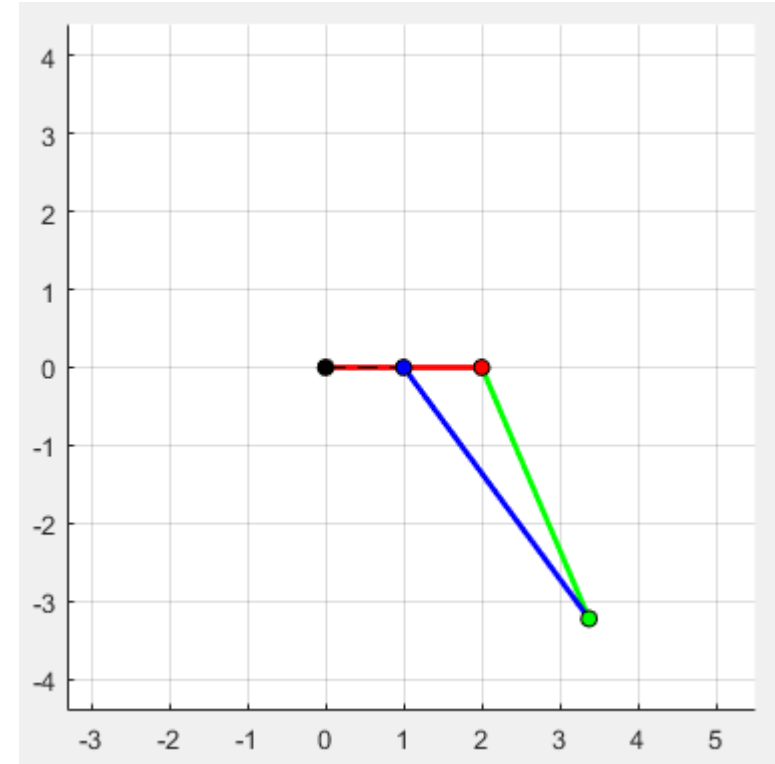


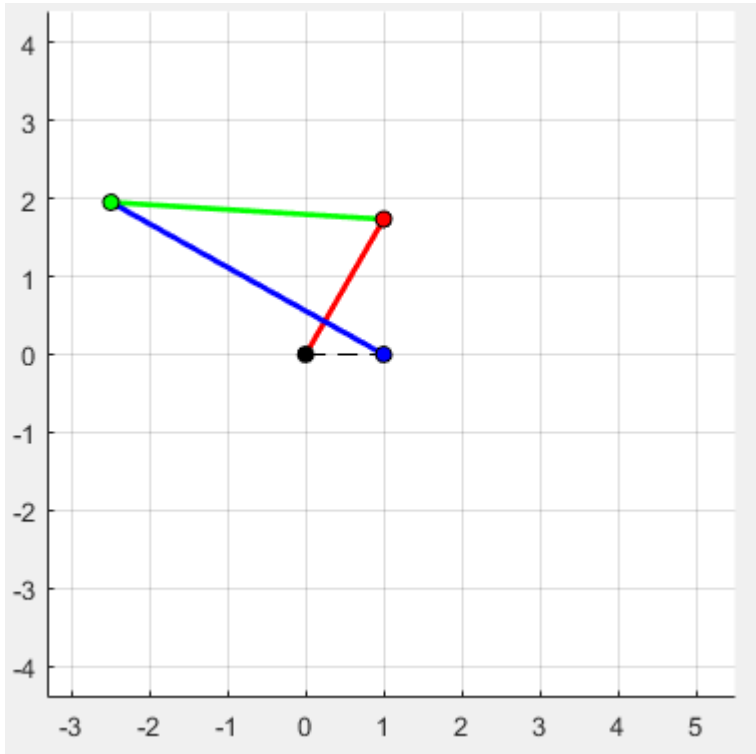
FIGURE 5.9 The linkage for Example 5.1, 5.2, and 5.3.



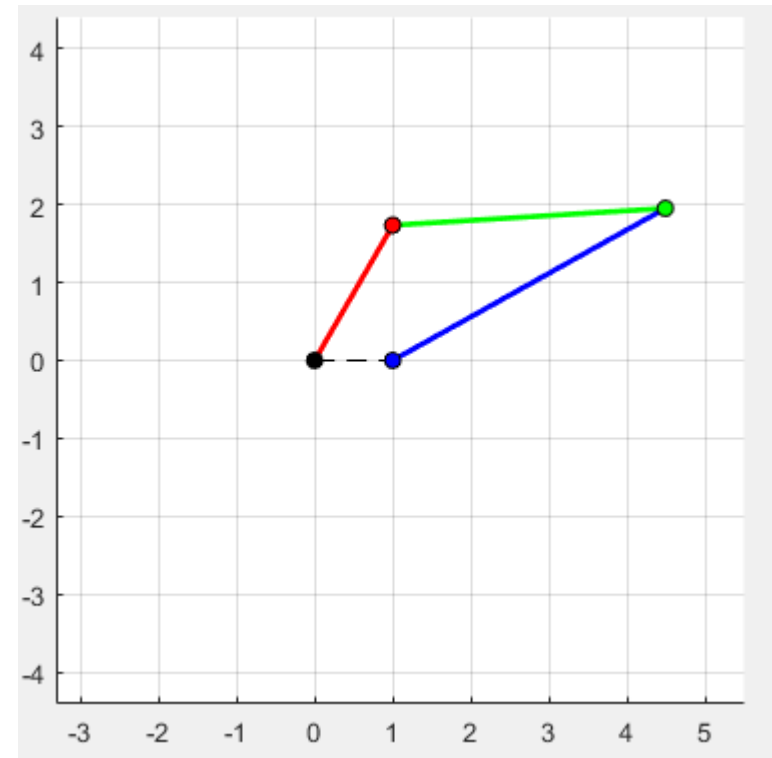
Solution 1 with $\sigma = 1$



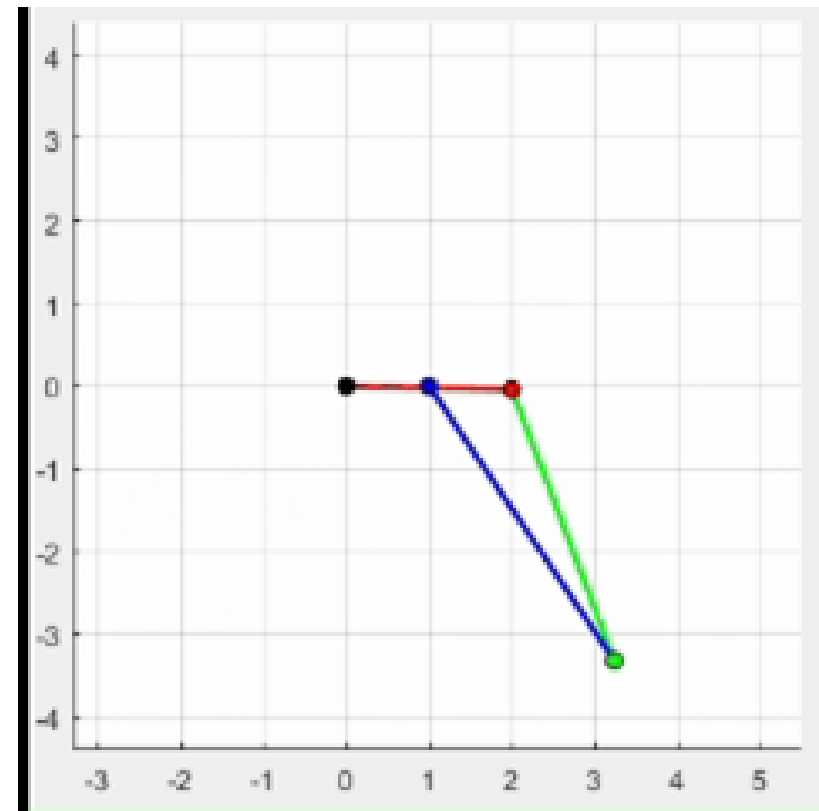
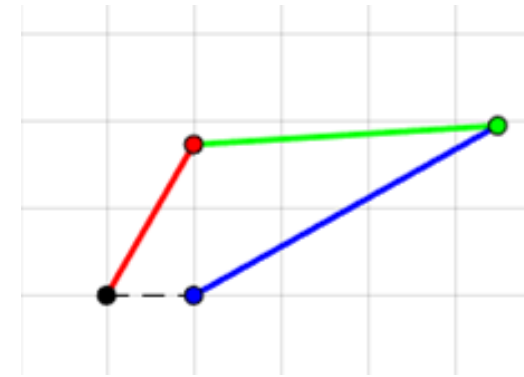
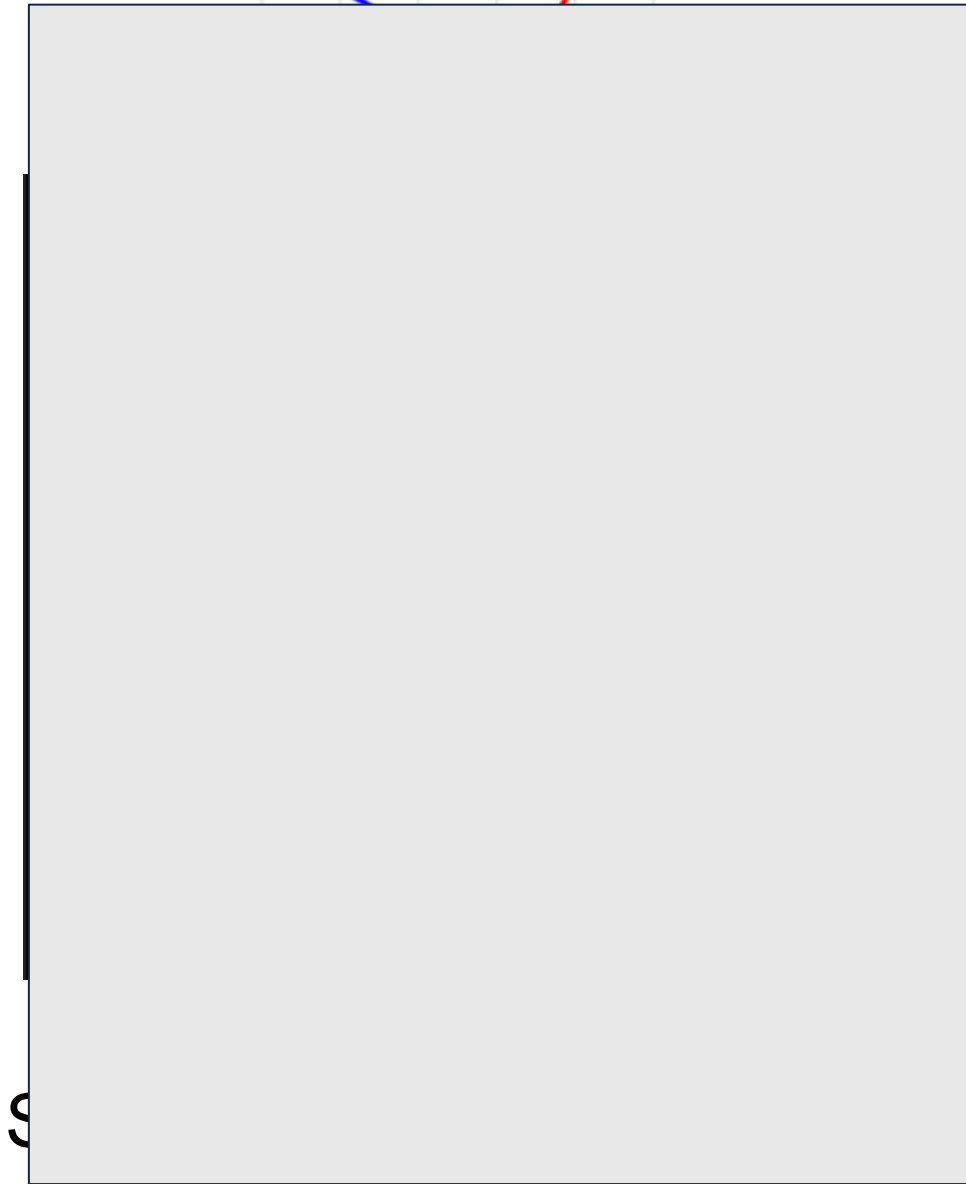
Solution 2 with $\sigma = -1$



Solution 1 with $\sigma = 1$



Solution 2 with $\sigma = -1$



Solution 2 with $\sigma = -1$

Instantaneous center of velocity

The instantaneous center of velocity is defined as the instantaneous location of a pair of coincident points of two different rigid bodies in planar motion for which the absolute velocities of the two points are equal.

The instantaneous center of velocity may also be defined as the location of a pair of coincident points of two different rigid bodies in planar motion for which the apparent velocity of one of the points is zero, as seen by an observer on the other body.

Intuition for the existence of instantaneous center of velocity

The instantaneous center of rotation/ velocity between a fixed and a moving body will be a point on the (possibly extended) body of zero velocity. We can easily see that such a point will exist as follows:

We know that the velocity of any point (B) on the body can be computed if we know the velocity any point (A) and the angular velocity of the body in planar motion.

Velocity Analysis

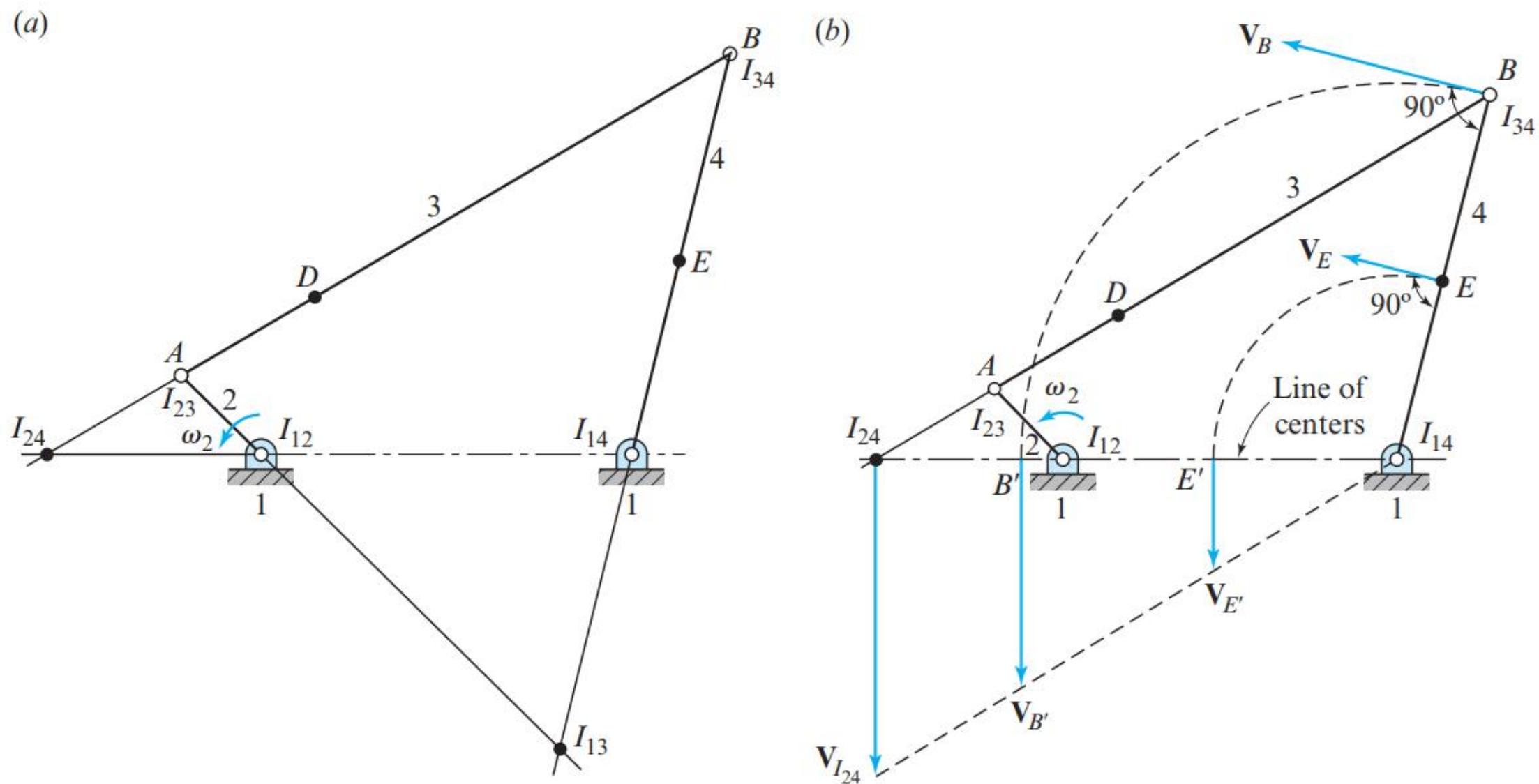


Figure 3.29 Graphic velocity analysis using instant centers.

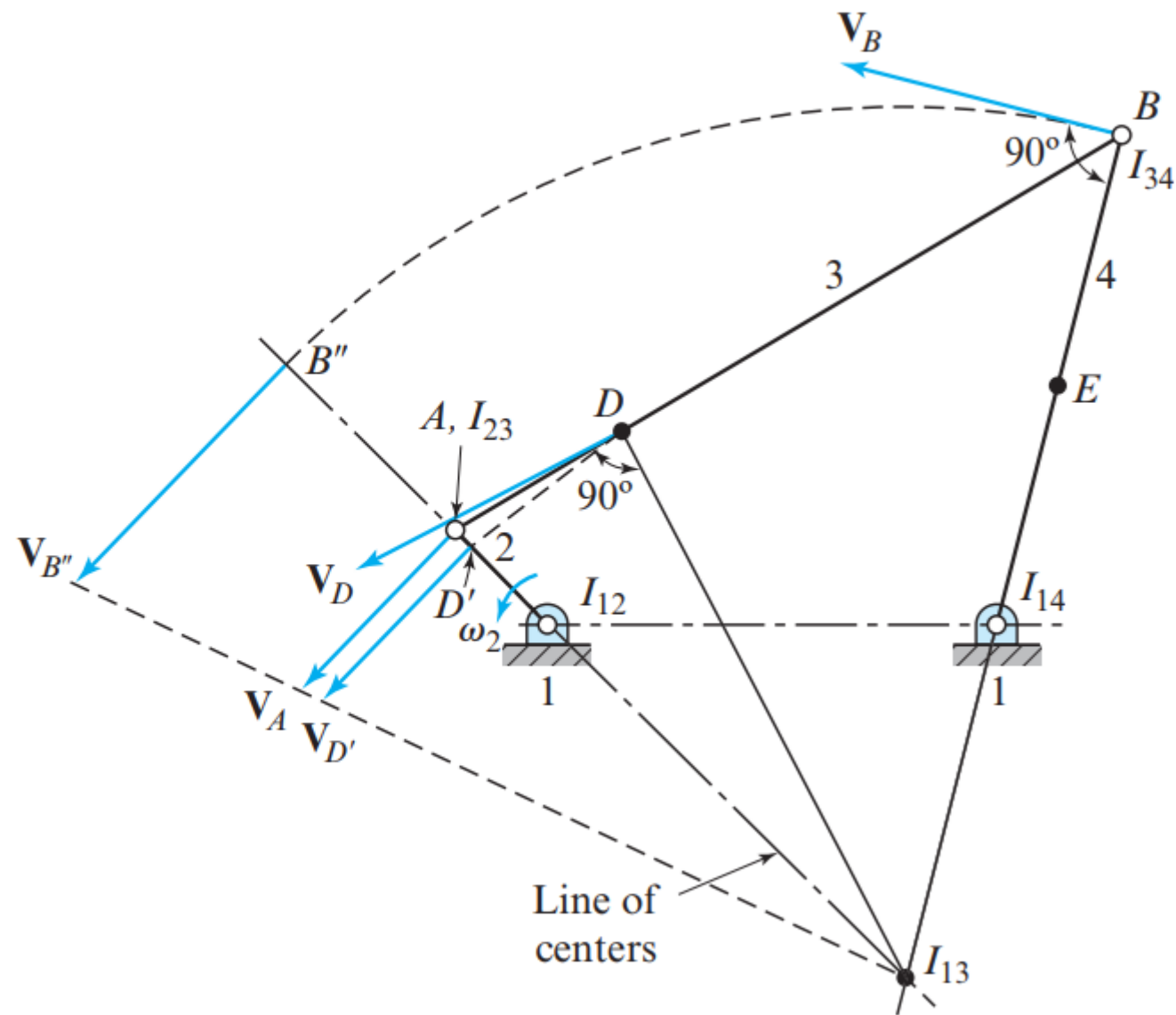
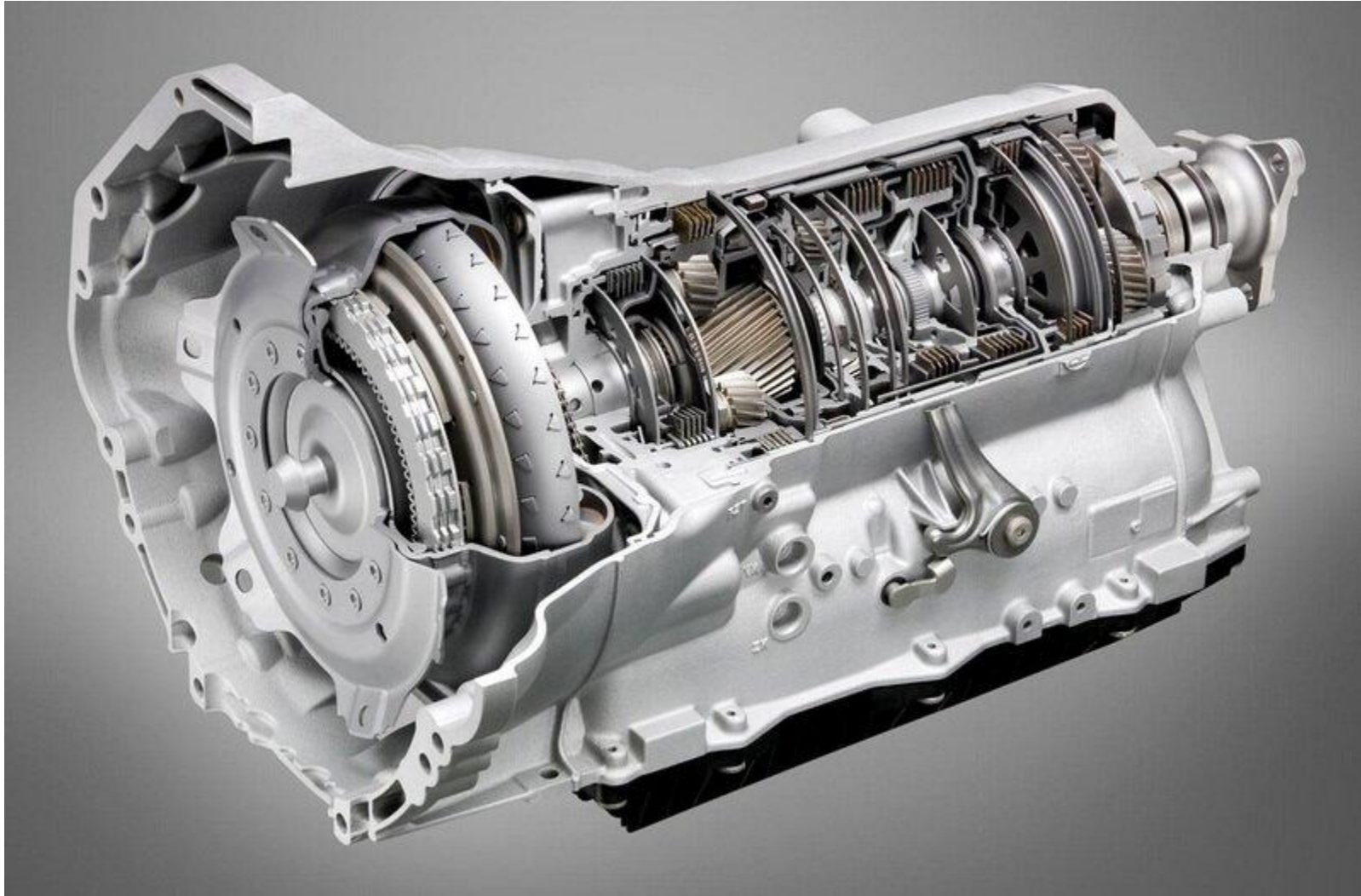


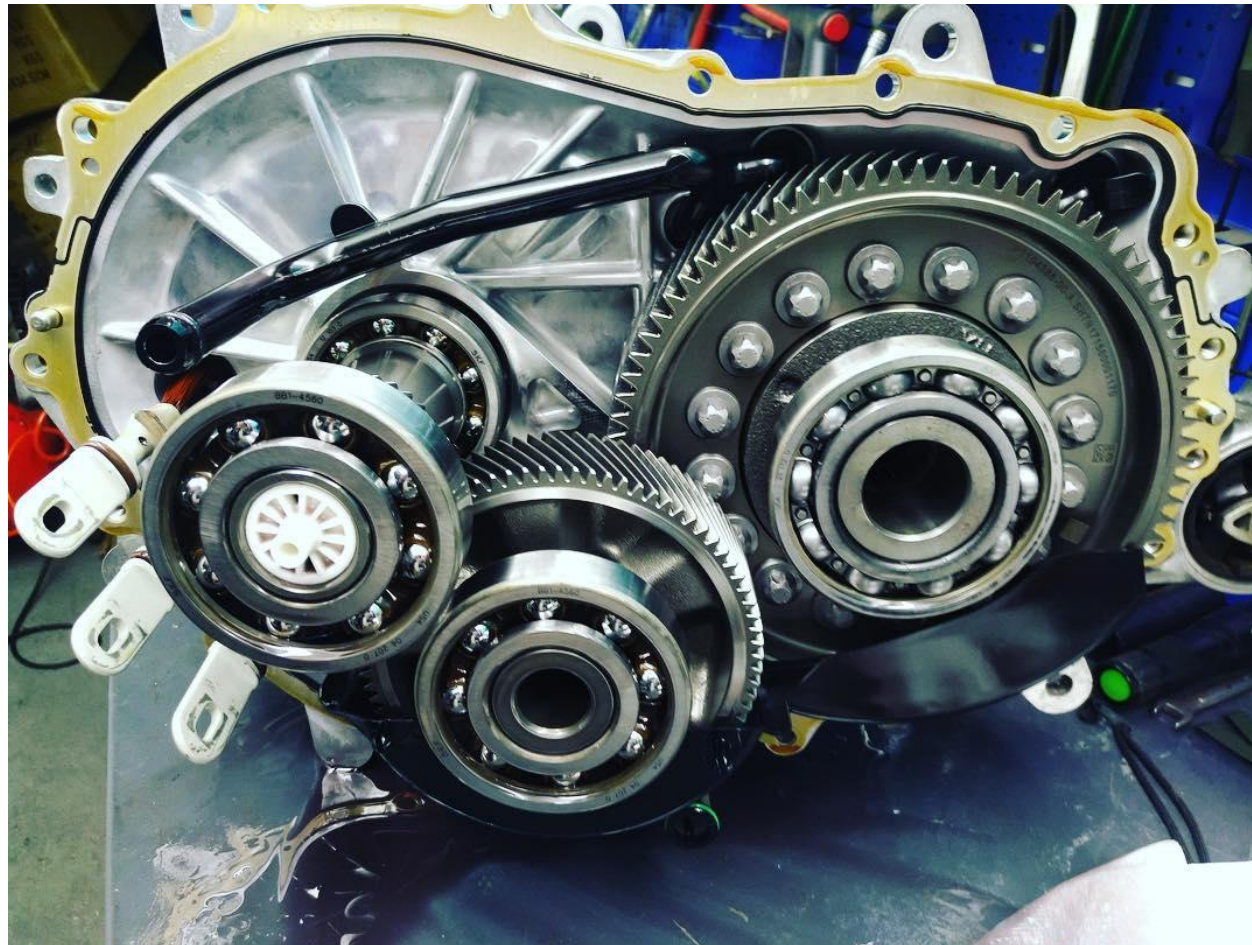
Figure 3.30 Velocities of pin B and point D .

Gear and Gear Train

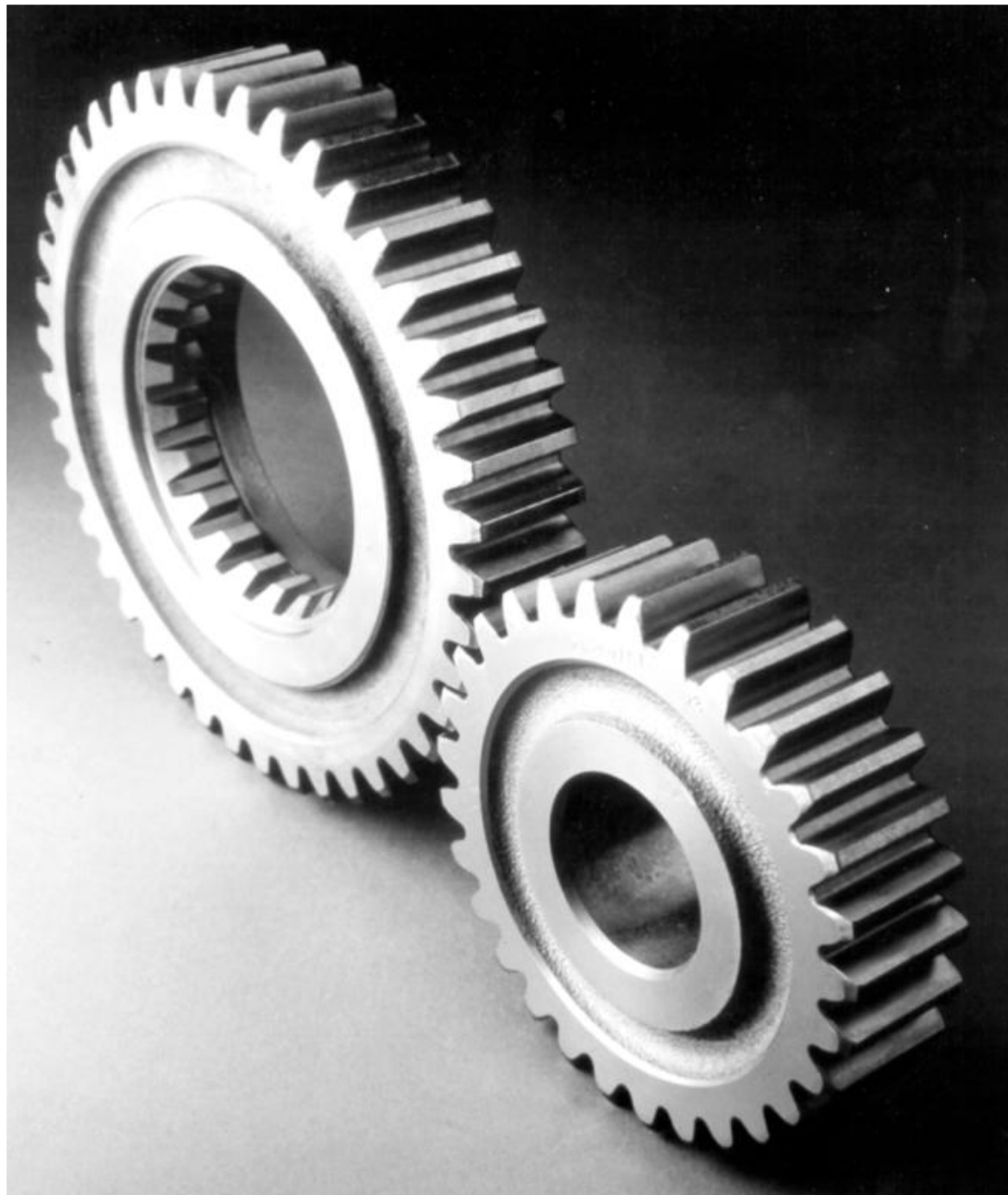
Car Gearbox

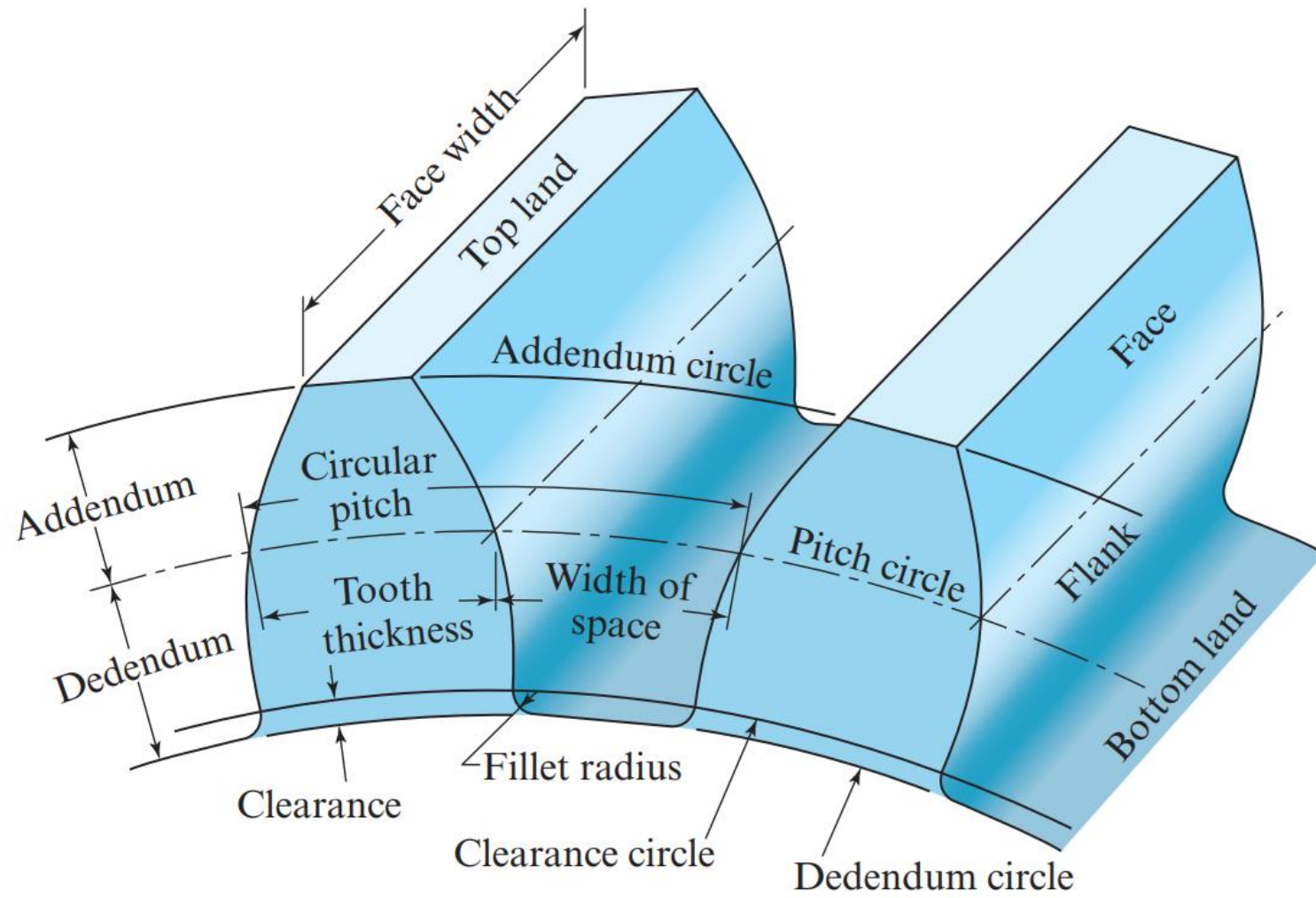


Single speed gearbox from a Tesla car









Terminology

- Spur Gears: Axes of two shafts are parallel and the teeth are straight and parallel to the axes of rotation of the shafts.
- Pinion: Smaller of the two mating gears.
- Pair of gears -> gearset.
- Pitch circle: the theoretical circle on which all the calculations are based. Pitch circles of mating gears are tangent to each other. Gears can be considered to be rolling on each other on the pitch circles without slip. Diameter of the pitch circle is the pitch diameter.
- Diametral pitch P is the ratio of the number of teeth on the gear to its pitch diameter. N is the number of teeth and R is the pitch circle radius. Used to indicate the tooth size in US customary units: teeth per inch.
- A pair of mating gears has the same diametral pitch.
- As the value of the diametral pitch become larger, teeth become smaller.

$$P = \frac{N}{2R},$$

Terminology

The module m is the ratio of pitch diameter of the gear to its number of teeth

$$m = \frac{2R}{N}.$$

It must be same for two meshing gears.

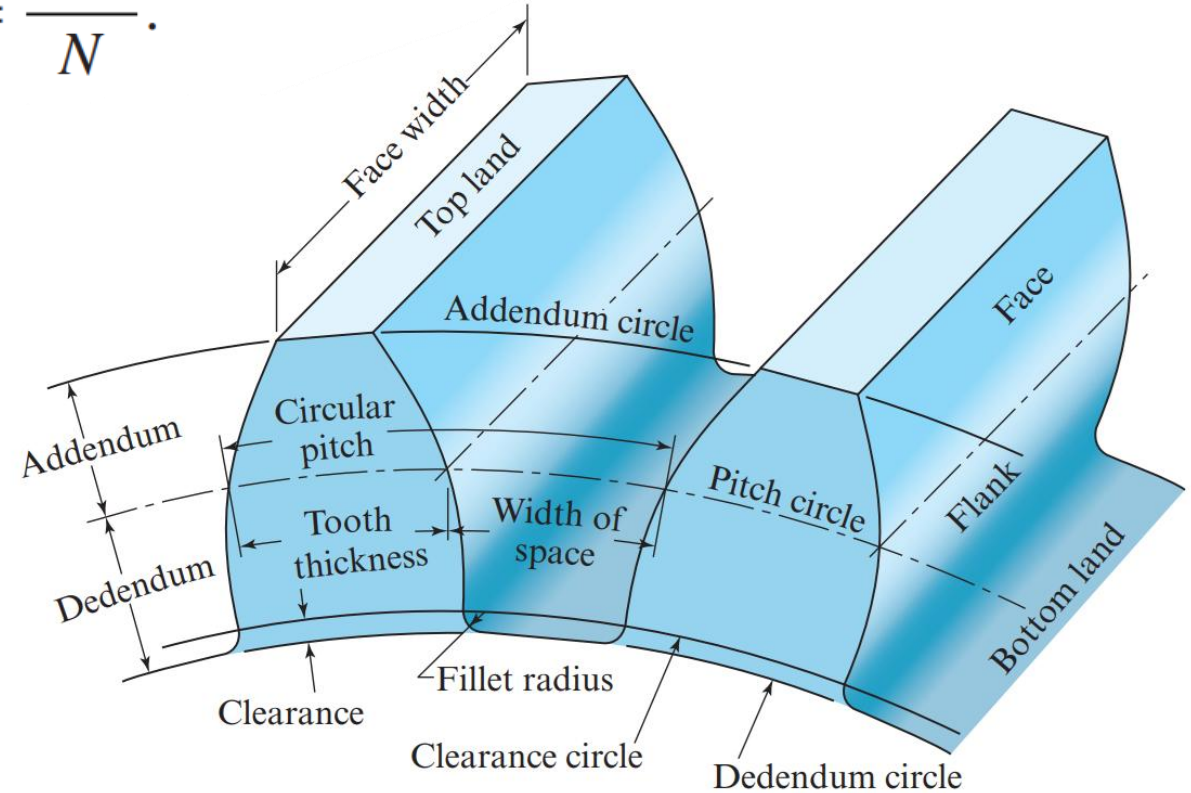
The module is the usual unit for indicating tooth size in International System (SI) units, and it customarily has units of millimeters per tooth. Note that the module is the reciprocal of the diametral pitch, and the relationship can be written as

$$m = \frac{25.4 \text{ (mm/in)}}{P \text{ (teeth/in)}} = \frac{25.4}{P} \text{ mm/tooth.}$$

Terminology

The *circular pitch*, p , is the distance from one tooth to the adjacent tooth, measured along the pitch circle. Therefore, it can be determined from

$$p = \frac{2\pi R}{N}.$$



Terminology

Circular pitch is related to the previous definitions, depending on the units, by

$$p = \frac{\pi}{P} = \pi m.$$

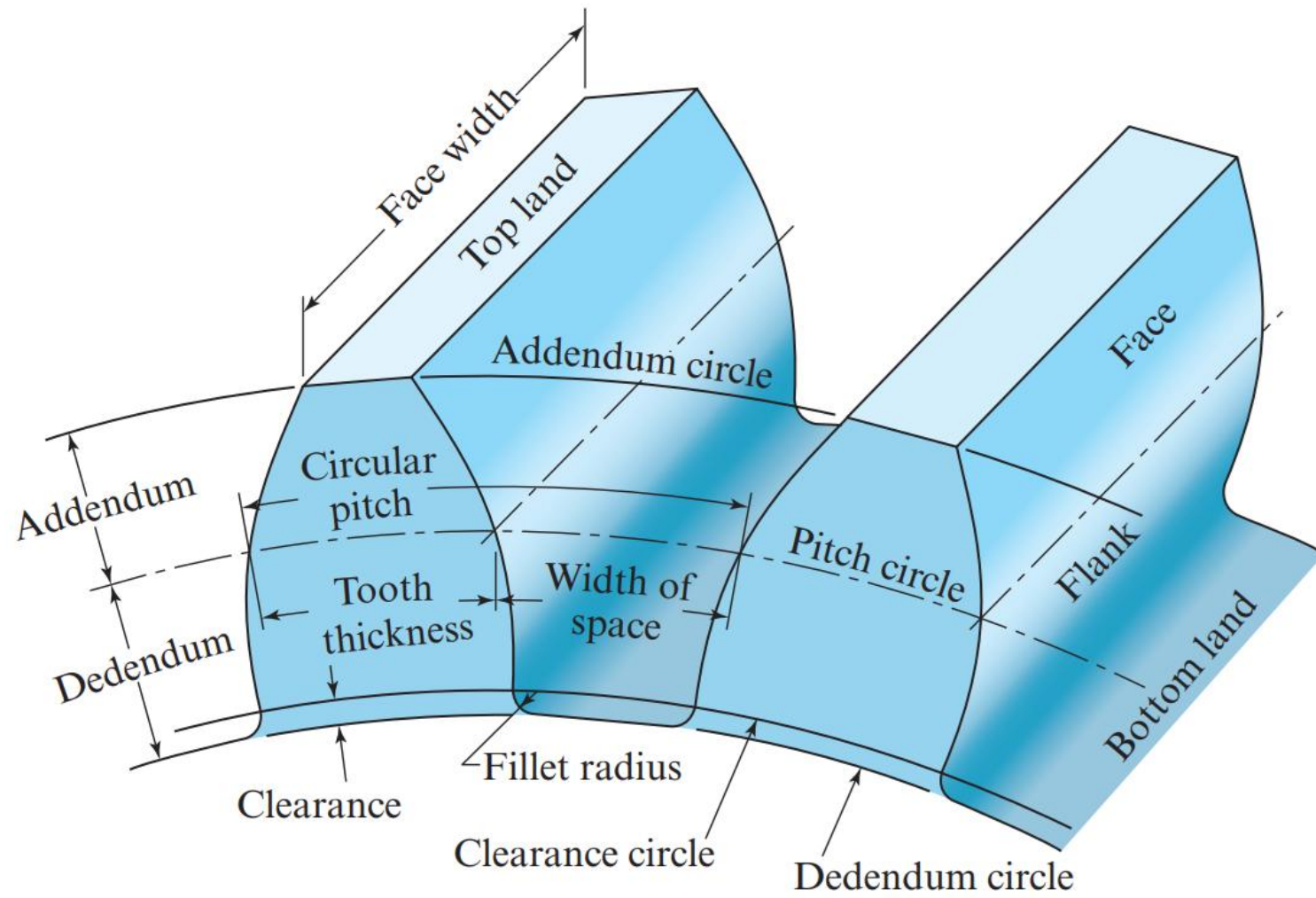
The *addendum*, a , is the radial distance from the pitch circle to the top land of each tooth.

The *dedendum*, d , is the radial distance from the pitch circle to the bottom land of each tooth.

The *whole depth* is the sum of the addendum and dedendum.

The *clearance*, c , is the amount by which the dedendum of a gear exceeds the addendum of the mating gear.

The *backlash* is the amount by which the width of a tooth space exceeds the thickness of the engaging tooth measured along the pitch circles.



Terminology

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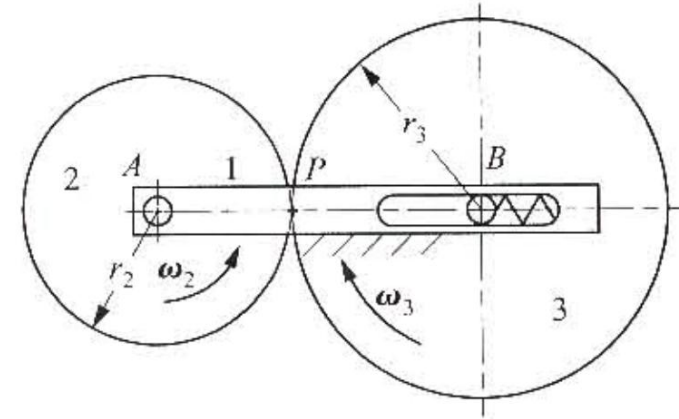
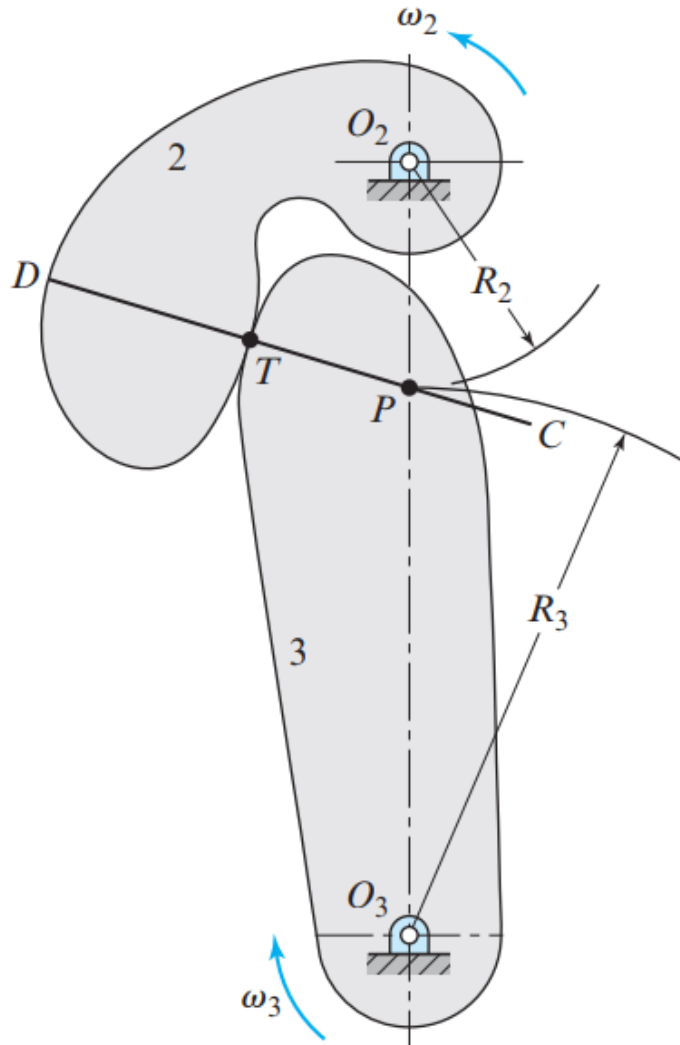
The *dedendum*, d , is the radial distance from the pitch circle to the bottom land of each tooth.

The *whole depth* is the sum of the addendum and dedendum.

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The *backlash* is the amount by which the width of a tooth space exceeds the thickness of the engaging tooth measured along the pitch circles.

Fundamental Law of Toothed Gearing



- Point P is the instant center for link 2 and link 3. It is also called the pitch point. It is the point of rolling for link 32 and link 3.
- The point can be located by observing the motion of link 3 from a frame fixed to link 2. If we can find the velocity any two points on link 3 relative to the frame fixed to link 2 with its origin at O_2 , we can determine the instant center by drawing the perpendicular to the velocity vectors and taking the intersection.
- If we observe the cam contact, we can determine the direction of velocity of point T and point O_3 on link 3 relative to a frame fixed on link 2 with origin at O_2 .
- Velocity of point T will be along the common tangent at the contact location and the velocity of point O_3 will be perpendicular to O_2O_3 . The point P will lie at the intersection of lines perpendicular to these velocity vectors. Thus on O_2O_3 .
- An easy rule is that **the pitch point lies on the intersection of common normal at the contact location and the line connection rotation axes.**

Fundamental Law of Toothed Gearing

A single pair of mating gear teeth must be shaped such that, as they pass through their entire period of contact, the ratio of the angular velocity of the driven gear to that of the driving gear *must remain constant*.

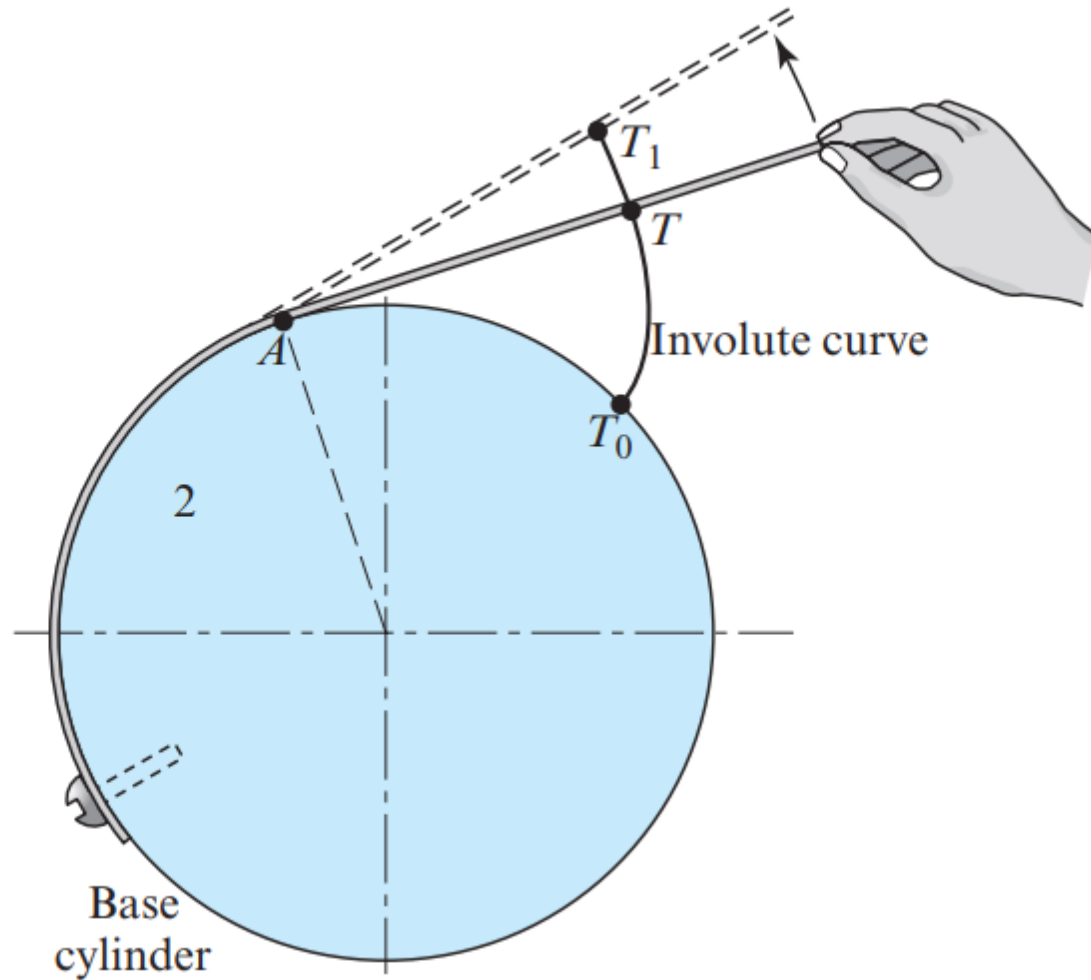
$$\frac{\omega_2}{\omega_3} = \frac{R_3}{R_2}.$$

This equation is frequently used to define the fundamental law of gearing, which states that as gears go through their mesh, *the pitch point must remain stationary on the line of centers* so that the speed ratio remains constant.

Conjugate tooth profile

- Tooth profiles (or cam and follower profile) that are shaped to produce a constant angular velocity between the two shafts are said to be **conjugate**.
- It is possible to specify an arbitrary profile for one tooth and then to find a profile for the mating tooth so that the two surfaces are conjugate.
- One possible choice for such conjugate profiles is the **involute profile**, which, with few exceptions, is in universal use for gear teeth.

Involute profile properties



- An *involute* curve is the path generated by a tracing point on a taut cord as the cord is unwrapped from a cylinder called the *base cylinder*.
- The cord is the generating line, and it is always normal to the involute.
- If the two mating tooth profiles both have the shapes of involute curves, the condition that the pitch point, P, remain stationary is satisfied.

Involute profile properties

If the two mating tooth profiles both have the shapes of involute curves, the condition that the pitch point, P , remain stationary is satisfied.

Involute profile properties

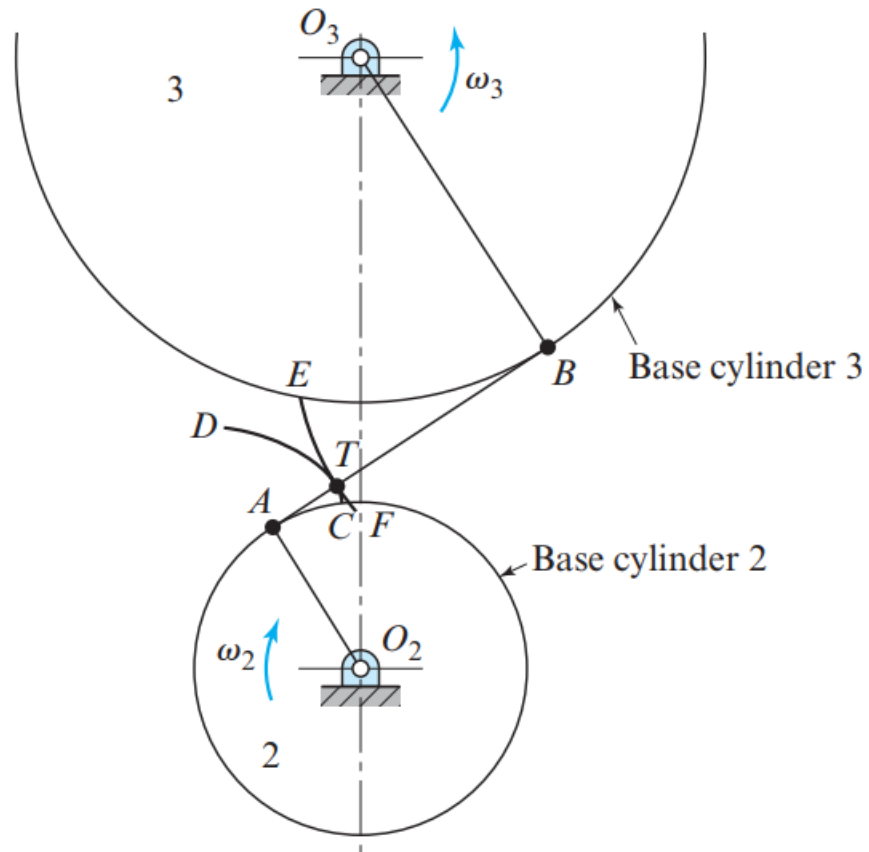


Figure 7.6 Conjugate involute curves.

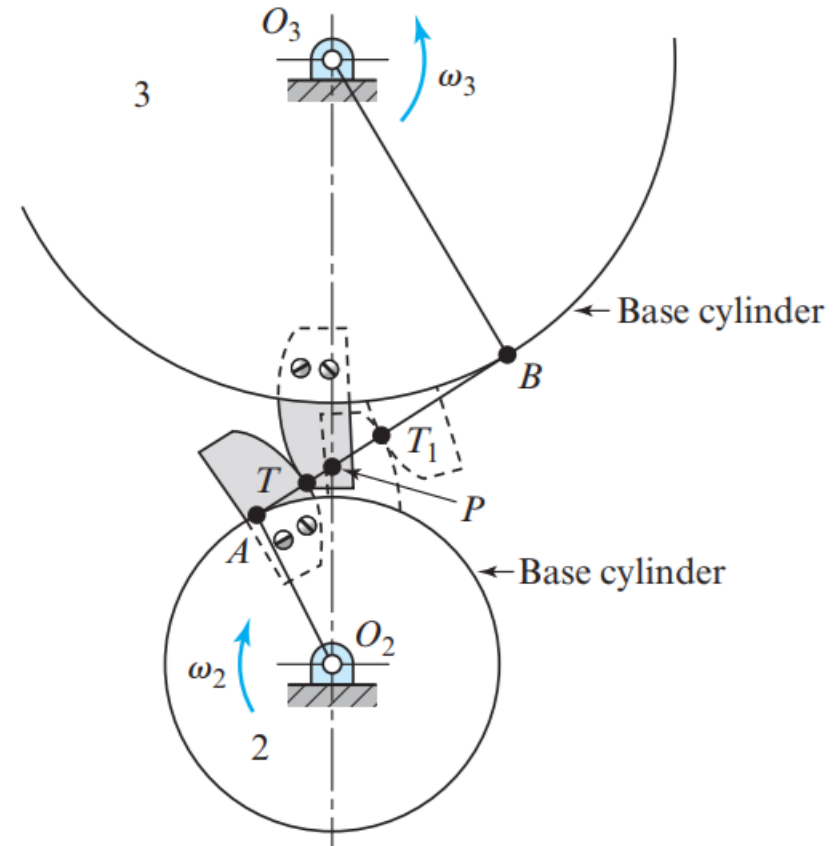
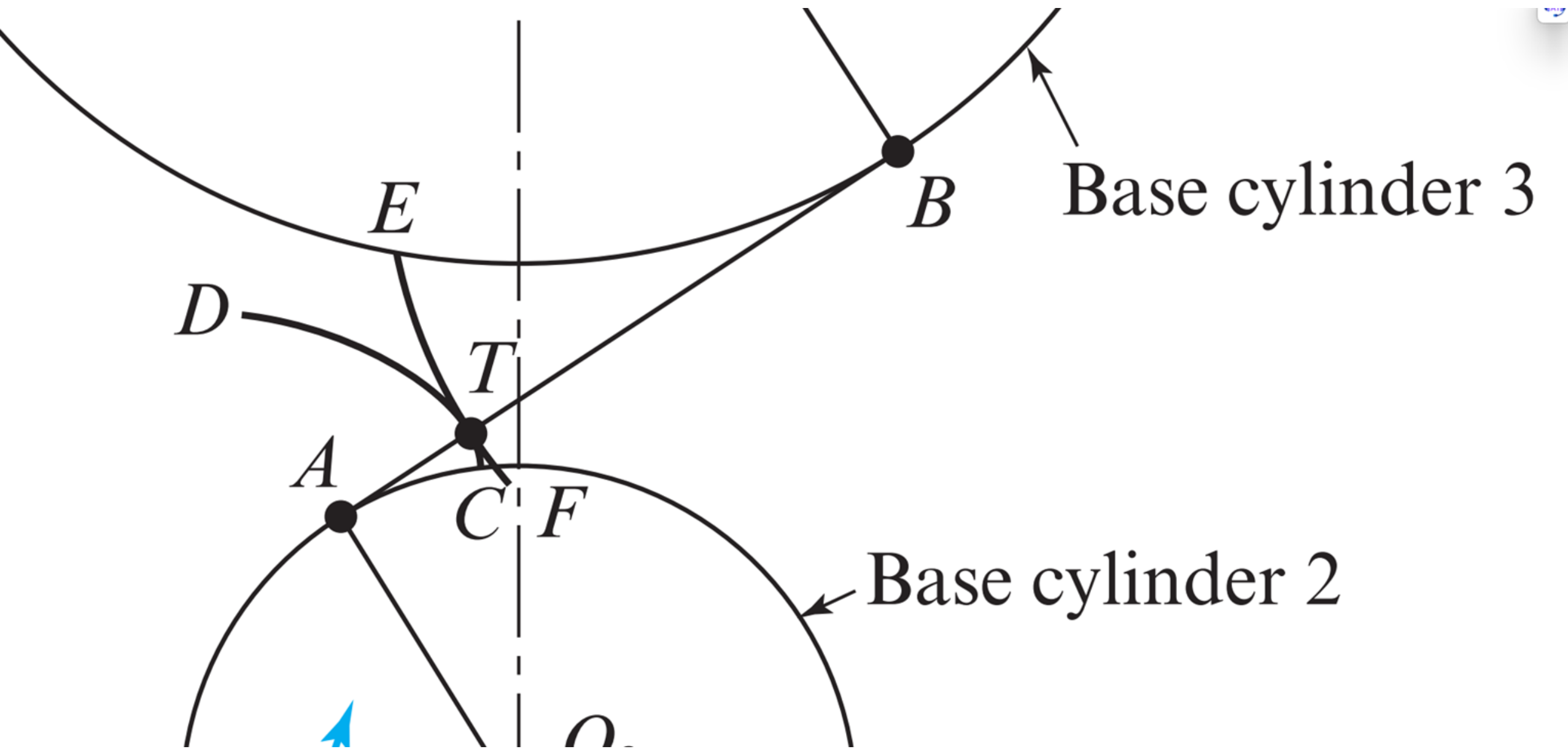
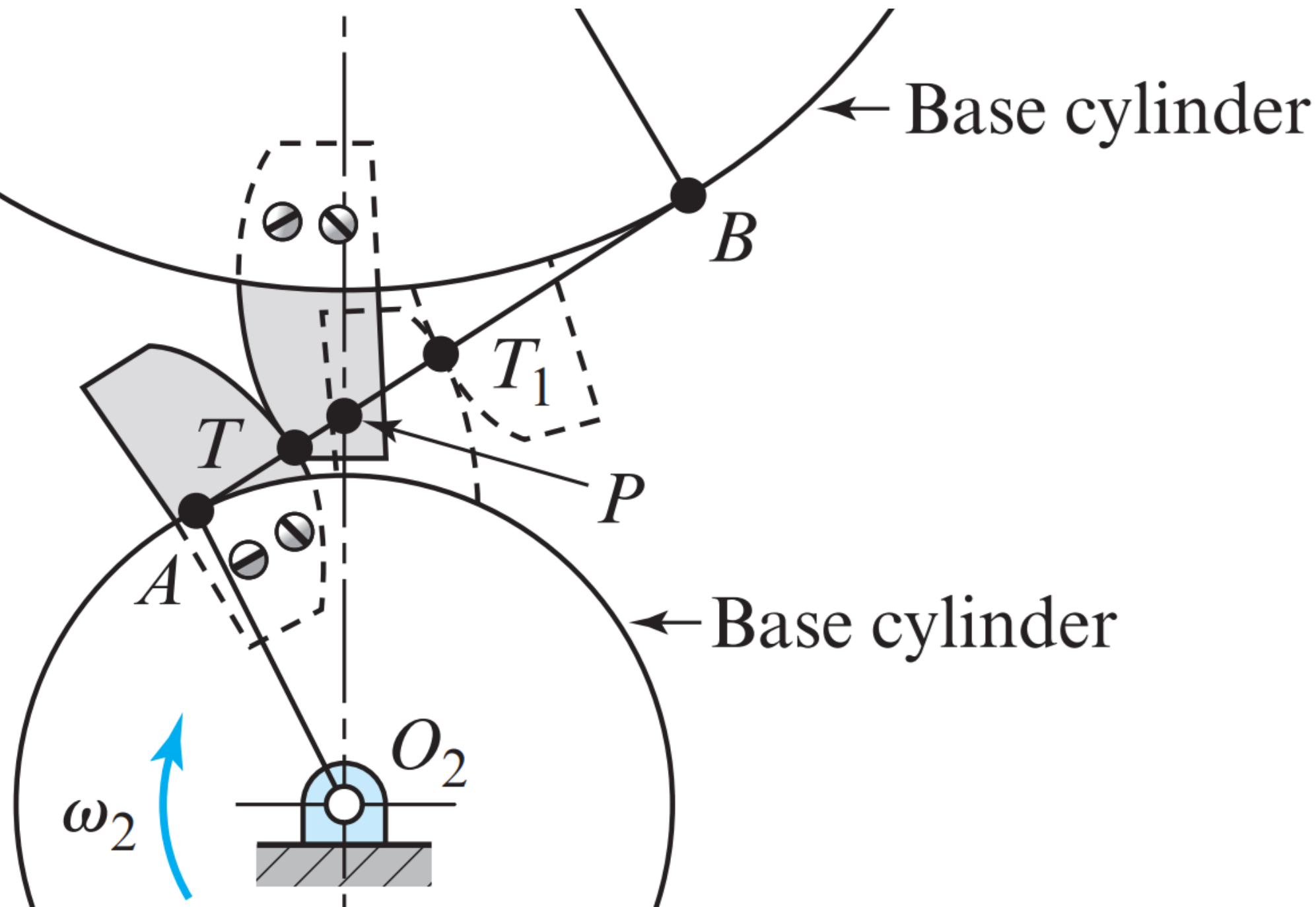


Figure 7.7 Involute action.



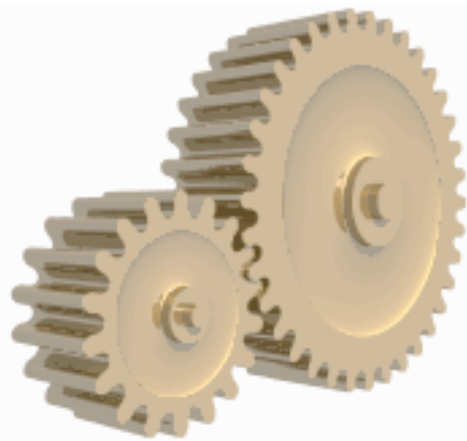


Arguments for the fact that the pitch point remains fixed for involute profiles for meshing gears

- In the involute, the cord remains perpendicular to the involute profile. The cord is also tangent to the base cylinder.
- AT is perpendicular to the involute EF and TB is perpendicular to the involute DC. AT and TB are tangents to the corresponding base circles.
- AT and TB lie in a straight line and AB is the common tangent for the base circles.
- As the gears rotate on each other, the point T moves along the common tangent AB. Another way of saying this is that the contact location T always lies on the common tangent AB.
- This common tangent to the base circles AB is also the common normal at the contact location.
- The pitch point will lie on the intersection of the common normal at the contact point (AB) and the line connecting the axes of rotations of the gears (O2-O3). Since the line AB and O2-O3 are always the same, the pitch point will be fixed on O2-O3.
- **Contact normal is the common tangent to the base circles.**

Involute profile properties

1.	The pressure angle remains constant throughout the engagement. This ensures smooth running
2.	Involute tooth profile consists of a single (involute) curve and the method of manufacture is simpler.
3.	Small variation in centre distance does not lead to violation of fundamental law of tooth gearing.



Example

Two parallel shafts, separated by a distance (commonly referred to as the *center distance*) of 3.5 in, are to be connected by a gear pair so that the output shaft rotates at 40% of the speed of the input shaft. Design a gearset to fit this situation.

SOLUTION

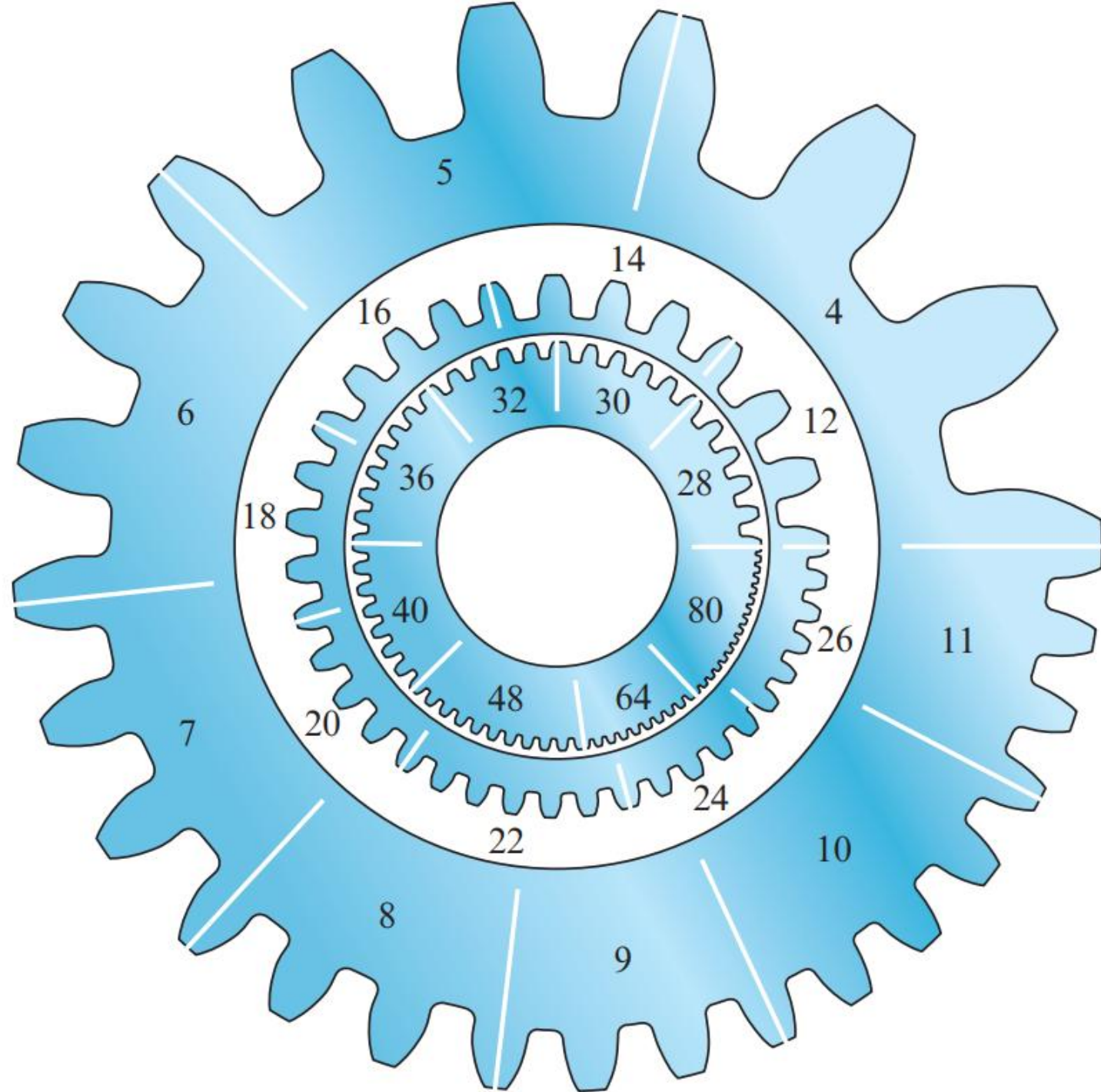
The center distance can be written as $R_2 + R_3 = 3.5$ in, and substituting the given information into Eq. (7.5), we have $\omega_3/\omega_2 = R_2/R_3 = 0.40$. Then, substituting the second equation into the first equation and rearranging, we find that $R_2 = 1.0$ in and $R_3 = 2.5$ in. Next, we must choose the size of the teeth by picking a value for the diametral pitch or module. From Eq. (7.1), we find the number of teeth on the two gears to be $N_2 = 2PR_2 = 2P$ and $N_3 = 2PR_3 = 5P$. The choice of P or m for tooth size is often iterative. First, we might choose a value of $P = 6$ teeth/in; this gives the numbers of teeth as $N_2 = 12$ teeth and $N_3 = 30$ teeth; if we choose $P = 10$ teeth/in, then we get $N_2 = 20$ teeth and $N_3 = 50$ teeth. At this time, either choice appears acceptable, and we choose $P = 10$ teeth/in. However, this choice of P (or m) must later be checked for possible undercutting, as we will study in Sec. 7.7; for contact ratio, which we will study in Sec. 7.8; and for strength and wear of the teeth [4].

Interchangeable Gears; AGMA Standards

- Gears cut to standard dimensions are interchangeable in the sense that any two standard gears can be meshed together.
- Gear manufacturers produce gears with standard dimensions.
- For gears to mesh, the following conditions are required:
 1. The **pressure angles** must be the same.
 2. The **diametral pitches** or module must be the same.
 3. The dedendum on a given gear must be greater than the addendum on the mating gear.
 4. The tooth thickness must be equal to one-half the circular pitch.
- A commonly used standard **pressure angle** is **20 degrees**.

Table 7.1 Standard Gear Tooth Sizes

Standard diametral pitches P, US customary, teeth/in	
Coarse	1, 1 $\frac{1}{4}$, 1 $\frac{1}{2}$, 1 $\frac{3}{4}$, 2, 2 $\frac{1}{2}$, 3, 4, 5, 6, 8, 10, 12, 14, 16, 18
Fine	20, 24, 32, 40, 48, 64, 72, 80, 96, 120, 150, 200
Standard modules m, SI, mm/tooth	
Preferred	1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12, 16, 20, 25, 32, 40, 50
Next choice	1.125, 1.375, 1.75, 2.25, 2.75, 3.5, 4.5, 5.5, 7, 9, 11, 14, 18, 22, 28, 36, 45



Interchangeable Gears; AGMA Standards

- Gears cut to standard dimensions are interchangeable in the sense that any two standard gears can be meshed together.
- Gear Specification: Number of teeth, pressure angle, diametral pitch or module, addendum, dedendum, clearance
- A ***tooth system*** is the name given to a *standard* that specifies the relationships among addendum, dedendum, clearance, tooth thickness, and fillet radius to attain interchangeability of gears of different tooth numbers but of the same pressure angle and the same diametral pitch or module.
- A relationship should be defined between the various quantities because all the quantities cannot be independently determined.
- For a pair of spur gears to properly mesh, they must share the same pressure angle and the same tooth size, as specified by the diametral pitch or module.
- The numbers of teeth and the pitch diameters of the two gears in mesh need not match but are chosen to give the desired speed ratio.

Table 7.2 Standard Tooth Systems for Spur Gears

System	Pressure angle, ϕ (deg)	Addendum, a	Dedendum, d
Full depth	20°	$1/P$ or $1m$	$1.25/P$ or $1.25m$
Full depth	$22\frac{1}{2}^\circ$	$1/P$ or $1m$	$1.25/P$ or $1.25m$
Full depth	25°	$1/P$ or $1m$	$1.25/P$ or $1.25m$
Stub teeth	20°	$0.8/P$ or $0.8m$	$1/P$ or $1m$

Internal Gear

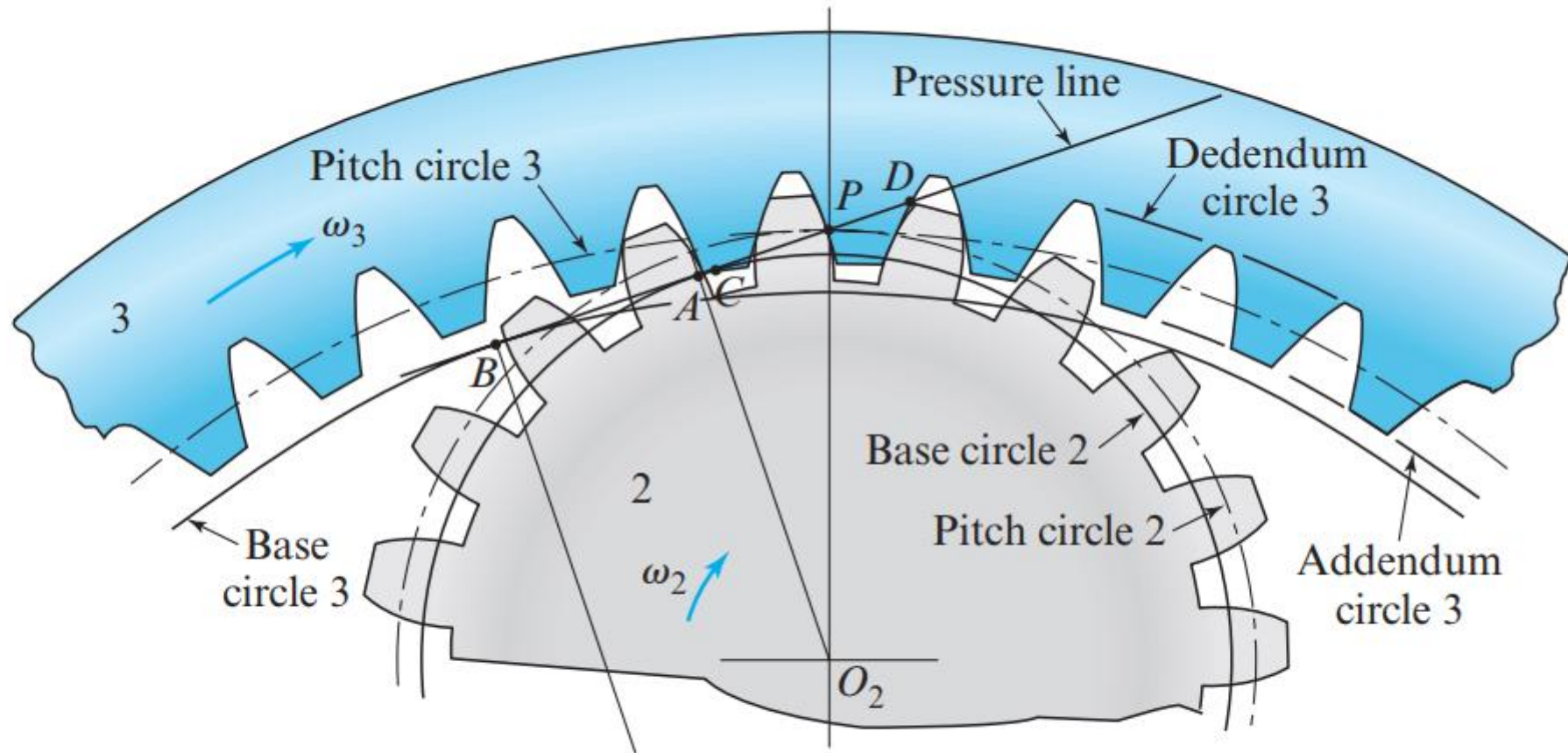
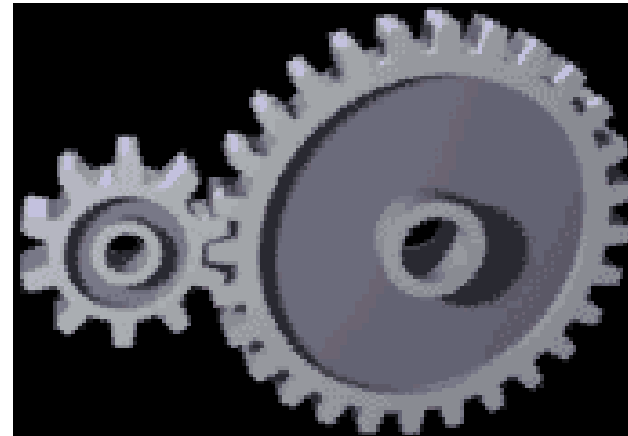
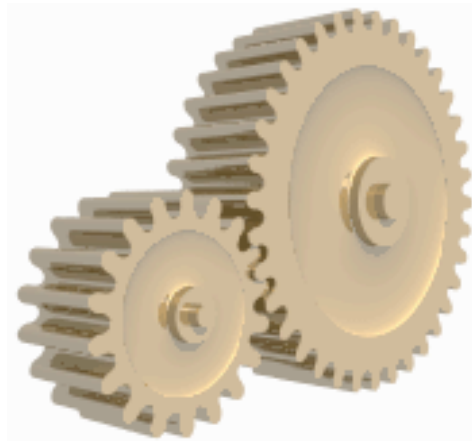


Figure 7.11 Involute pinion and internal gear.

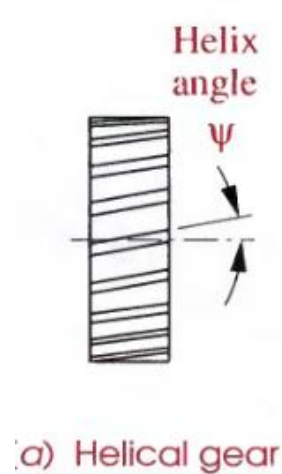
Spur gear

- The most common type of gear.
- Teeth run perpendicular to the face of the gear.
- Problem of noise and vibration at high speed
- Efficiency: 98 – 99%



Helical Gear

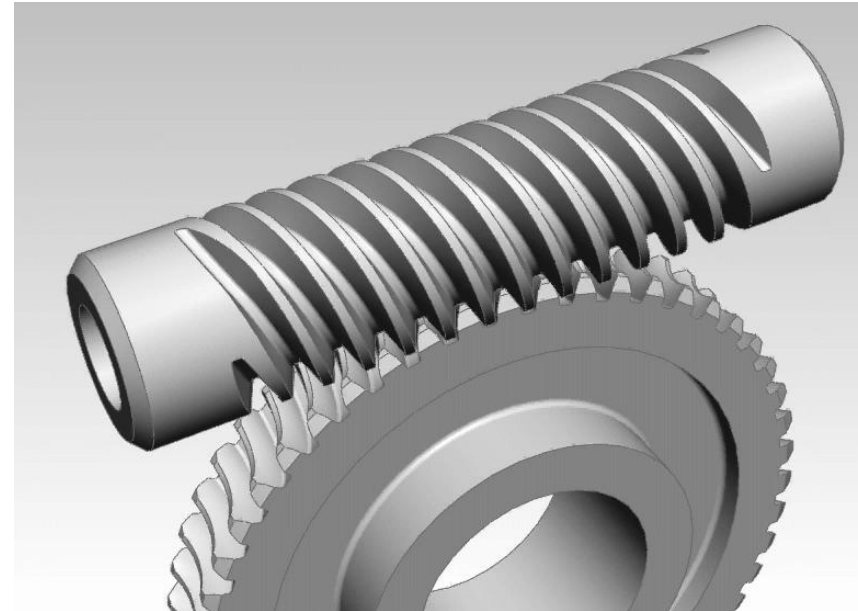
- Teeth are at an angle to the face
- Teeth engage gradually causing them to run more smoothly and quietly
- Used on non-parallel shafts to transmit motion.
- Disadvantage: resultant thrust along the axis of the gear
- Efficiency: 96 – 98%, 50 – 90% (crossed)



Herringbone

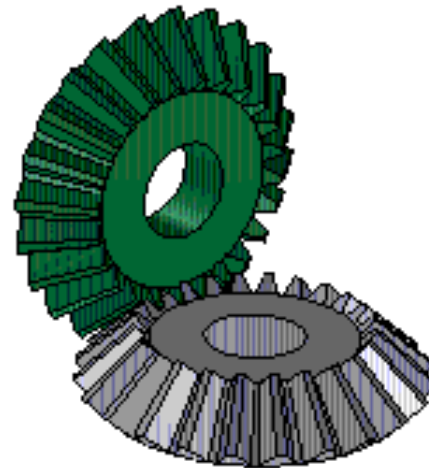
Worm and worm wheel

- Worm gears are used to transmit power at 90° and where high reductions are required.
- Single tooth rapped on a cylinder
- High gear ratio and non back drivable
- High sliding and thrust
- Efficiency: 40 – 85%



Bevel Gear

- Based on principle of rolling cones
- When power to be transmitted at right angles.
- Can have different angles of application but tend to be 90° .
- Efficiency: 40 – 85%



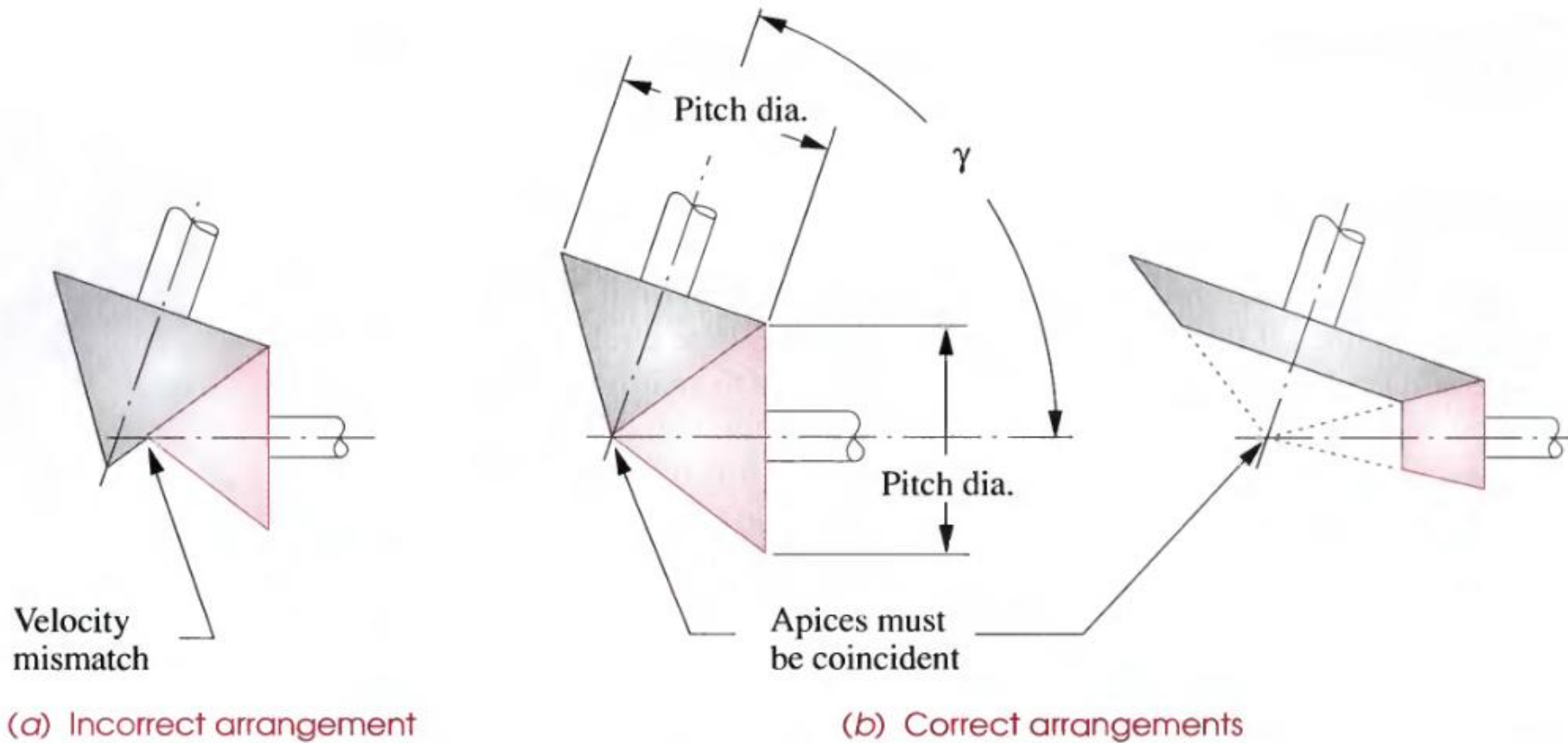
Straight



Spiral

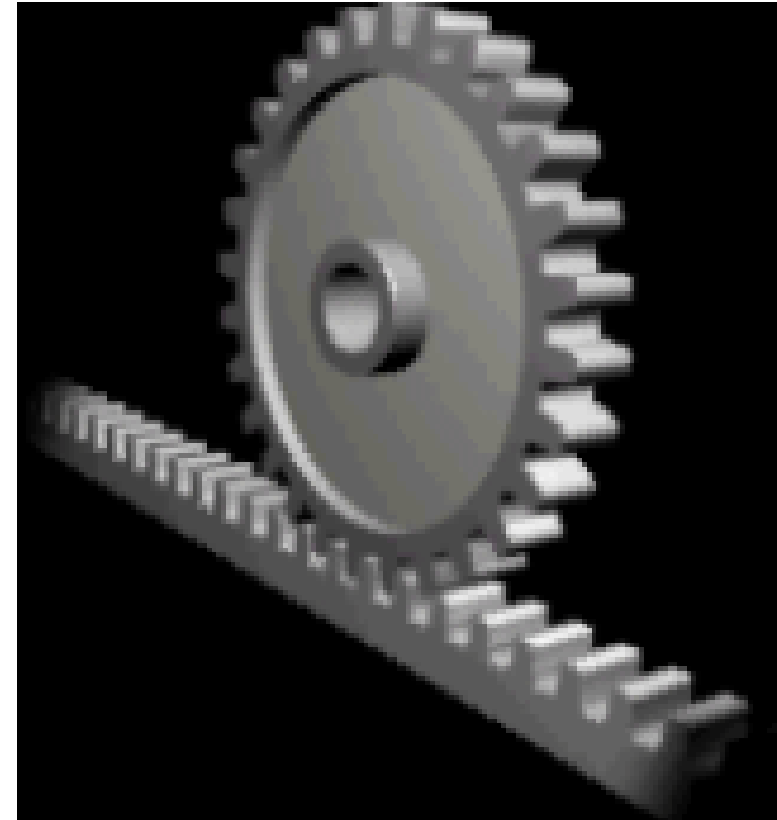
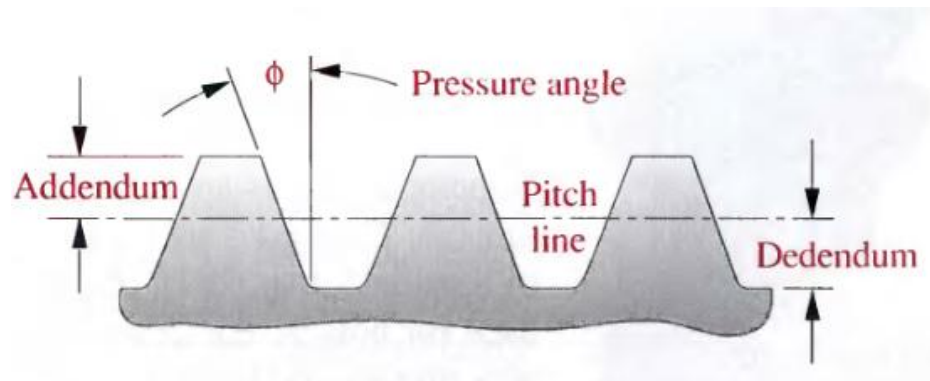
Bevel Gear

- Apices must be coincident for Constant ratio



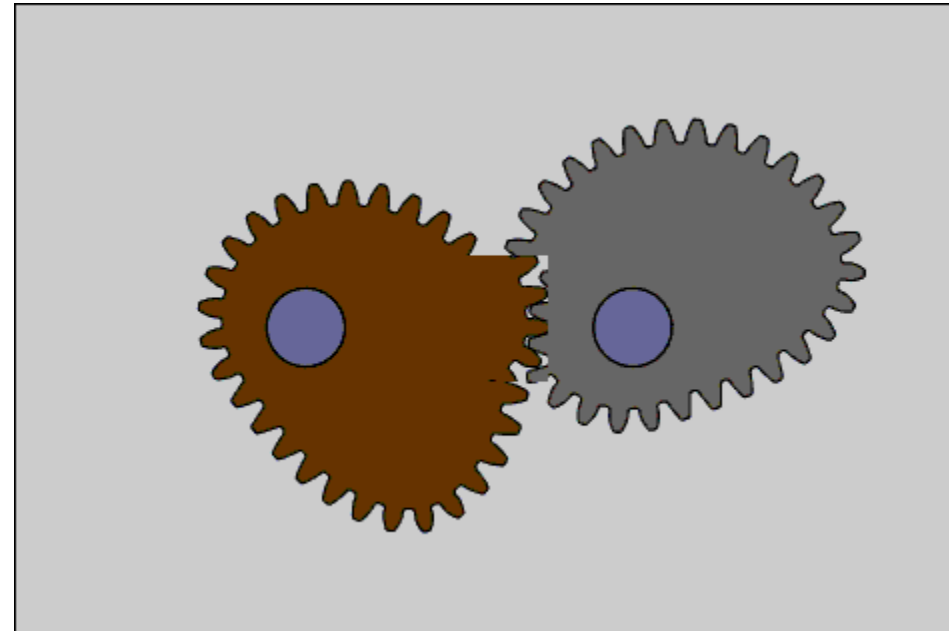
Rack and pinion

- Straight gear used to transmit power and motion in a linear movement.
- Automobile Steering



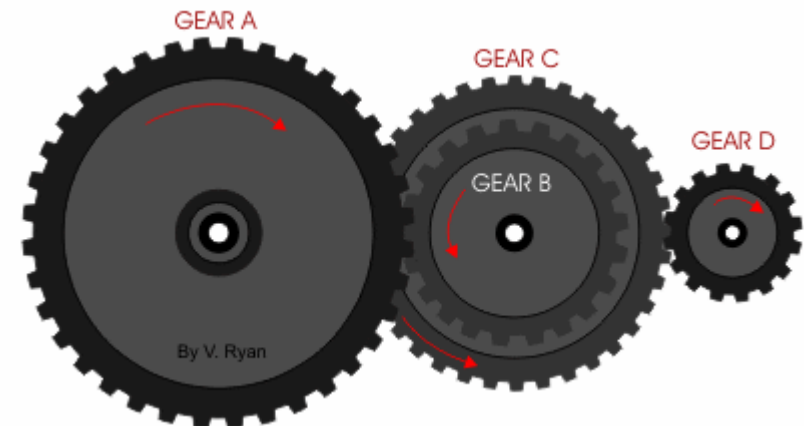
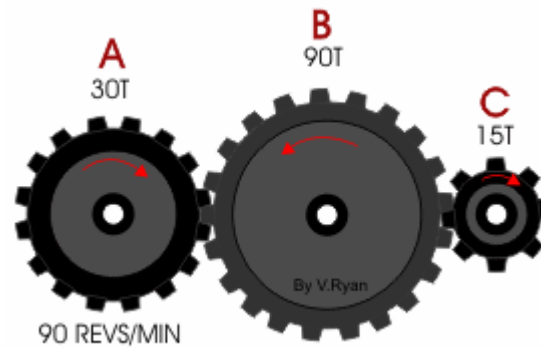
Non Circular Gears

- Based on rolling centrodes of a double-crank fourbar linkage
- Velocity ratio is not constant

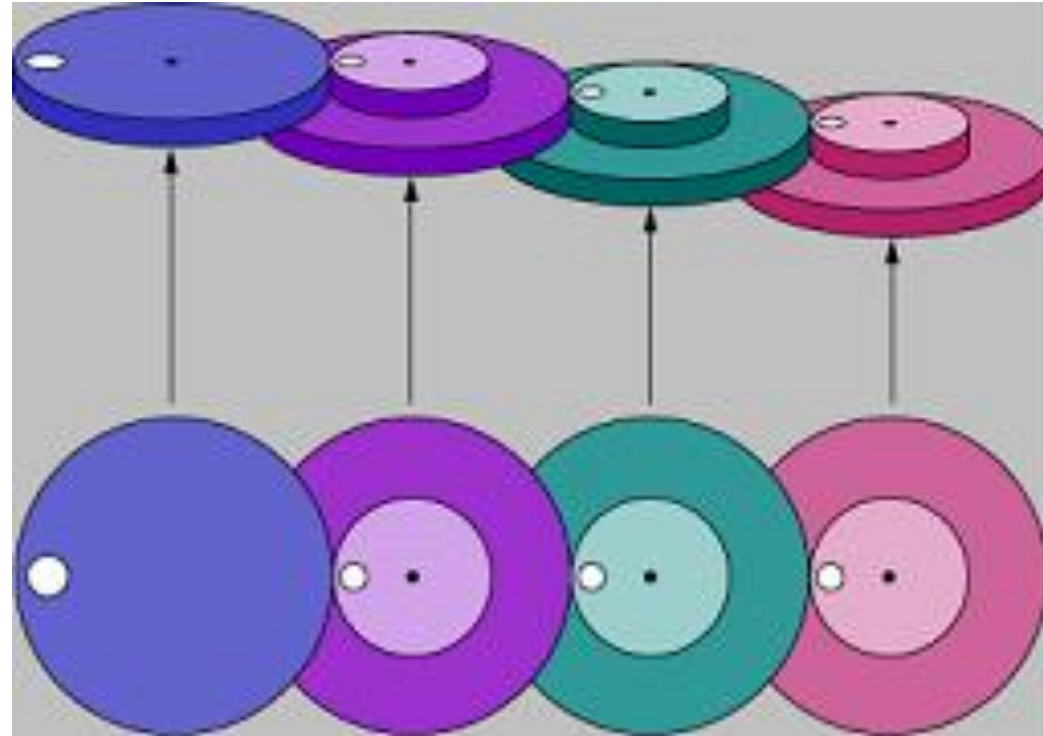


Gear Trains

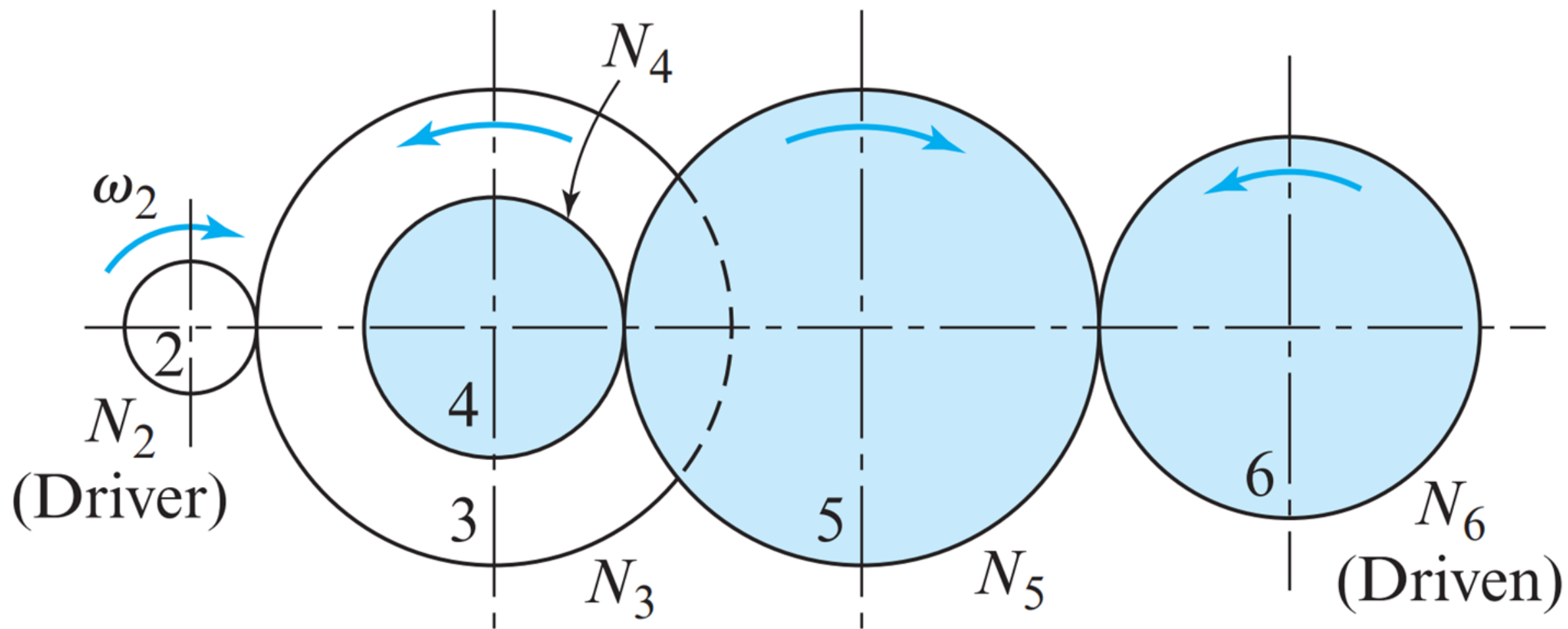
- To create larger gear ratios, gears are often connected in **gear trains**
- Simple
- Compound



Compound Gear Trains

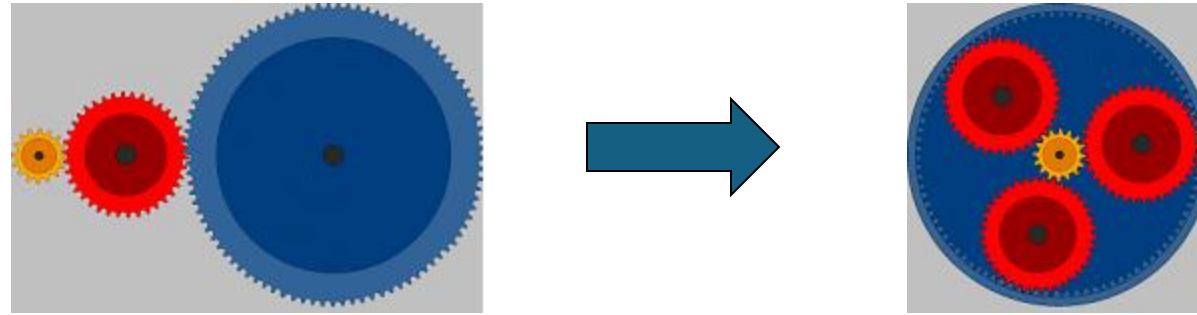


Each gear turns at twice the speed of the one to its left, which means that the red gear turns at $2 \times 2 \times 2 = 8$ times the speed of the blue gear.



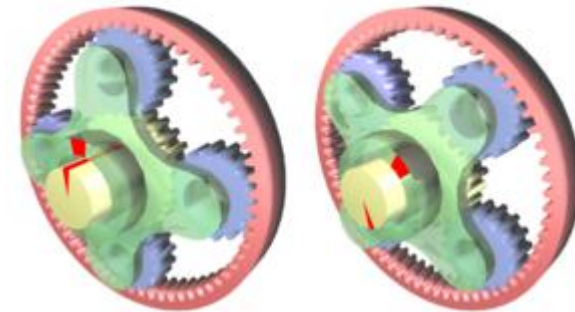
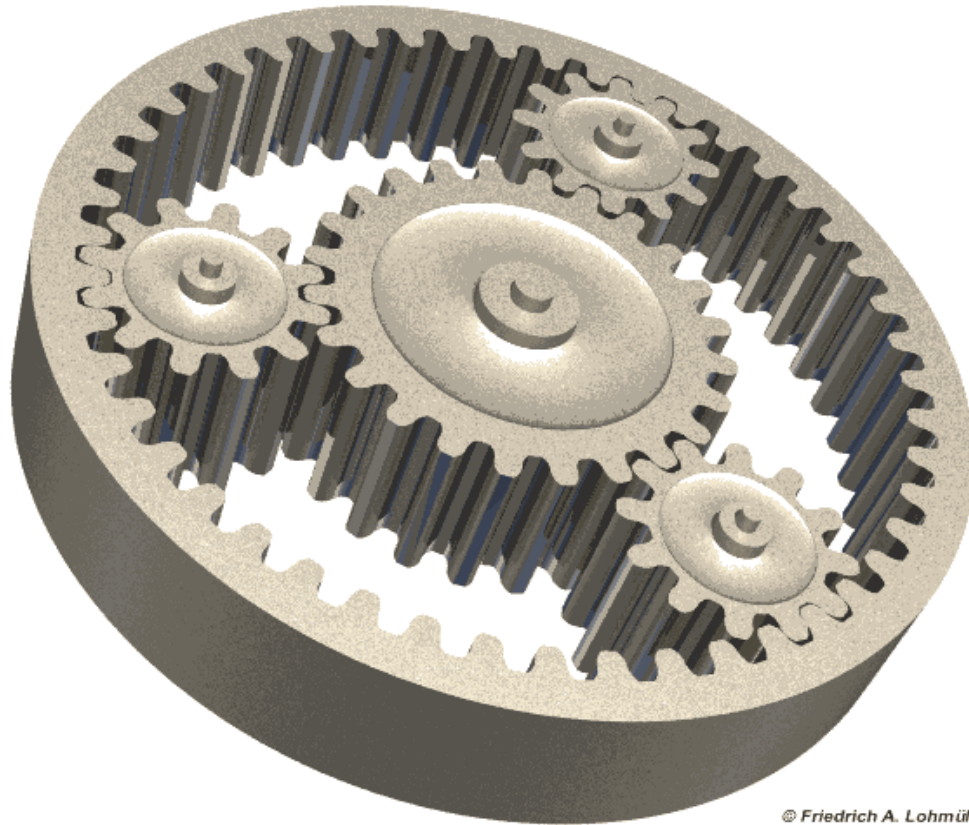
Epicyclic gearing or planetary gearing

Planetary gear trains allow the output shaft to be along the same axis as the input shaft:



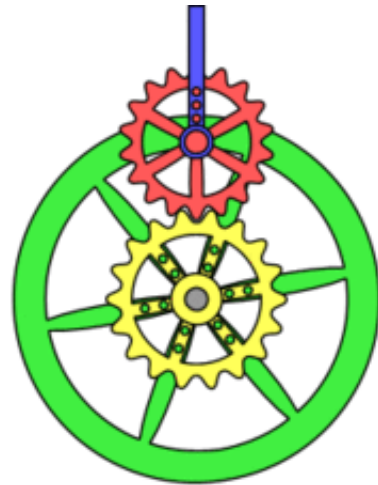
Such a gearing system would be used in a device such as an electric screwdriver.

Epicyclic gear train



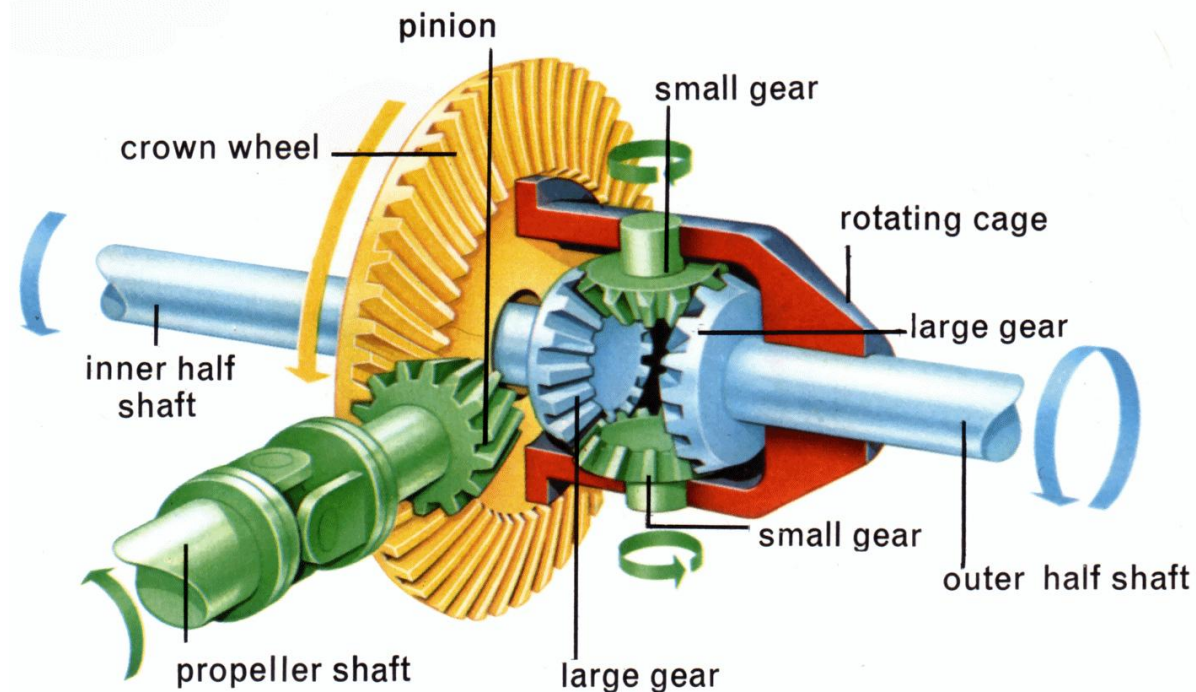
Sun and planet

- Used by James Watt on his early steam
- Reciprocating motion into rotary motion
- Sun is yellow, the planet red, the reciprocating arm is blue, the flywheel is green and the driveshaft is grey.



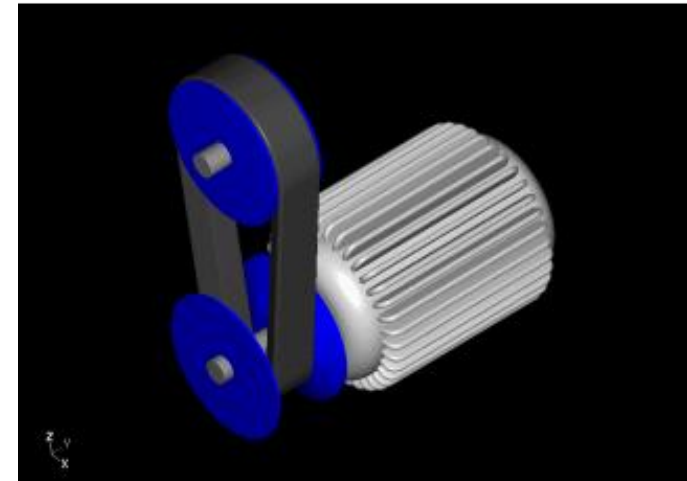
Automobile differential gear

- It is used to help cars turn corners.
- It allows wheels to turn at different speeds while providing same torque to both wheels.



Belt drive

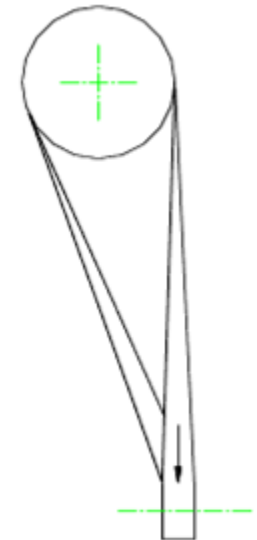
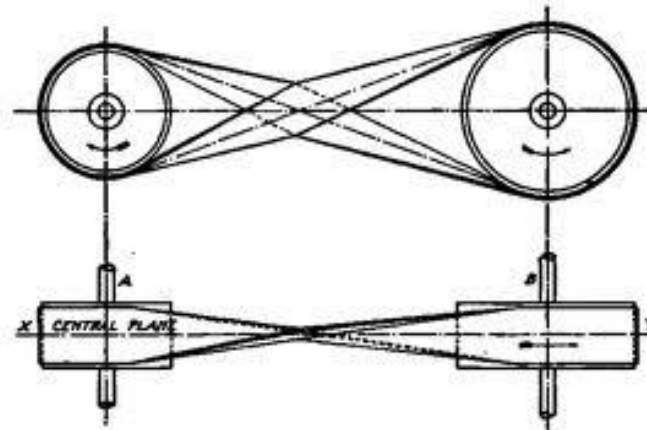
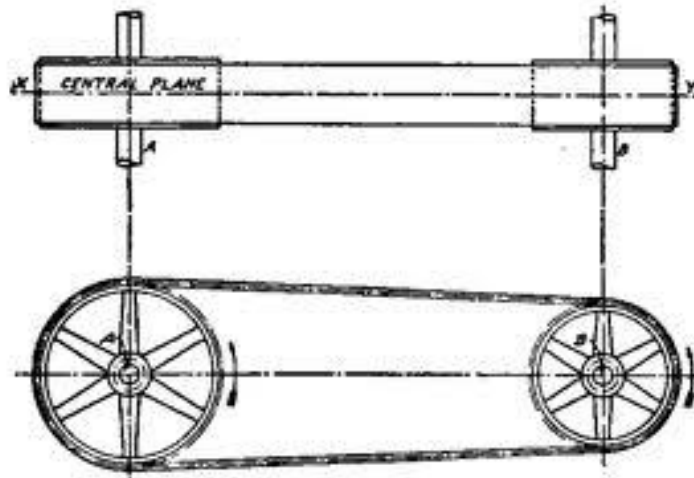
- Replacement for gears as power transmission device.
- Suited for applications where the center distance between rotating shafts are large.
- In contrast to other system that friction is no good, here we rely on the friction to transmit power
- Efficiency: 95 – 98 % (V-belt), 93%



Belt drive



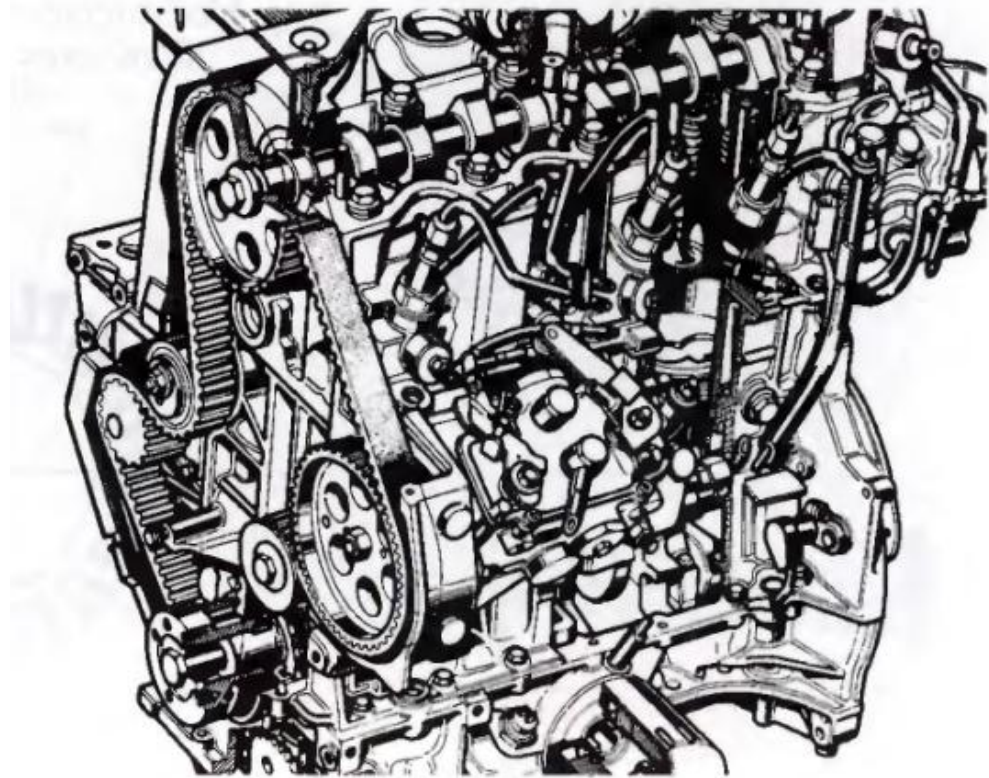
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Crossed Pulley

Timing belt

- Toothed belt



(b) Engine valve camshaft drive

*Courtesy of Chevrolet Division,
General Motors Co., Detroit, MI*

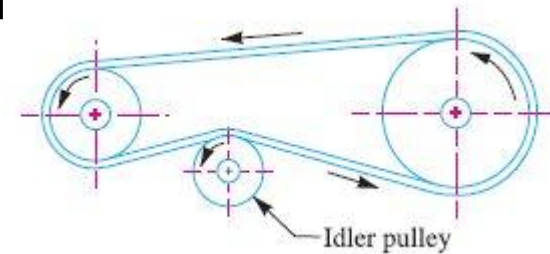
Advantages

- They are simple and economical and highly efficient .
- Economical when shafts are separated by large distances.
- Parallel shafts are not required can tolerate small misalignment
- Prolonged life due to cushioning of noise and vibration
- They are lubrication-free requiring low



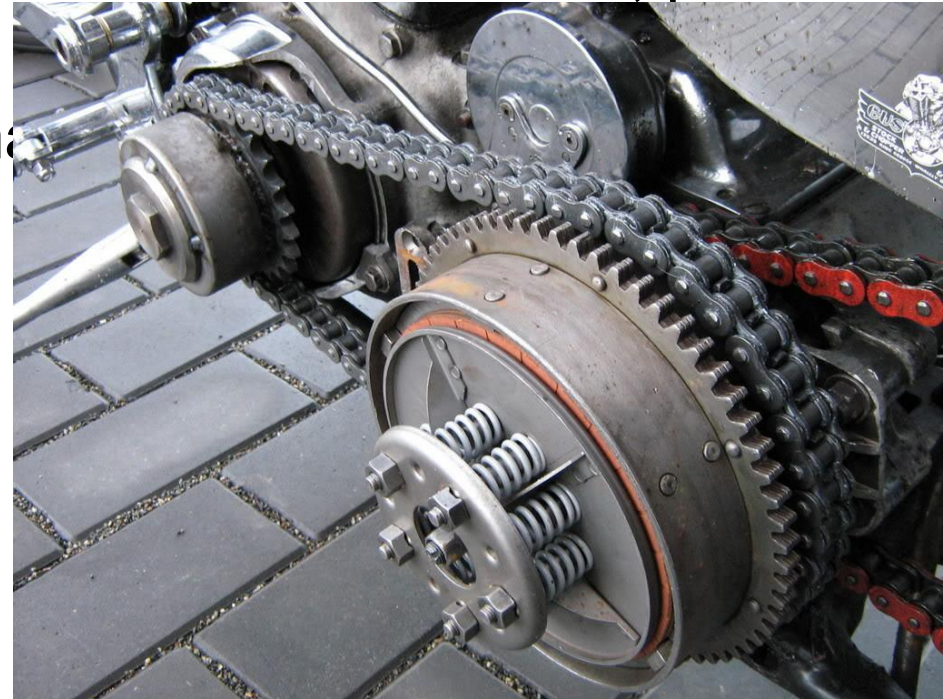
Disadvantages

- The torque capacity is limited by the coefficient of friction and pressure between belt and pulley
- Because of slip and/or creep, the angular velocity ratio will not remain constant
- Low speed reduction ratio, up to 3:1.
- Belt tension needs to be adjusted periodically.
- Idler pulley and stretch compensation



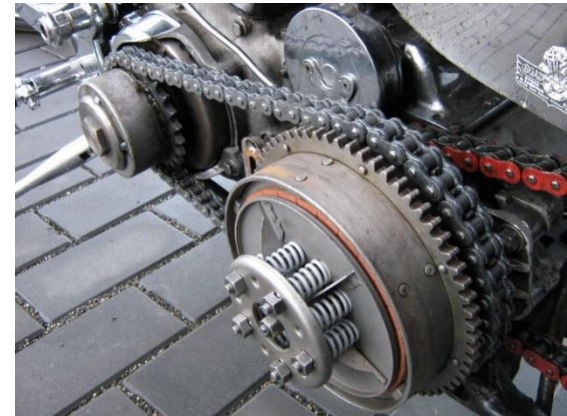
Chain drive

- Chain drive is a way of transmitting mechanical power from one place to another.
- It is often used to convey power to the wheels of a vehicle, particularly bicycles and motorcycles.
- It is also used in a wide variety of machines.



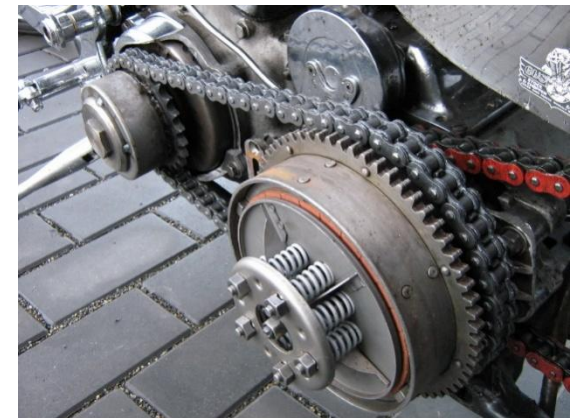
Advantages

- No slip hence perfect velocity ratio is obtained.
- Occupy less space in width than a belt drive.
- Suitable for both long as well as short distances.
- It transmits more power than belts.
- It permits high speed ratio of 8 to 10 in one step.
- Works under adverse temperature and atmospheric conditions.



Disadvantages

- Production cost of chains is relatively high.
- Accurate mounting and careful maintenance, particularly lubrication and slack adjustment.
- Velocity fluctuations when unduly stretched.
- Use is limited to slower speed than belt drive



Inertia forces and D'Alembert's (dull-un-bear) Principle

- Inertia Force
- Inertia Moment
- D'Alembert's principle is summarized as follows: The vector sum of all external forces and inertia forces acting upon a system of rigid bodies is zero. The vector sum of all external moments and inertia moments acting upon a system of rigid bodies is also separately zero.
- The inertia force should be applied at the center of mass.
- The moment can be considered about any point by including the inertia moment and also the moment due to inertia force. This gives great flexibility in solving complex problems.

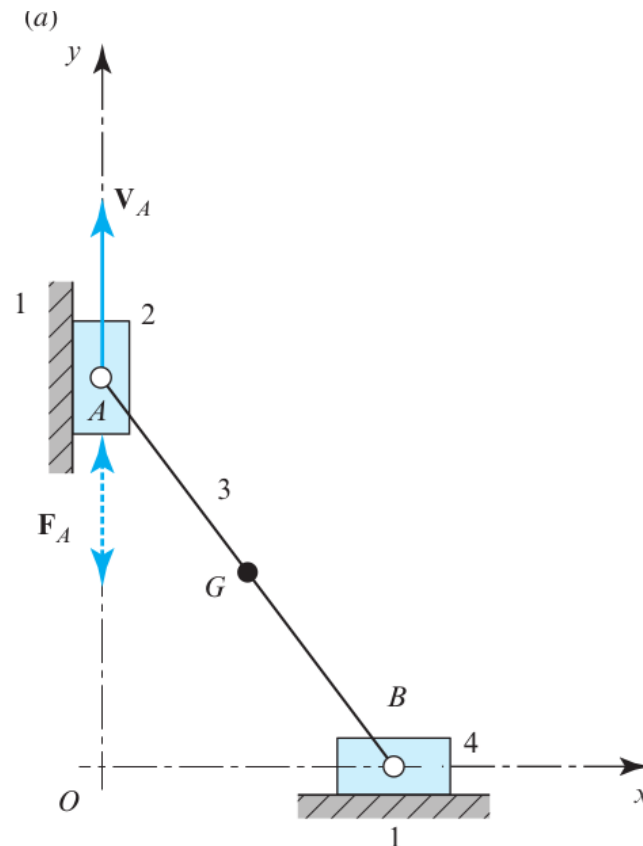
$$\sum \mathbf{F} + (-m\mathbf{A}_G) = \mathbf{0}$$

$$\sum \mathbf{F} = \mathbf{0} \quad \text{and} \quad \sum \mathbf{M} = \mathbf{0},$$

$$\sum \mathbf{M}_G + (-I_G\alpha) = \mathbf{0}.$$

Inertia forces and D'Alembert's (dull-un-bear) Principle

For the elliptic-trammel linkage shown in Fig. 12.8a, determine the force F_A that ensures that point A has the constant velocity $\mathbf{V}_A = 12.6\hat{\mathbf{j}}$ ft/s. The linkage is in the horizontal plane with gravity normal to the plane of motion. The weight and mass moment of inertia of coupler link 3 are $w_3 = 2.20$ lb and $I_{G_3} = 0.0479$ in $\cdot$ lb $\cdot$ s², respectively.



EXAMPLE 12.4

Determine the reaction forces and the driving torque for the four-bar linkage in the posture shown in Fig. 12.9. The input crank is operating at a constant angular velocity $\omega_2 = 48 \text{ rad/s}$ ccw, and the applied force acting at point C is $\mathbf{F}_C = -0.8\hat{\mathbf{j}} \text{ kN}$. The masses of links 3 and 4 are $m_3 = 1.5 \text{ kg}$ and $m_4 = 5 \text{ kg}$, respectively, and the second moments of mass of links 2, 3, and 4 are $I_{G_2} = 0.025 \text{ kg} \cdot \text{m}^2$, $I_{G_3} = 0.012 \text{ kg} \cdot \text{m}^2$, and $I_{G_4} = 0.054 \text{ kg} \cdot \text{m}^2$, respectively.

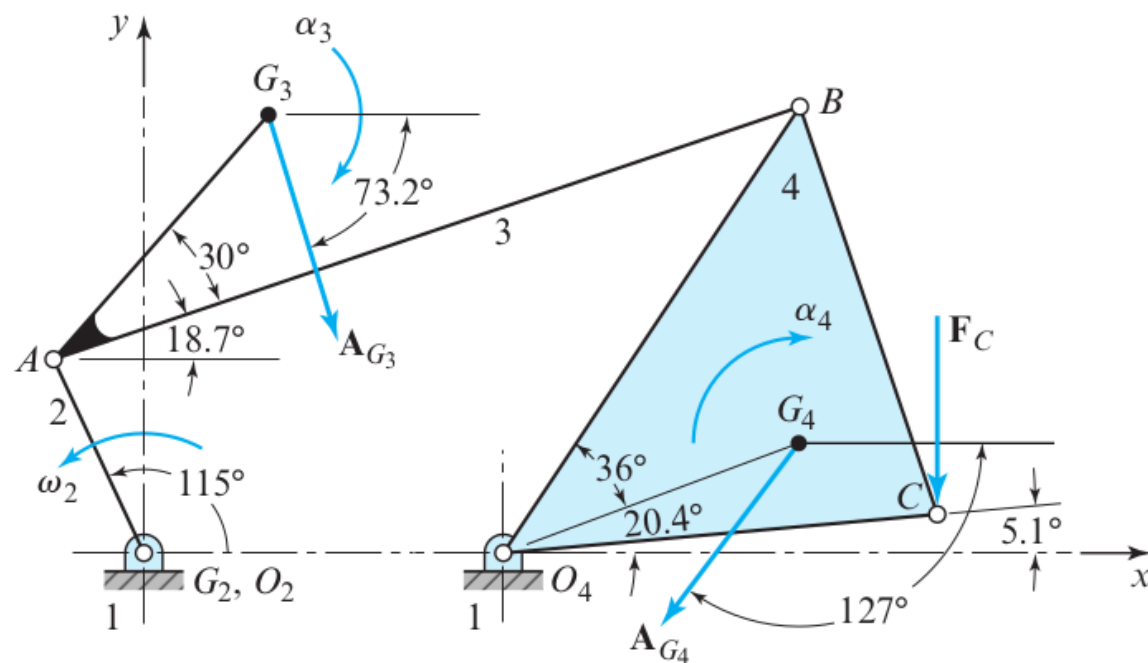


Figure 12.9 $R_{AO_2} = 60 \text{ mm}$, $R_{O_4O_2} = 100 \text{ mm}$, $R_{BA} = 220 \text{ mm}$, $R_{BO_4} = 150 \text{ mm}$, $R_{CO_4} = R_{CB} = 120 \text{ mm}$, $R_{G_3A} = 90 \text{ mm}$, and $R_{G_4O_4} = 90 \text{ mm}$.

ASSUMPTIONS

Gravity and friction effects are negligible.

SOLUTION

The information given in the figure caption in vector form is:

$$\mathbf{R}_{AO_2} = 60 \text{ mm} \angle 115^\circ = -25.4\hat{\mathbf{i}} + 54.4\hat{\mathbf{j}} \text{ mm},$$

$$\mathbf{R}_{G_3A} = 90 \text{ mm} \angle 48.7^\circ = 59.4\hat{\mathbf{i}} + 67.6\hat{\mathbf{j}} \text{ mm},$$

$$\mathbf{R}_{BA} = 220 \text{ mm} \angle 18.7^\circ = 208.4\hat{\mathbf{i}} + 70.5\hat{\mathbf{j}} \text{ mm},$$

$$\mathbf{R}_{BO_4} = 150 \text{ mm} \angle 61.5^\circ = 71.6\hat{\mathbf{i}} + 131.8\hat{\mathbf{j}} \text{ mm},$$

$$\mathbf{R}_{G_4O_4} = 90 \text{ mm} \angle 25.5^\circ = 81.2\hat{\mathbf{i}} + 38.7\hat{\mathbf{j}} \text{ mm},$$

$$\mathbf{R}_{CO_4} = 120 \text{ mm} \angle 5.1^\circ = 119.5\hat{\mathbf{i}} + 10.7\hat{\mathbf{j}} \text{ mm},$$

$$\mathbf{R}_{O_4O_2} = 100 \text{ mm} \angle 0^\circ = 100.0\hat{\mathbf{i}} \text{ mm}.$$

A kinematic analysis of the linkage gives: $\alpha_3 = -119\hat{\mathbf{k}} \text{ rad/s}^2$, $\alpha_4 = -625\hat{\mathbf{k}} \text{ rad/s}^2$, $\mathbf{A}_{G_3} = 162 \text{ m/s}^2 \angle -73.2^\circ = 46.8\hat{\mathbf{i}} - 155\hat{\mathbf{j}} \text{ m/s}^2$, and $\mathbf{A}_{G_4} = 104 \text{ m/s}^2 \angle -127^\circ = -62.6\hat{\mathbf{i}} - 83.1\hat{\mathbf{j}} \text{ m/s}^2$.

Given: The system shown in Figure 12-3 has the following data:

$$m_1 = 1.2 \text{ kg}$$

$$m_2 = 1.8 \text{ kg}$$

$$m_3 = 2.4 \text{ kg}$$

$$R_1 = 1.135 \text{ m @ } \angle 113.4^\circ$$

$$R_2 = 0.822 \text{ m @ } \angle 48.8^\circ$$

$$R_3 = 1.04 \text{ m @ } \angle 251.4^\circ$$

The z distances in meters from plane A are:

$$l_1 = 0.854,$$

$$l_2 = 1.701,$$

$$l_3 = 2.396,$$

$$l_B = 3.097$$

Find: The mass-radius products and their angular locations needed to dynamically balance the system using the correction planes A and B.

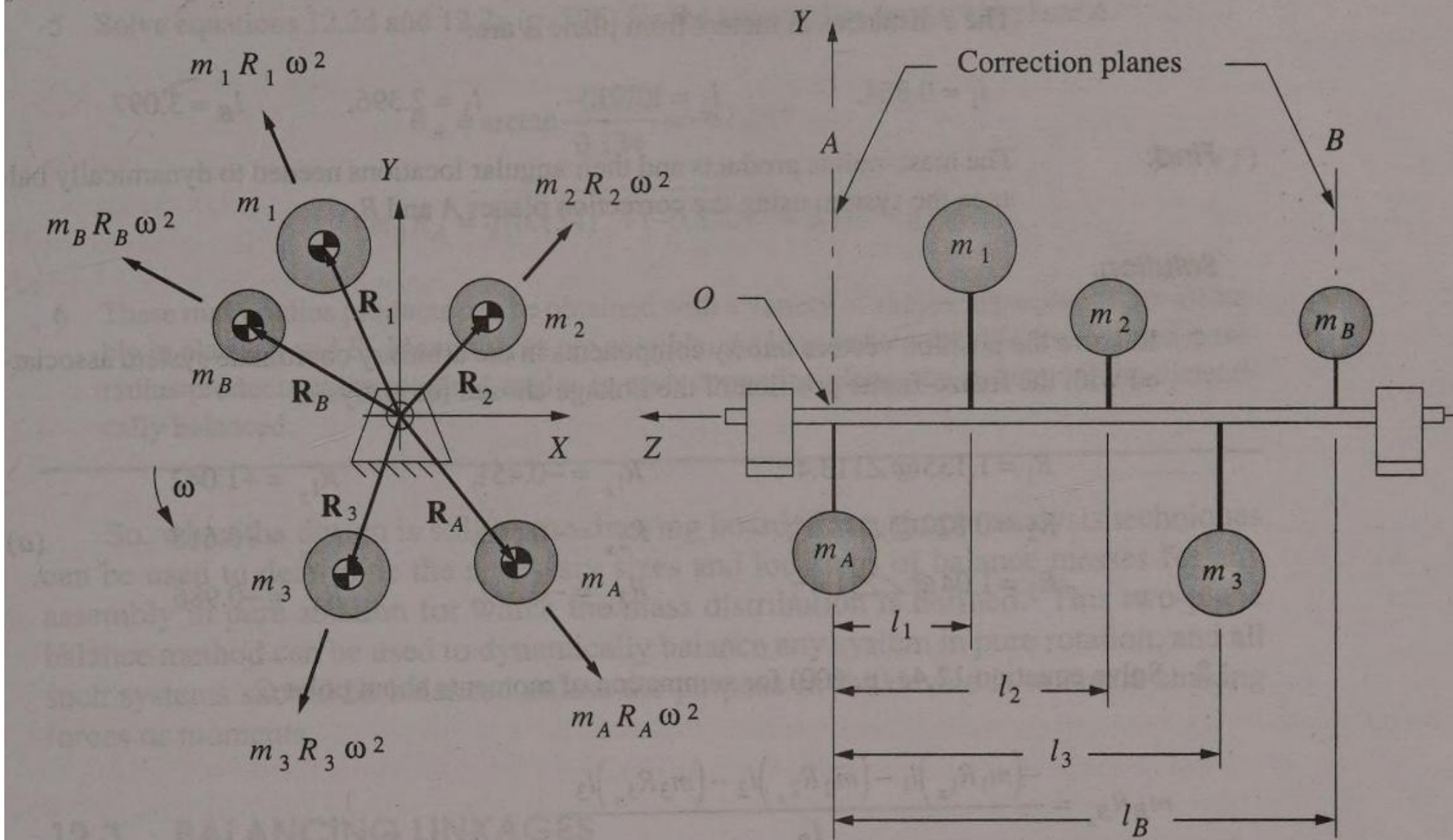
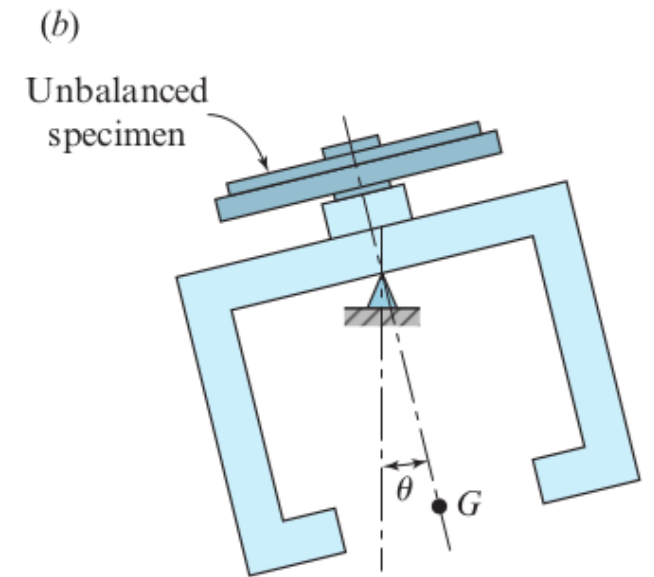
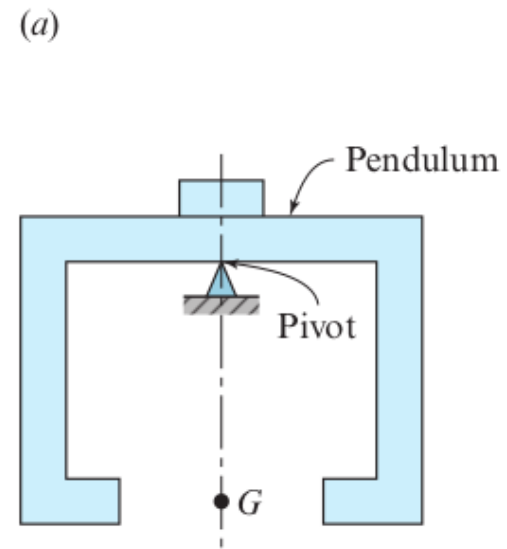


FIGURE 12-3

Two-plane dynamic balancing

Static Balancing Machines

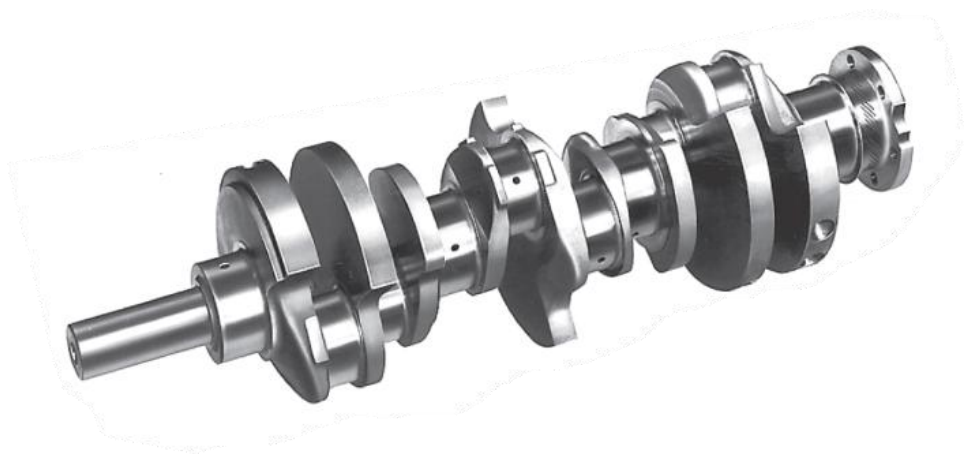


Dynamics of Reciprocating Engines

Classification according to

- Intended Application
 - Aircraft engines
 - Automotive engines
 - Marine engines
 - Stationary engines
- Combustion cycle
 - Otto cycle: fuel and air are mixed before combustion. No excess air.
 - Diesel cycle: Fuel is injected near the end of compression. Excess air.
 - Two stroke cycle
 - Four stroke cycle
- Number and arrangement of cylinders
 - Single and multi-cylinder engines
 - In-line engines and V engines

Dynamics of Reciprocating Engines



Crank-shaft/ crank



Piston and connecting rod

Dynamics of Reciprocating Engines

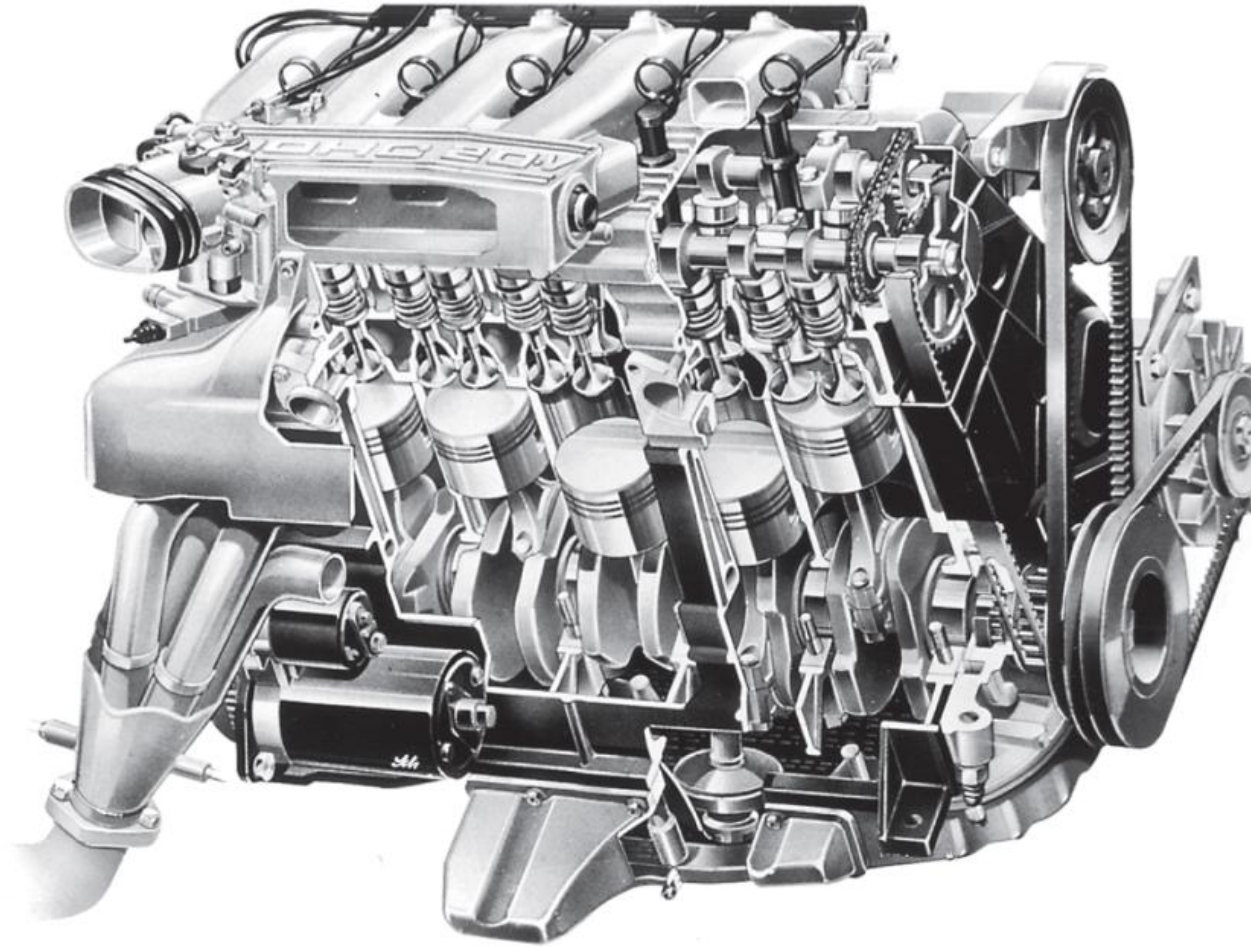


Figure 14.2 Five-cylinder in-line engine from an Audi Coupe Quattro. (Courtesy of Audi of America, Inc., Troy, MI.)

Dynamics of single cylinder engines– balancing of reciprocating masses

Approximation: Harmonic oscillation of the reciprocating mass. Harmonic oscillation at the fundamental frequency.

Higher harmonics are neglected because:

- Higher harmonics are of smaller amplitudes.
- Higher frequency vibrations are much more effectively damped out in most structures, and the fundamental frequency tends to dominate the transmitted vibration.

Approximation: For dynamic modelling of reciprocating machines, the mass of the connecting rod is modelled as two equivalent masses, one located on the crank pin axis and the other at the wrist pin axis. That is the distributed mass of the connecting rod is split into a reciprocating point mass, at the wrist pin, and a pure rotating point mass, at the crank pin.

This decomposition is done to preserve the correct location of the center of mass of the connecting rod.

Dynamics of single cylinder engines– balancing of reciprocating masses

The shaking force is the resultant of all forces acting on the frame of a mechanism due to inertia force only

Inertia of the connecting rod can be approximated by two concentrated masses, one at the crank pin and the other at the wrist pin.

Dynamics of single cylinder engines– balancing of reciprocating masses

The slider-crank system then becomes a combination of a mass assumed to be rotating at constant angular velocity (the crank mass together with the equivalent mass at the crank pin from the connecting rod) and a reciprocating mass (the piston and wrist pin mass, together with the equivalent mass from the connecting rod concentrated at the wrist pin).

Dynamics of single cylinder engines– balancing of reciprocating masses

To a first approximation, the piston motion can be regarded as a simple harmonic motion. This is equivalent to replacing a Fourier Series by its fundamental term; it is an approximation but a useful one.

Dynamics of single cylinder engines– balancing of reciprocating masses

The inertia force from the reciprocating mass is then a simple harmonic force acting along the cylinder axis.

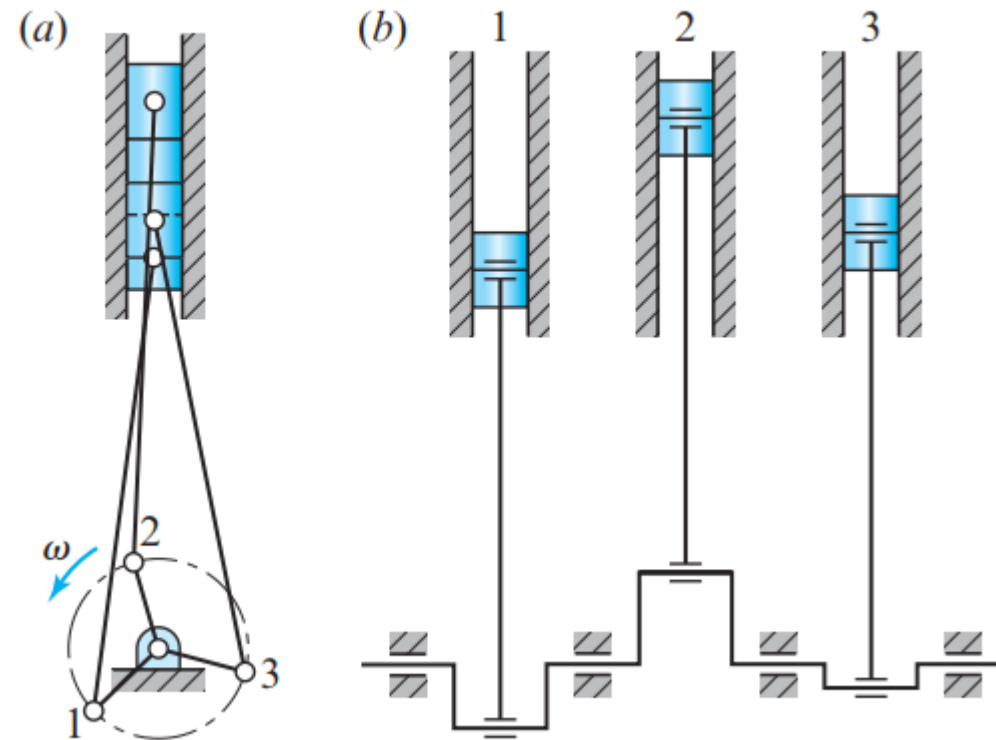
Dynamics of single cylinder engines– balancing of reciprocating masses

The inertia force from the rotating mass rotates with the crank. Its component along the cylinder axis is also harmonic, with the same period as the inertia force from the reciprocating mass.

Dynamics of single cylinder engines– balancing of reciprocating masses

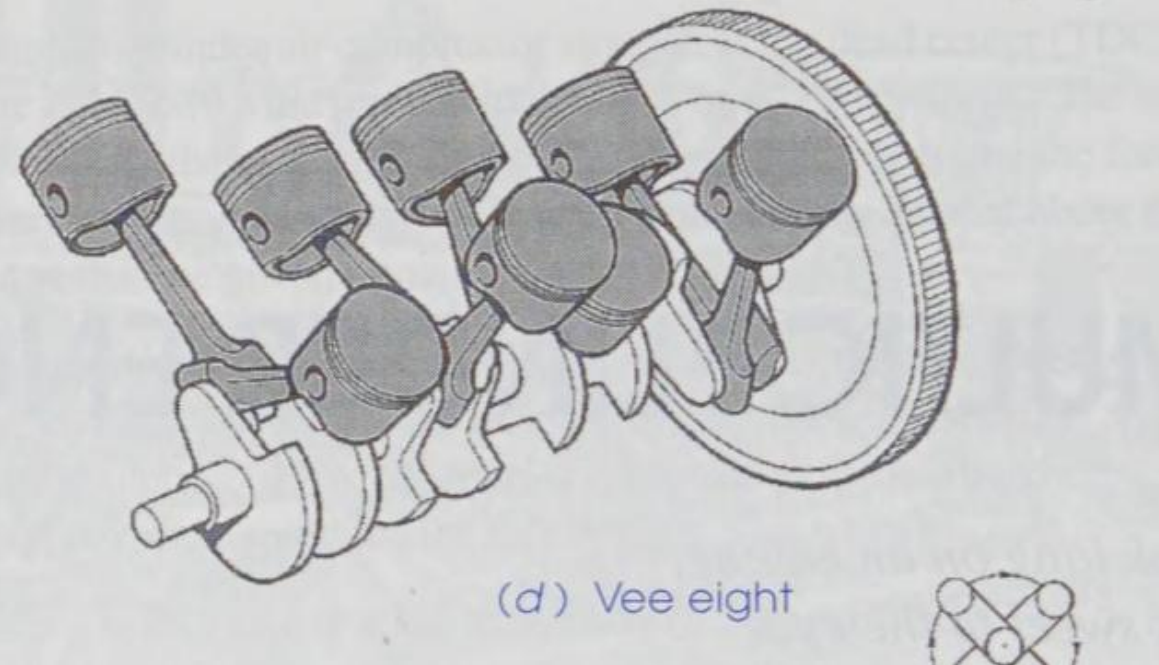
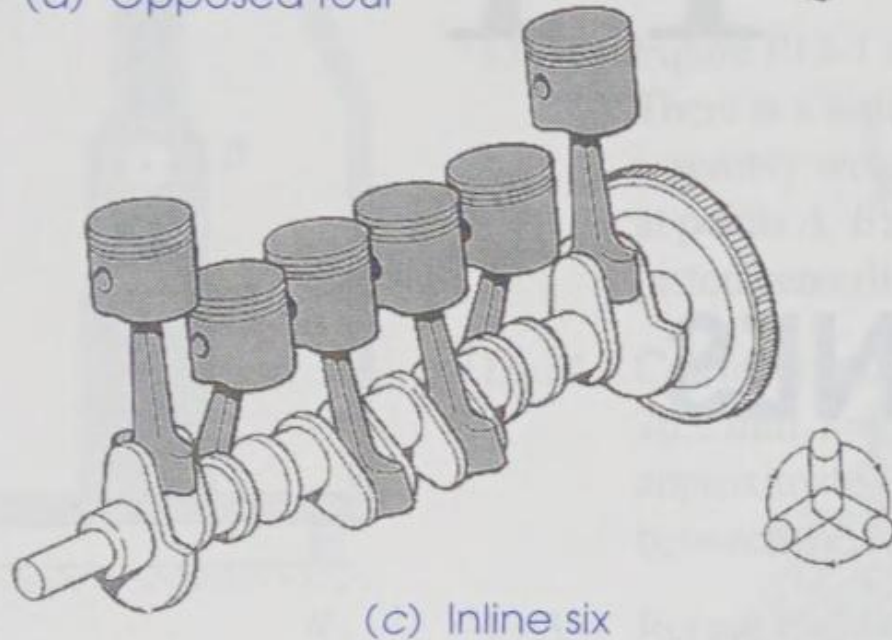
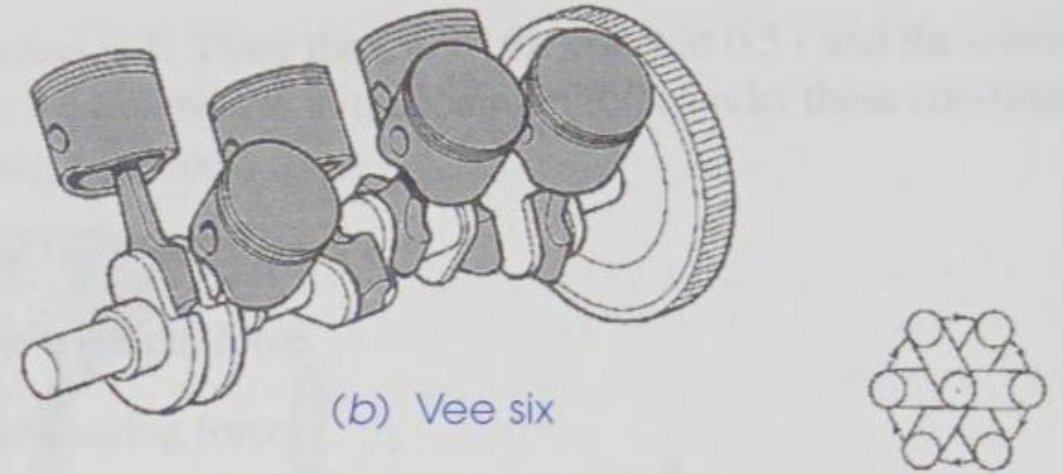
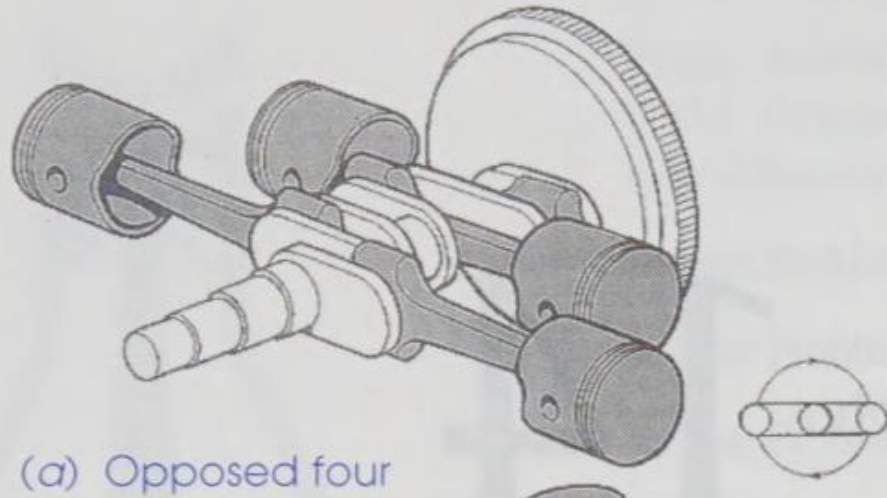
The inertia force of the reciprocating mass cannot be completely removed.

Balancing of multi-cylinder machines – engines/ compressors etc.

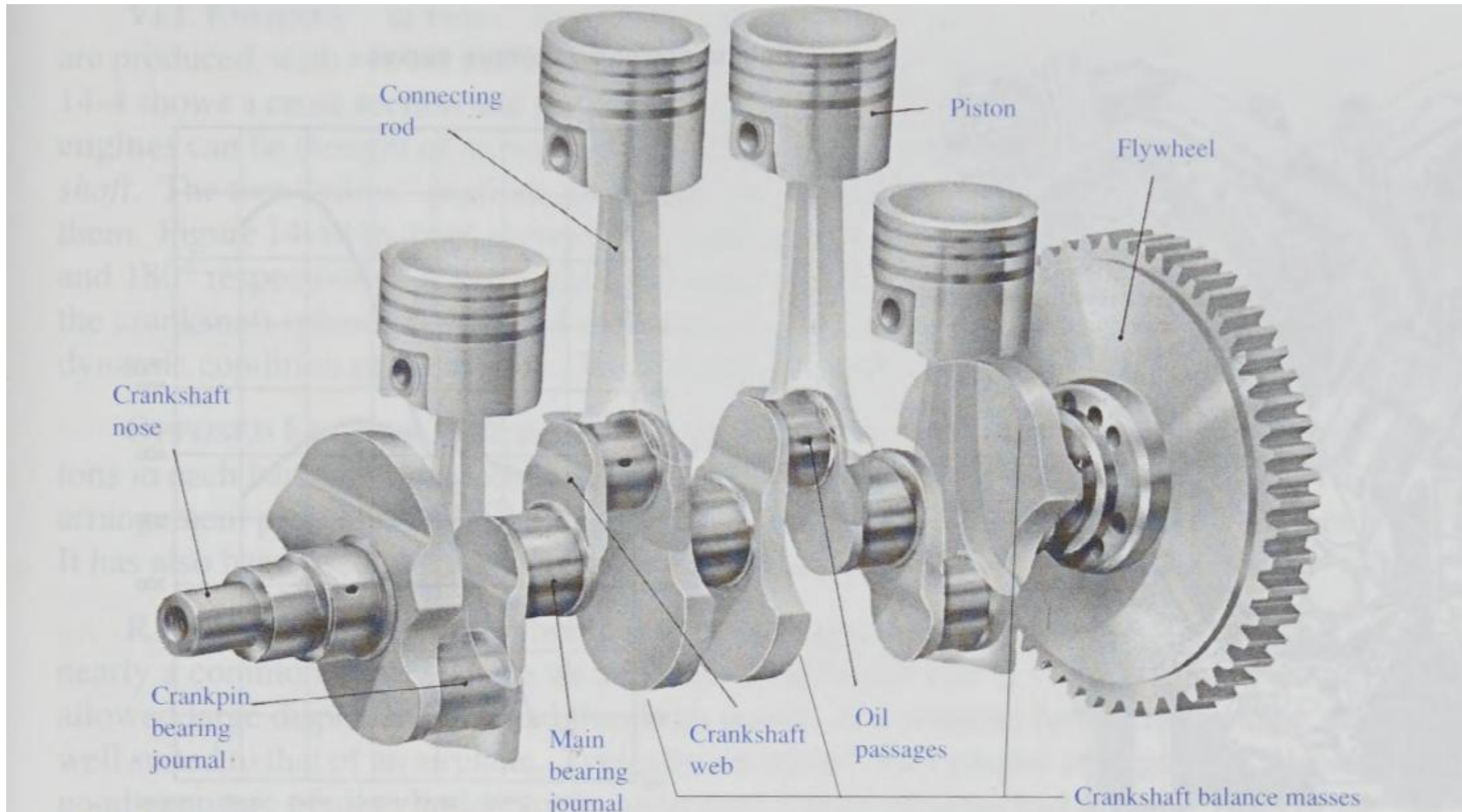


Three cylinder **in-line engine**

Balancing of multi-cylinder machines – engines/ compressors etc.

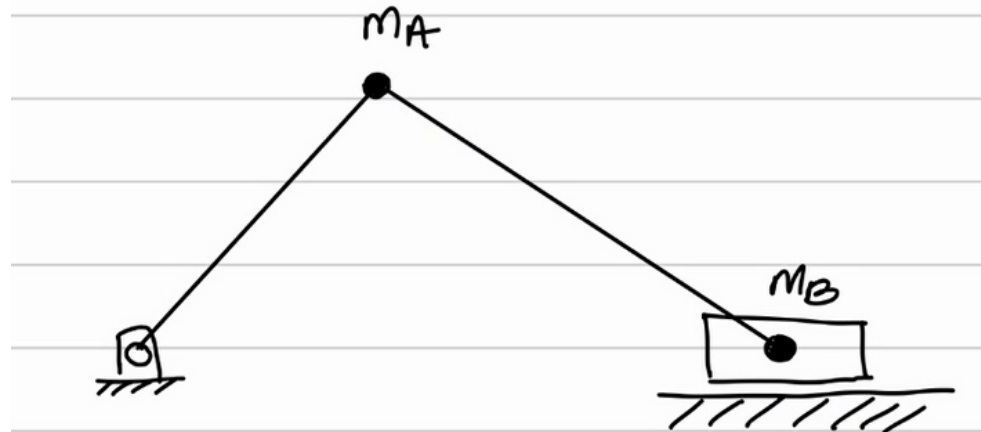


Balancing of multi-cylinder machines – engines/ compressors etc.



Balancing of multi-cylinder machines – engines/ compressors etc.

In multi-cylinder engines and compressors, the weight of the crankshaft and the part of the weight of the connecting rod that is concentrated at the crank pin can be counter-balanced exactly. Therefore, only the balancing of the reciprocating weight will be considered here.



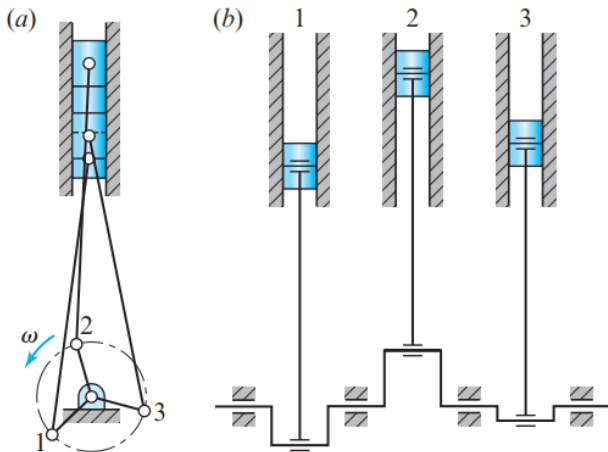
$$\Rightarrow \vec{f}_S = \vec{f}_A + \vec{f}_B = R\omega^2 \left[(m_A + m_B - m_{cb}) \cos\theta + m_B \frac{R}{L} \cos 2\theta \right] \hat{i} \\ + R\omega^2 (m_A - m_{cb}) \sin\theta \hat{j}$$

Balancing of multi-cylinder in-line engines

$$f_{B_i} = m_B R \omega^2 \left[\cos(\theta + \phi_i) + \frac{R}{L} \cos 2(\theta + \phi_i) \right], \quad i=1, 2, \dots, n$$

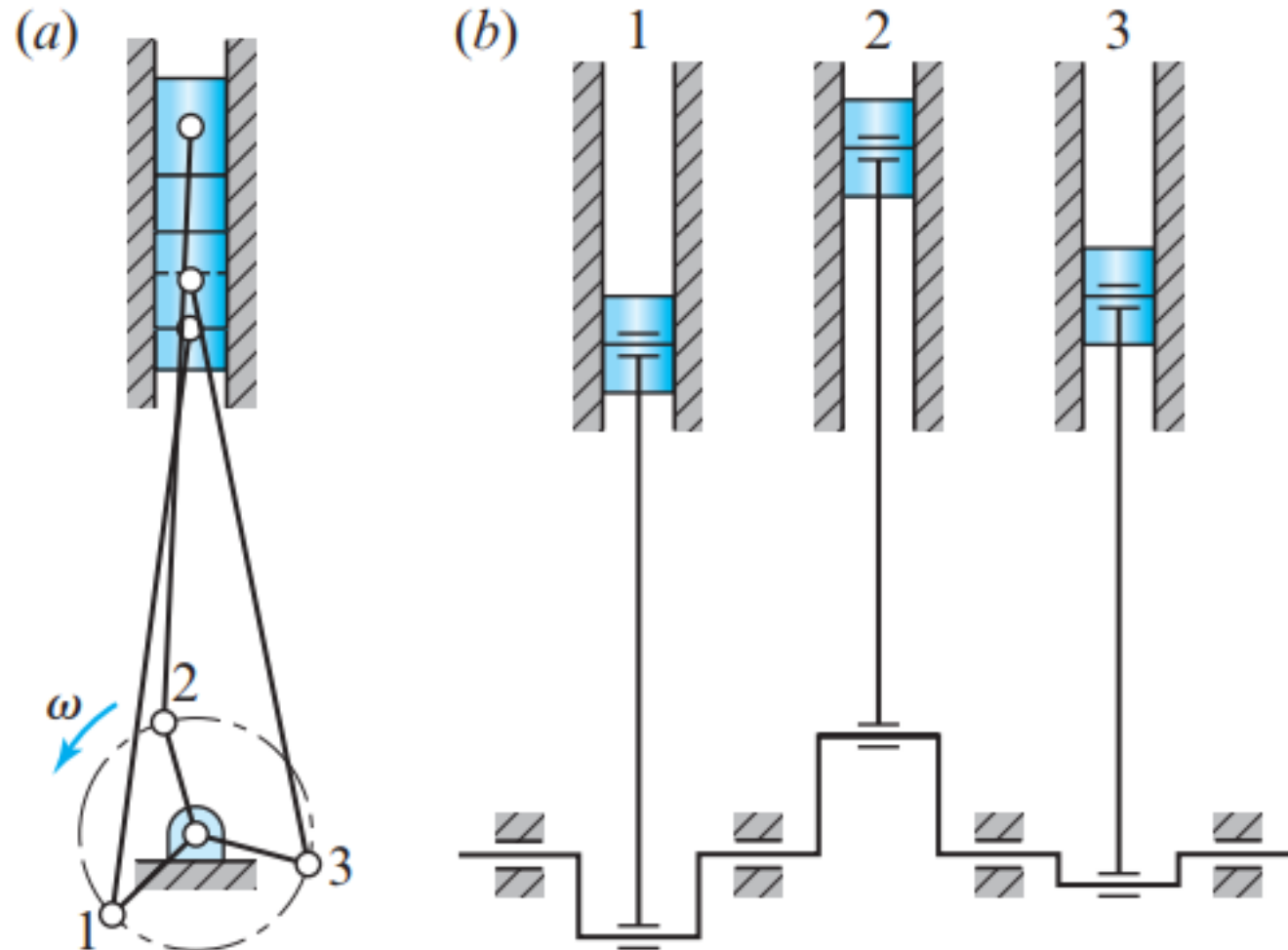
$\phi_i \rightarrow$ phase angle of the i^{th} reciprocating mass.

$\theta \rightarrow$ crank angle (for the first cylinder).



Three cylinder in-line engine

Example– balancing of three-cylinder in-line engine



Example— balancing of three-cylinder in-line engine

Phase angles $\phi_2 = 120^\circ$, $\phi_3 = 240^\circ$ ($\phi_1 = 0$)

$$f_{B_1} + f_{B_2} + f_{B_3} = R\omega^2 \left(\cos \theta + \cos(\theta + 120^\circ) + \cos(\theta + 240^\circ) \right) \\ + R\omega^2 \frac{R}{L} \left(\cos 2\theta + \cos 2(\theta + 120^\circ) + \cos 2(\theta + 240^\circ) \right)$$

$$\Rightarrow \cos \theta + \cos(\theta + 120^\circ) + \cos(\theta + 240^\circ) = \cos(\theta) + 2 \cos\left(\frac{2\theta + 360^\circ}{2}\right) \cos\left(\frac{120^\circ}{2}\right)$$

$$= \cos \theta + 2 \cos(\theta + 180^\circ) \cos 60^\circ = \cos \theta - \cos \theta = 0$$

Example— balancing of three-cylinder in-line engine

Similarly,

$$\cos 2\theta + \cos 2(\theta + 120^\circ) + \cos 2(\theta + 240^\circ) = 0$$

Using the identity $\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

For 3-cylinder in-line engine, the first and second harmonics are inherently balanced.

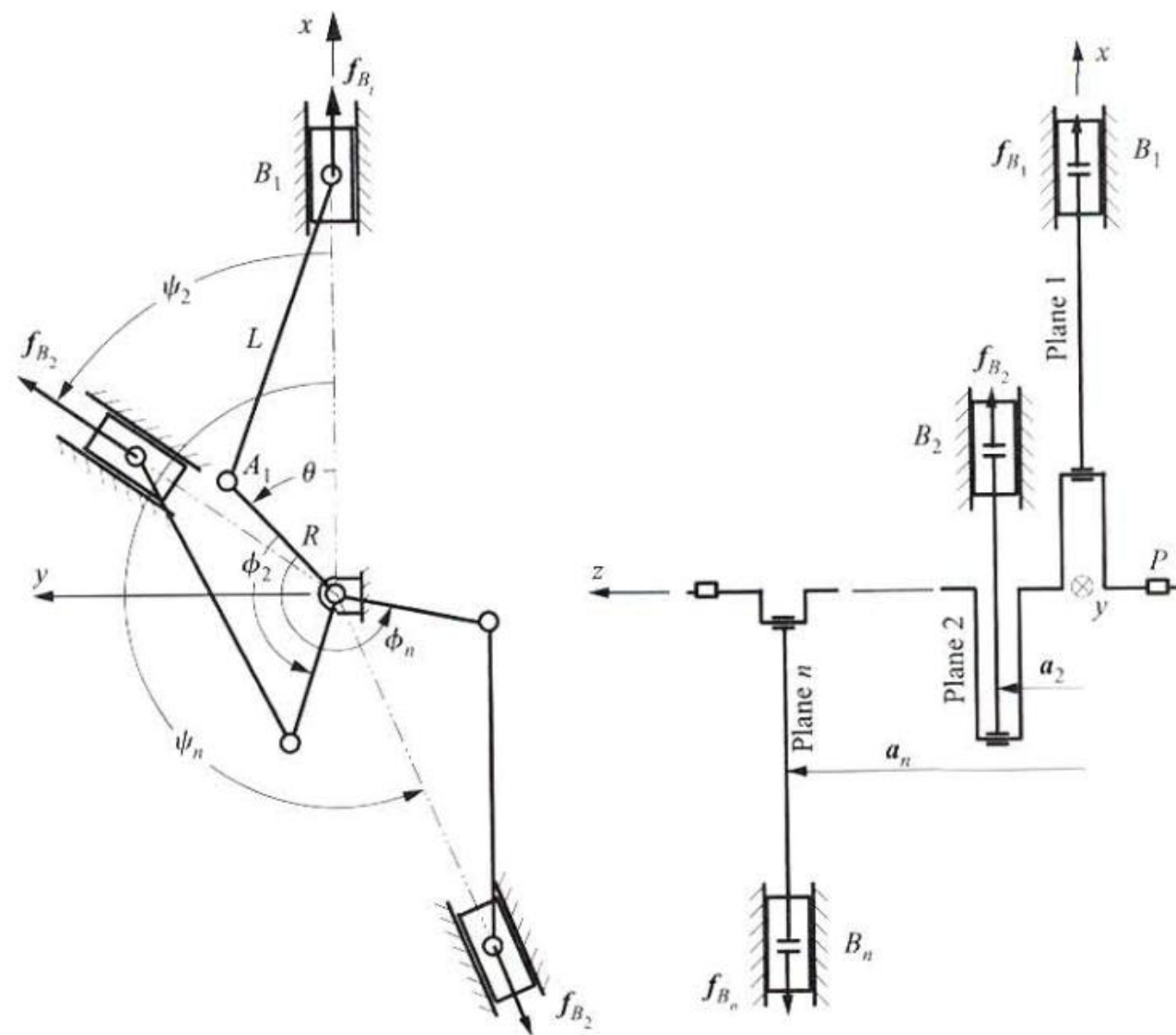


FIGURE 15.14 Schematic diagram of shaking forces in a multicylinder engine.

Inertia force for non-inline engines

$$f_B = m_B R \omega^2 \left[\cos(\theta + \phi_i - \psi_i) + \frac{R}{L} \cos 2(\theta + \phi_i - \psi_i) \right], \quad i = 1, 2, \dots, n$$

$\phi_i \rightarrow$ phase angle / angle of crank from the first crank ($\phi_1 = 0$).

$\psi_i \rightarrow$ angle the cylinder axis makes with the x-axis.

The x -component of the shaking force from the i^{th} cylinder is

$$\Rightarrow f_{B_{x_i}} = f_{B_i} \cos \psi_i = m_B R \omega^2 \left[\cos(\theta + \phi_i - \psi_i) + \frac{R}{L} \cos 2(\theta + \phi_i - \psi_i) \right] \cos \psi_i$$

the y component is

$$\Rightarrow f_{B_{y_i}} = f_{B_i} \sin \psi_i = m_B R \omega^2 \left[\cos(\theta + \phi_i - \psi_i) + \frac{R}{L} \cos 2(\theta + \phi_i - \psi_i) \right] \sin \psi_i$$

$$f_x = m_B R \omega^2 \left[\sum_{i=1}^n \cos(\theta + \phi_i - \psi_i) \cos \psi_i + \frac{R}{L} \sum \cos 2(\theta + \phi_i - \psi_i) \cos \psi_i \right]$$

and

$$f_y = m_B R \omega^2 \left[\sum_{i=1}^n \cos(\theta + \phi_i - \psi_i) \sin \psi_i + \frac{R}{L} \sum_{i=1}^n \cos 2(\theta + \phi_i - \psi_i) \sin \psi_i \right]$$

using, $\cos(\theta + \phi_i - \psi_i) = \cos \theta \cos(\phi_i - \psi_i) - \sin \theta \sin(\phi_i - \psi_i)$

To balance the forces, it is necessary for f_x and f_y to be zero.

This happens for all values of θ when

$$\sum_{i=1}^n \cos(\phi_i - \psi_i) \cos \psi_i = 0$$

$$\sum_{i=1}^n \cos(\phi_i - \psi_i) \sin \psi_i = 0$$

$$\sum_{i=1}^n \sin(\phi_i - \psi_i) \sin \psi_i = 0$$

$$\sum_{i=1}^n \sin(\phi_i - \psi_i) \cos \psi_i = 0$$

For balancing primary shaking forces.

$$\sum_{i=1}^n \cos 2(\phi_i - \psi_i) \cos \psi_i = 0$$

$$\sum_{i=1}^n \cos 2(\phi_i - \psi_i) \sin \psi_i = 0$$

For balancing secondary shaking forces.

$$\sum_{i=1}^n \sin 2(\phi_i - \psi_i) \sin \psi_i = 0$$

$$\sum_{i=1}^n \sin 2(\phi_i - \psi_i) \cos \psi_i = 0$$

For balancing moments

$$M_x = \sum_{i=1}^n f_{By_i} a_i = 0$$

$$M_y = \sum_{i=1}^n f_{Bx_i} a_i = 0$$

From this, we get

$$\sum_{i=1}^n a_i \cos(\phi_i - \psi_i) \cos \psi_i = 0$$

$$\sum_{i=1}^n a_i \cos(\phi_i - \psi_i) \sin \psi_i = 0$$

$$\sum_{i=1}^n a_i \sin(\phi_i - \psi_i) \sin \psi_i = 0$$

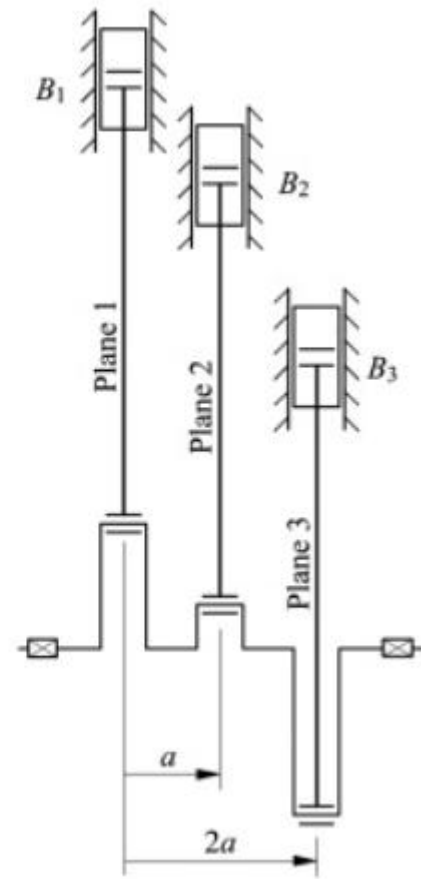
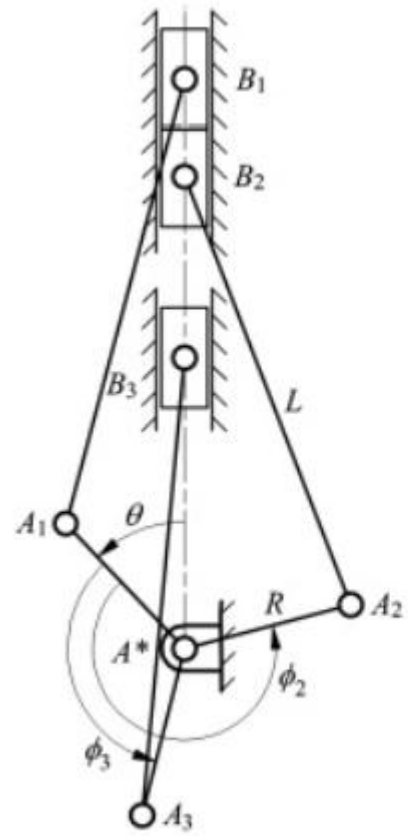
$$\sum_{i=1}^n a_i \sin(\phi_i - \psi_i) \cos \psi_i = 0$$

$$\sum_{i=1}^n a_i \cos 2(\phi_i - \psi_i) \cos \psi_i = 0$$

$$\sum_{i=1}^n a_i \cos 2(\phi_i - \psi_i) \sin \psi_i = 0$$

$$\sum_{i=1}^n a_i \sin 2(\phi_i - \psi_i) \sin \psi_i = 0$$

$$\sum_{i=1}^n a_i \sin 2(\phi_i - \psi_i) \cos \psi_i = 0$$



Moment components for the three-cylinder engine.

$$\psi_i = 0, \sin \psi_i = 0$$

$$\sum_{i=1}^3 a_i \cos \phi_i = 0 + a \cos 240^\circ + 2a \cos 120^\circ = 0 - \frac{a}{2} - \frac{2a}{2} = -\frac{3a}{2}$$

$$\sum_{i=1}^3 a_i \sin \phi_i = 0 + a \sin 240^\circ + 2a \sin 120^\circ = 0 - \frac{a\sqrt{3}}{2} + \frac{2a\sqrt{3}}{2} = \frac{a\sqrt{3}}{2}$$

$$\sum_{i=1}^3 a_i \cos 2\phi_i = 0 + a \cos 480^\circ + 2a \cos 240^\circ = 0 - \frac{a}{2} - \frac{2a}{2} = -\frac{3a}{2}$$

$$\sum_{i=1}^3 a_i \sin 2\phi_i = 0 + a \sin 480^\circ + 2a \sin 240^\circ = 0 + \frac{a\sqrt{3}}{2} - \frac{2a\sqrt{3}}{2} = -\frac{a\sqrt{3}}{2}$$

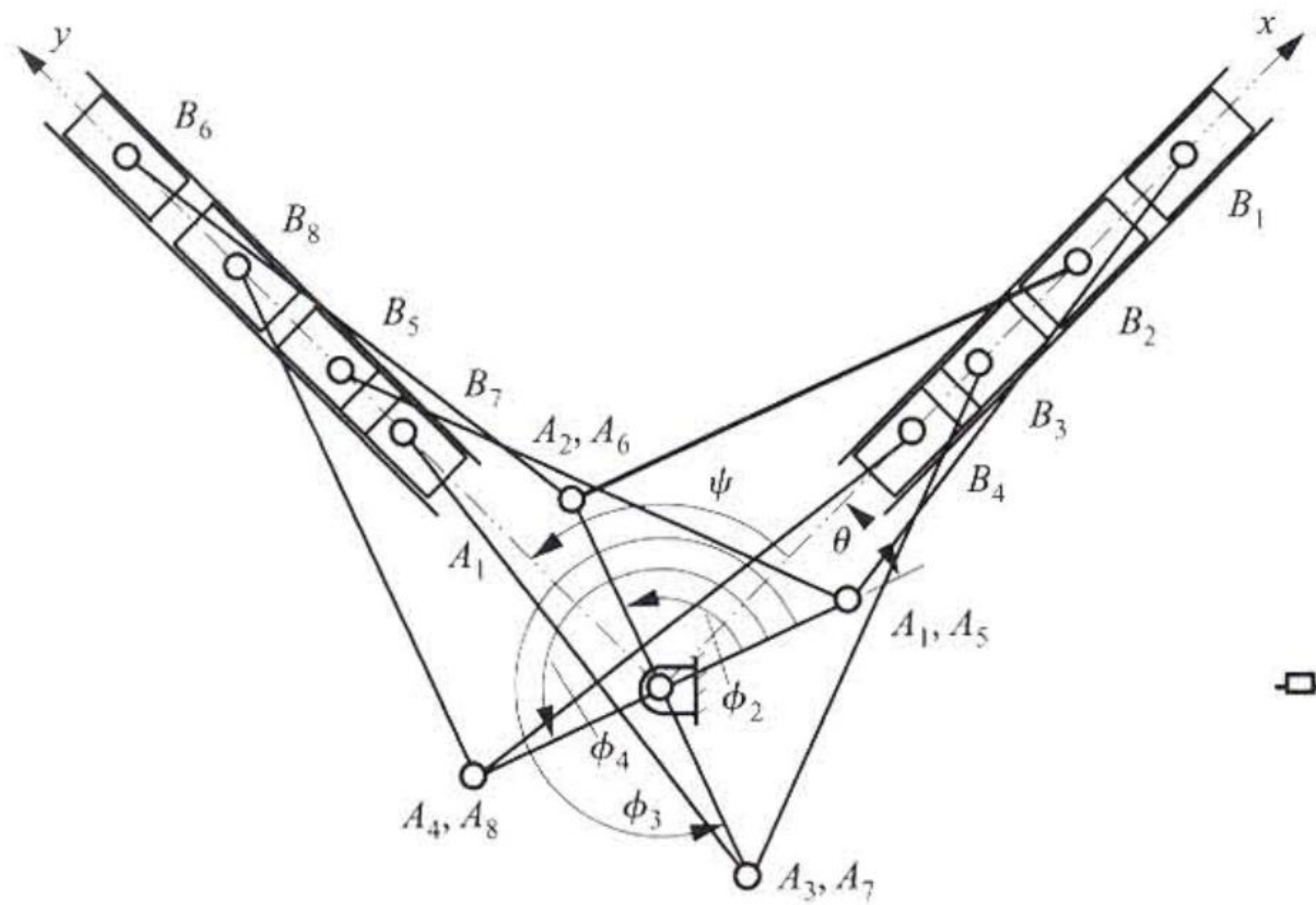
$$M = m_B R \omega^2 \left[-\frac{3a}{2} \cos \theta - \frac{a\sqrt{3}}{2} \sin \theta - \frac{R}{L} \frac{3a}{2} \cos 2\theta + \frac{R}{L} \frac{a\sqrt{3}}{2} \sin 2\theta \right]$$

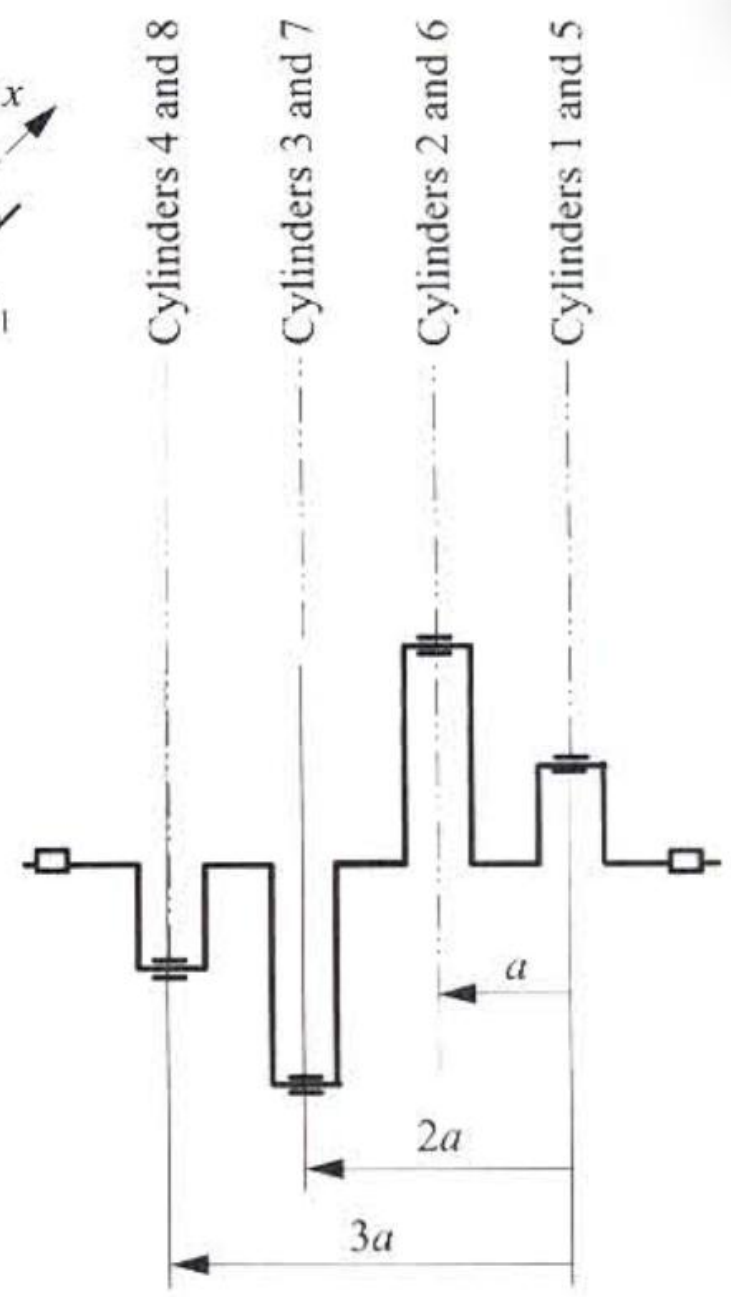
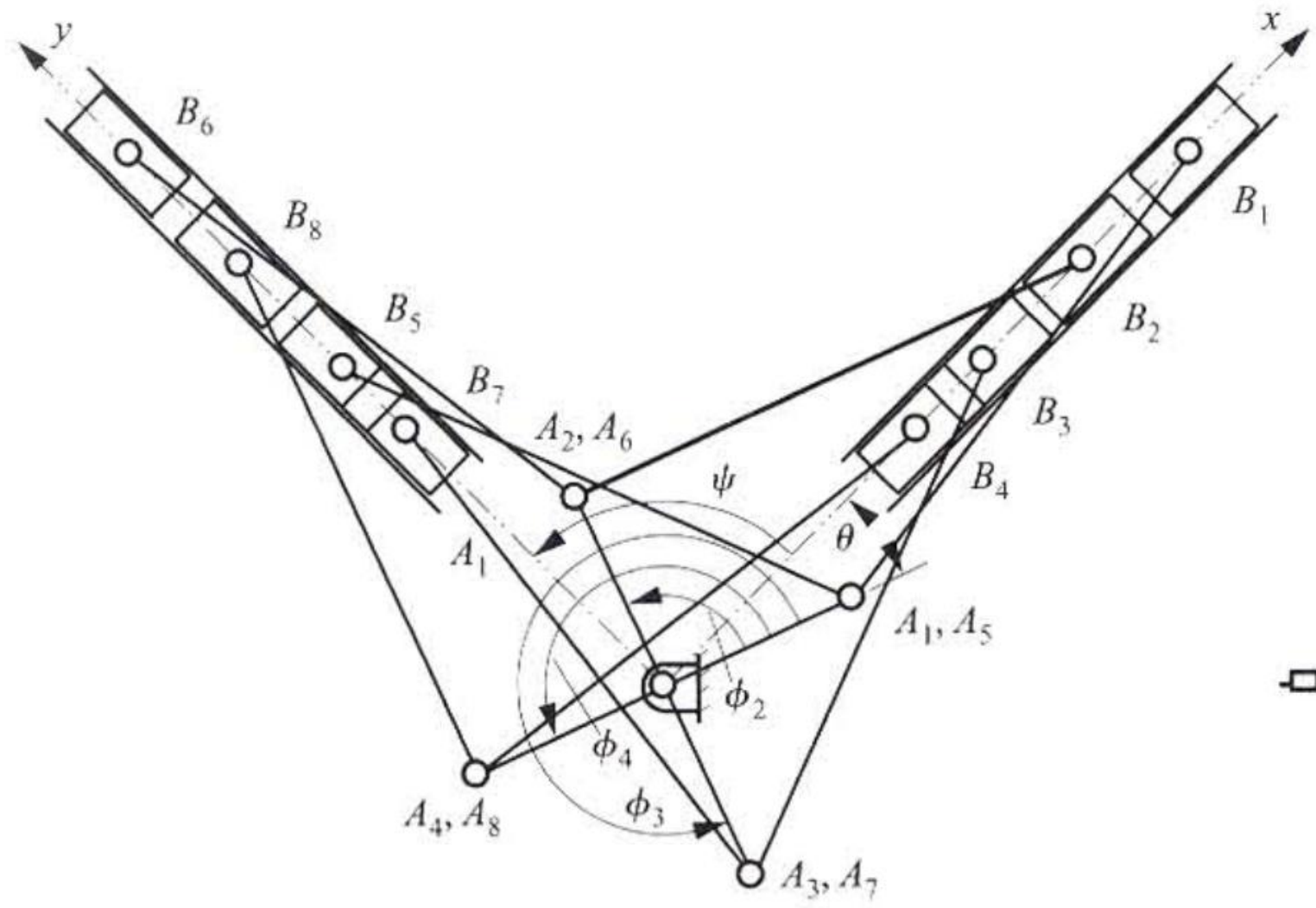
$$= m_B \frac{a R \omega^2}{2} \left[-3 \cos \theta - \sqrt{3} \sin \theta - 3 \frac{R}{L} \cos 2\theta + \frac{R}{L} \sqrt{3} \sin 2\theta \right]$$

$$M = m_B \frac{a R \omega^2}{2} \left[-2\sqrt{3} \sin(\theta + 60^\circ) - 2 \frac{R}{L} \sqrt{3} \sin(2\theta - 60^\circ) \right]$$

Primary shaking moment

Secondary shaking moment





Machine Vibration – Terminologies

- Any motion that exactly repeats itself after a certain interval of time is a periodic motion and is called a vibration.
- Free vibration or forced vibration.
- A mechanical element is said to have a **free vibration** if the periodic motion continues after the cause of the original disturbance is removed, but if a vibratory motion persists because of the continuing existence of a disturbing force, then it is called a **forced vibration**.
- Any free vibration of a mechanical system eventually ceases because of loss of energy. In vibration analysis, we often take account of these energy losses using a single factor called the **damping factor**.

Machine Vibration – Terminologies

- The **period of a vibration** is the time for a single cycle of the motion after which repetition begins; the **frequency** is the number of cycles or periods occurring in unit time.
- The **natural frequency** is the frequency of a free vibration.
- If the forcing frequency becomes equal to the natural frequency of a system, then **resonance** is said to occur.
- We shall also use the terms **steady-state** vibration to indicate that a motion is repeating itself exactly in each successive cycle and **transient vibration** to indicate a vibration-like motion that is changing in character.

Machine Vibration – Terminologies

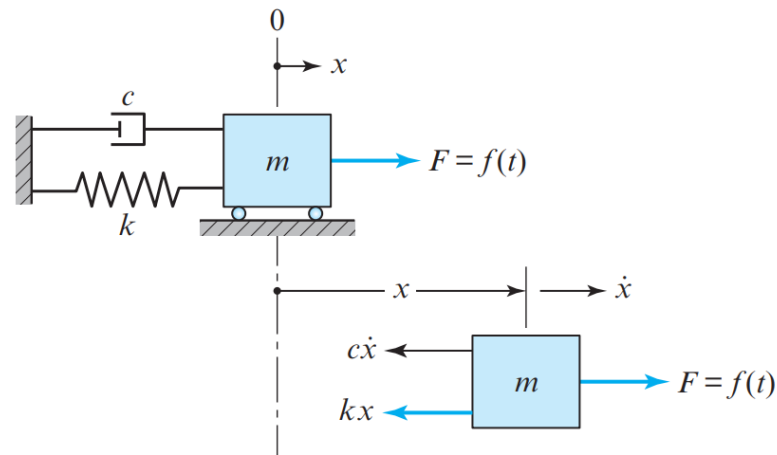
- If a **periodic force** operates on a mechanical system, the resulting motion is transient in character when the force first begins to act, but after an interval of time, the transient decays, owing to damping, and the resulting motion is termed a steady-state vibration.
- The word **response** is frequently used in discussing vibratory systems. The words response, behavior, and performance have roughly the same meaning when used in **dynamic analysis**. Thus, we can apply an external force having a sine-wave relationship with time to a vibrating system to determine how the system “responds,” or “behaves,” when the frequency of the force is varied. A plot showing the vibration amplitude along the ordinate and the forcing frequency along the abscissa is described as a **performance or response curve** for the system.

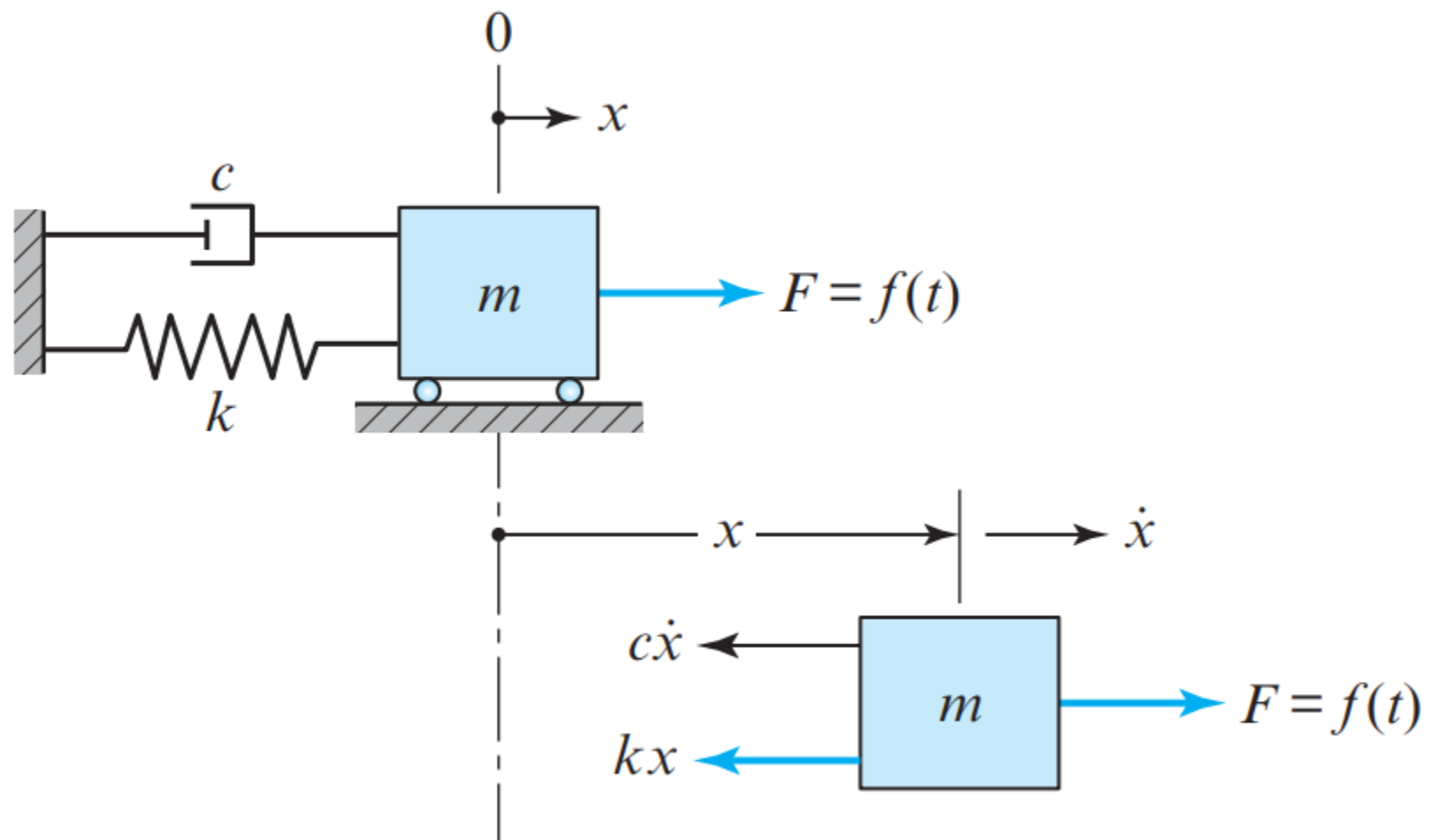
Machine Vibration – Terminologies

- Vibration analysis is sometimes called **elastic-body analysis** or **deformable-body analysis**, since, as we shall see, it is elasticity in a mechanical system that allows vibration.

Machine Vibration – Terminologies

- Figure 13.1 shows an idealized vibrating system having a mass m guided to move only in the x direction. The mass is connected to a fixed frame through the spring k , and the dashpot, c . The assumptions used are as follows:
 1. The spring and the dashpot are massless.
 2. The mass is absolutely rigid.
 3. All damping is concentrated in the dashpot.





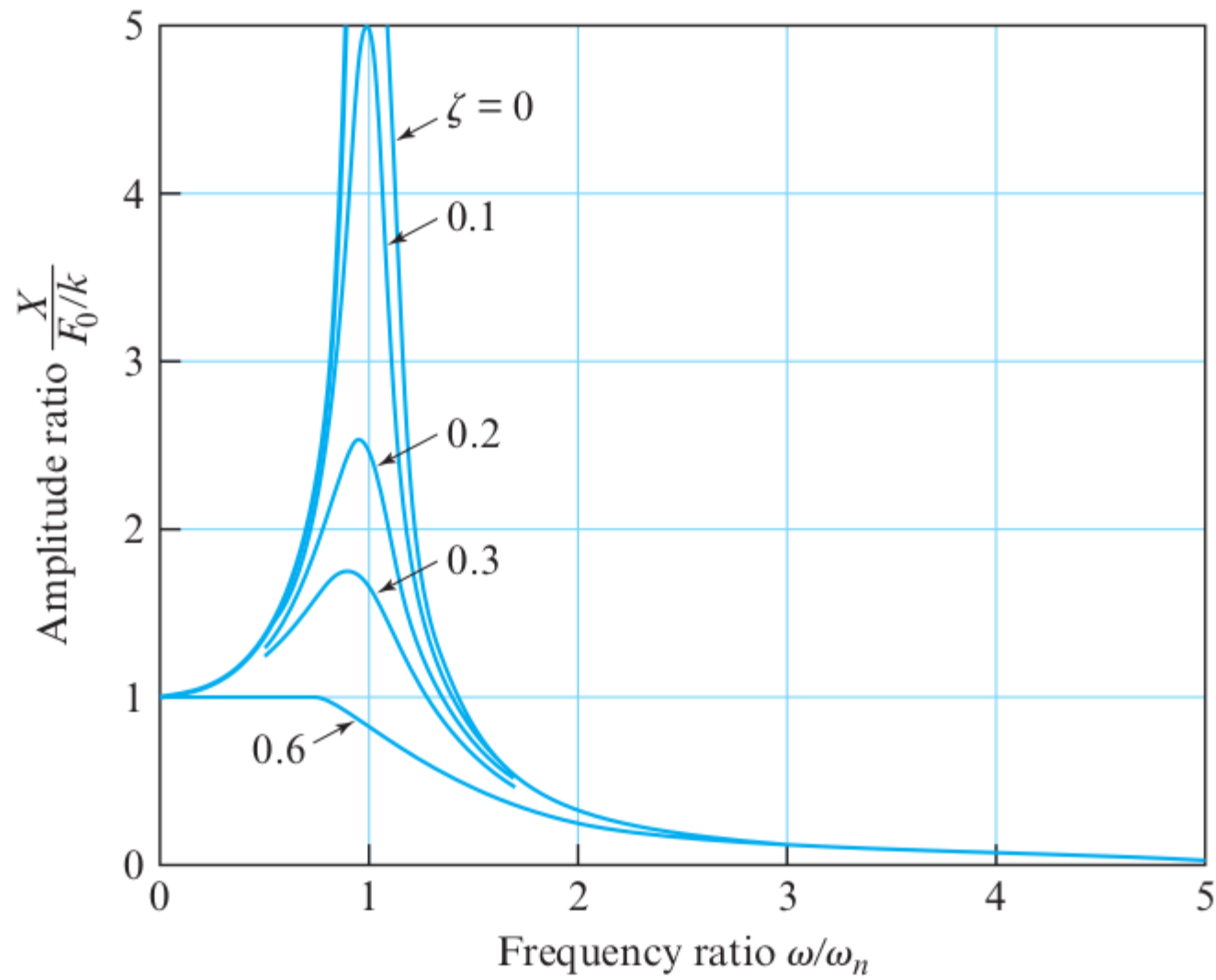


Figure 13.32 Relative displacement of a damped forced system as a function of frequency and damping ratios.

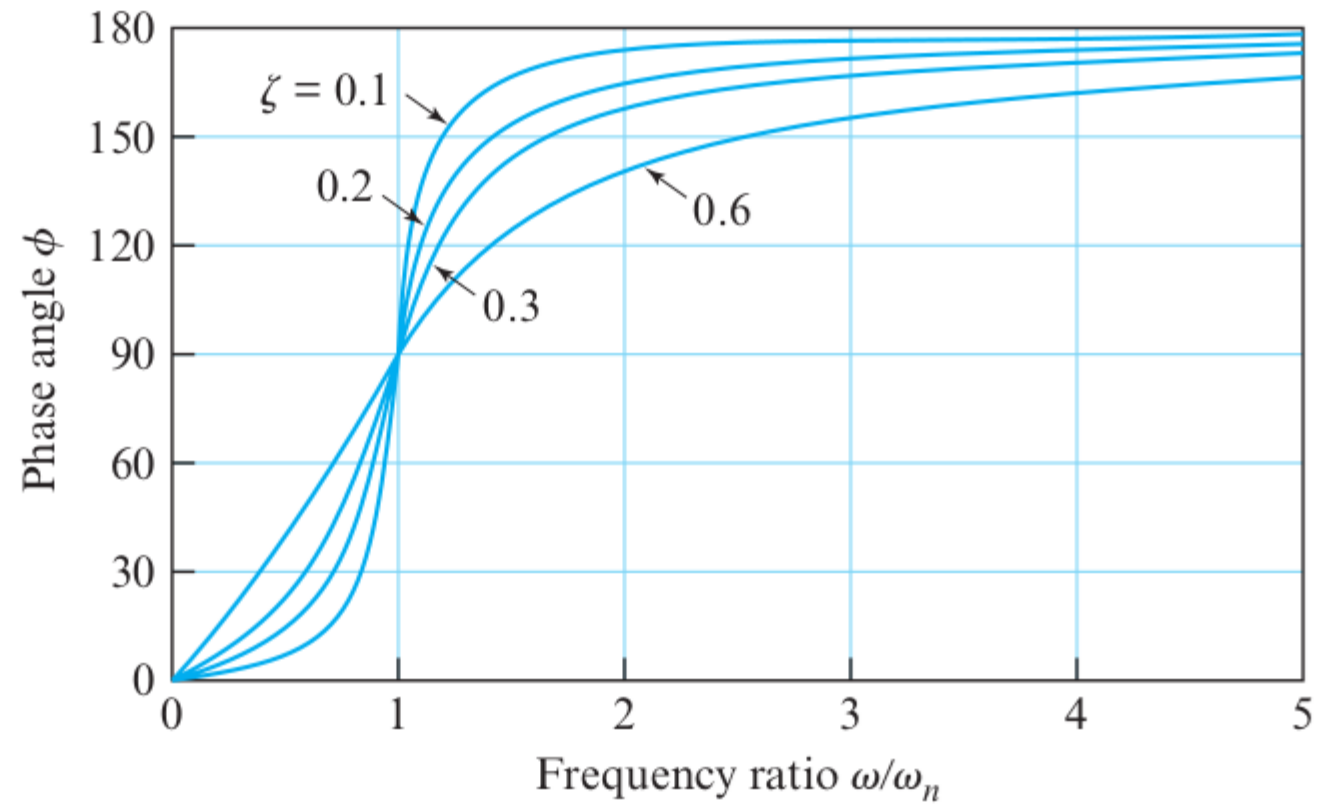


Figure 13.33 Relationship of the phase angle to frequency and damping ratios.

$$m\ddot{x} + c\dot{x} + kx = m_ue\omega^2 \cos \omega t, \quad (13.54)$$

where m = vibrating mass, m_u = unbalanced mass, and e = eccentricity. The solution to this equation is the same as that for Eq. (13.51) with F_0 replaced by $m_ue\omega^2$. Thus, the

motion of the vibrating mass is

$$x = \frac{m_u e \omega^2 \cos(\omega t - \phi)}{\sqrt{(k - m\omega^2)^2 + c^2 \omega^2}}. \quad (13.55)$$

If we make the same transformation we used to obtain Eq. (13.52), we get

$$\frac{mX}{m_u e} = \frac{\omega^2 / \omega_n^2}{\sqrt{[1 - (\omega^2 / \omega_n^2)]^2 + (2\zeta \omega / \omega_n)^2}}. \quad (13.56)$$

This is plotted in Fig. 13.34. In one case, the main mass is excited by a rotating unbalanced mass m_u . The plot gives the nondimensional absolute response of the mass, m . In another case, as shown in Fig. 13.35, the mass is excited by moving the far end of a spring and dashpot, which is connected to the mass, with harmonic motion. The plot then gives the nondimensional response Z/y_0 relative to the end of the spring-dashpot combination. This plot shows that for high speeds where the frequency ratio is greater than unity, we can reduce the amplitude only by reducing the mass and eccentricity of the rotating unbalance.

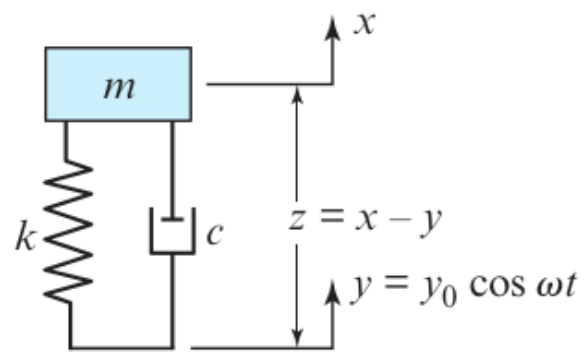


Figure 13.35

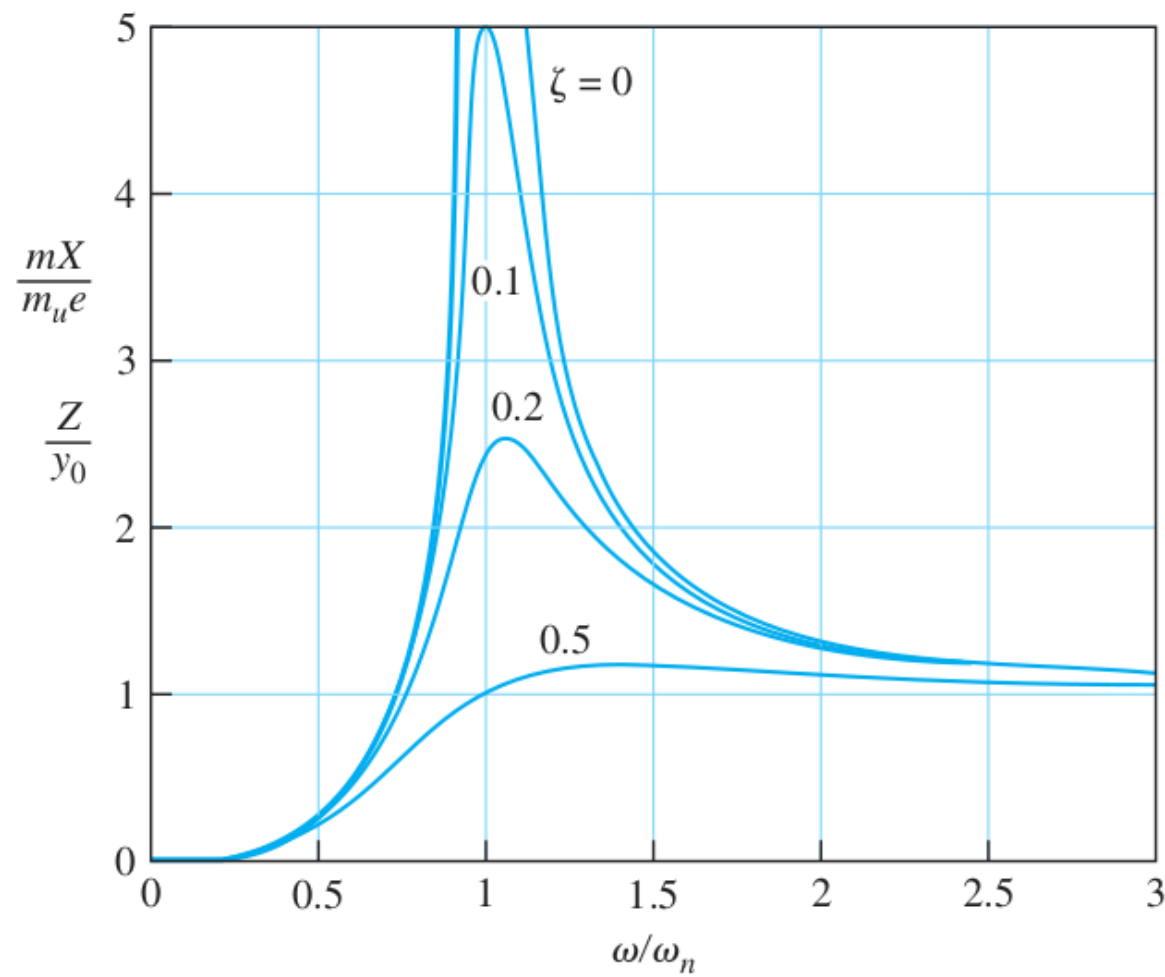


Figure 13.34 Plot of magnification factor versus frequency ratio.